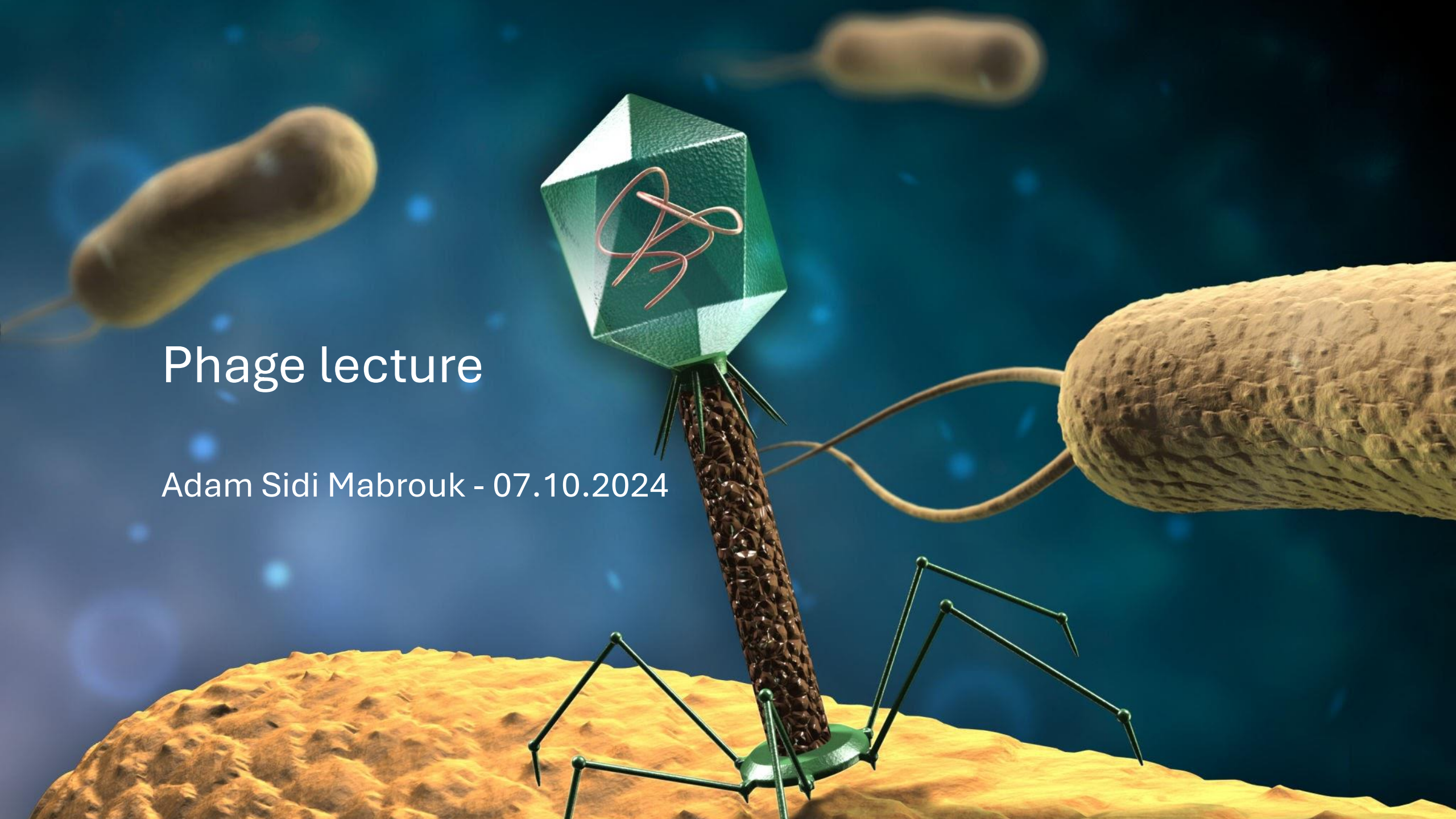


Phage lecture

Adam Sidi Mabrouk - 07.10.2024

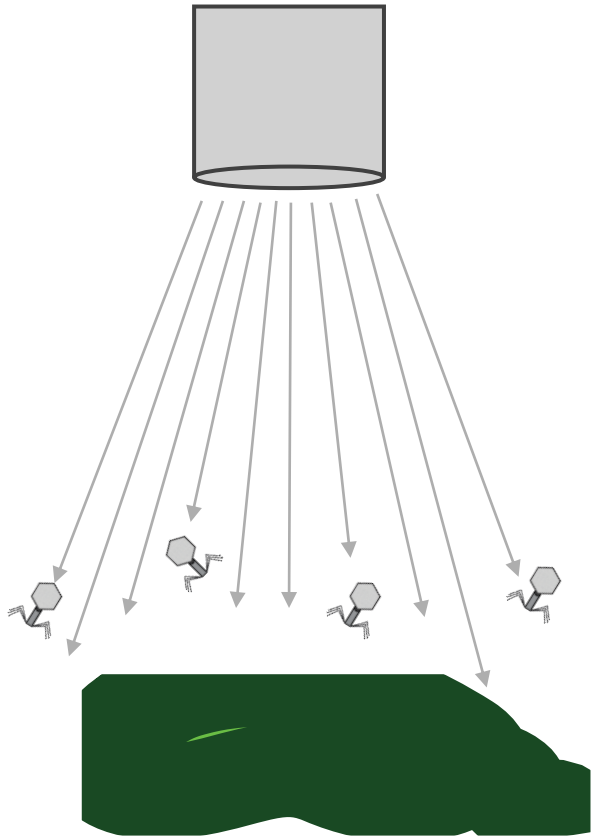
Who am I?

- Adam Sidi Mabrouk
- PhD student in the lab of Ariane Briegel
- Use Cryo-electron microscopy to image bacteriophages



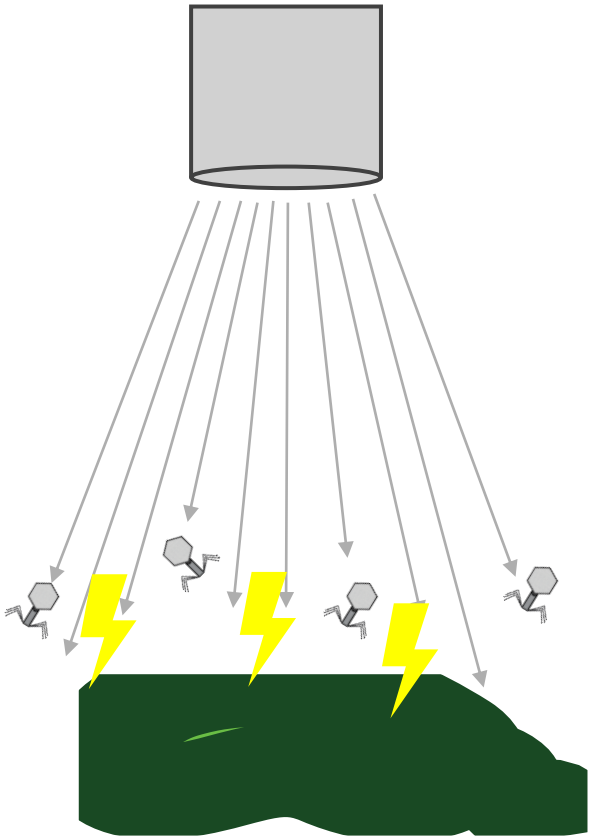




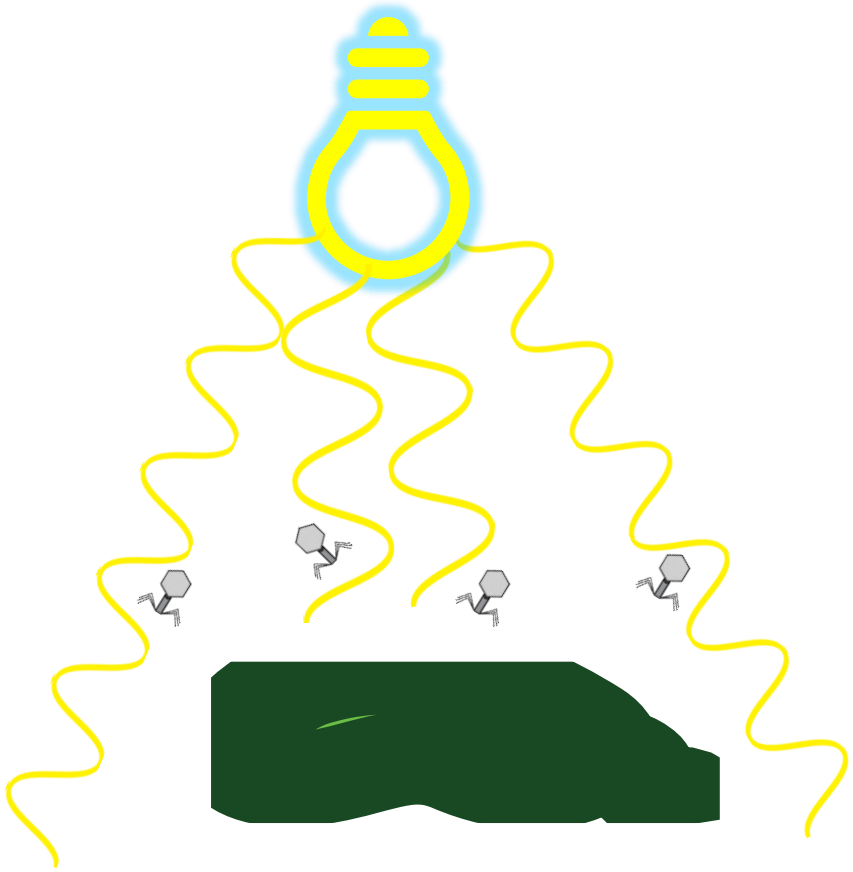


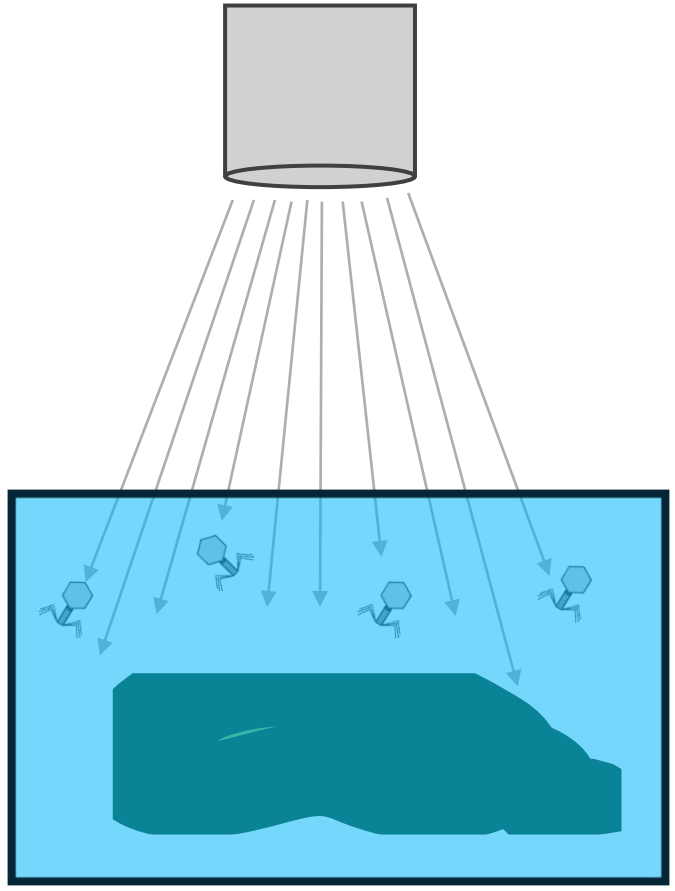




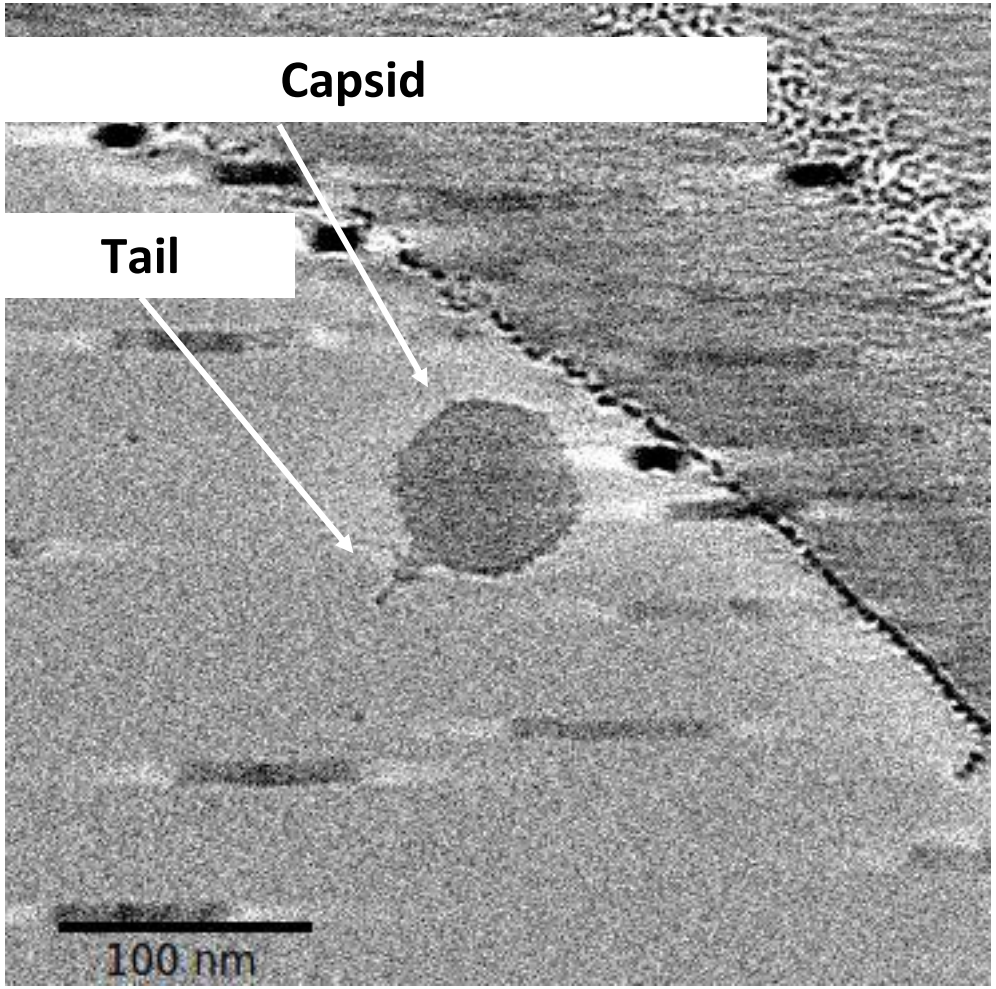


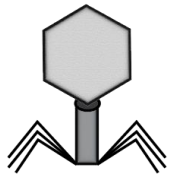


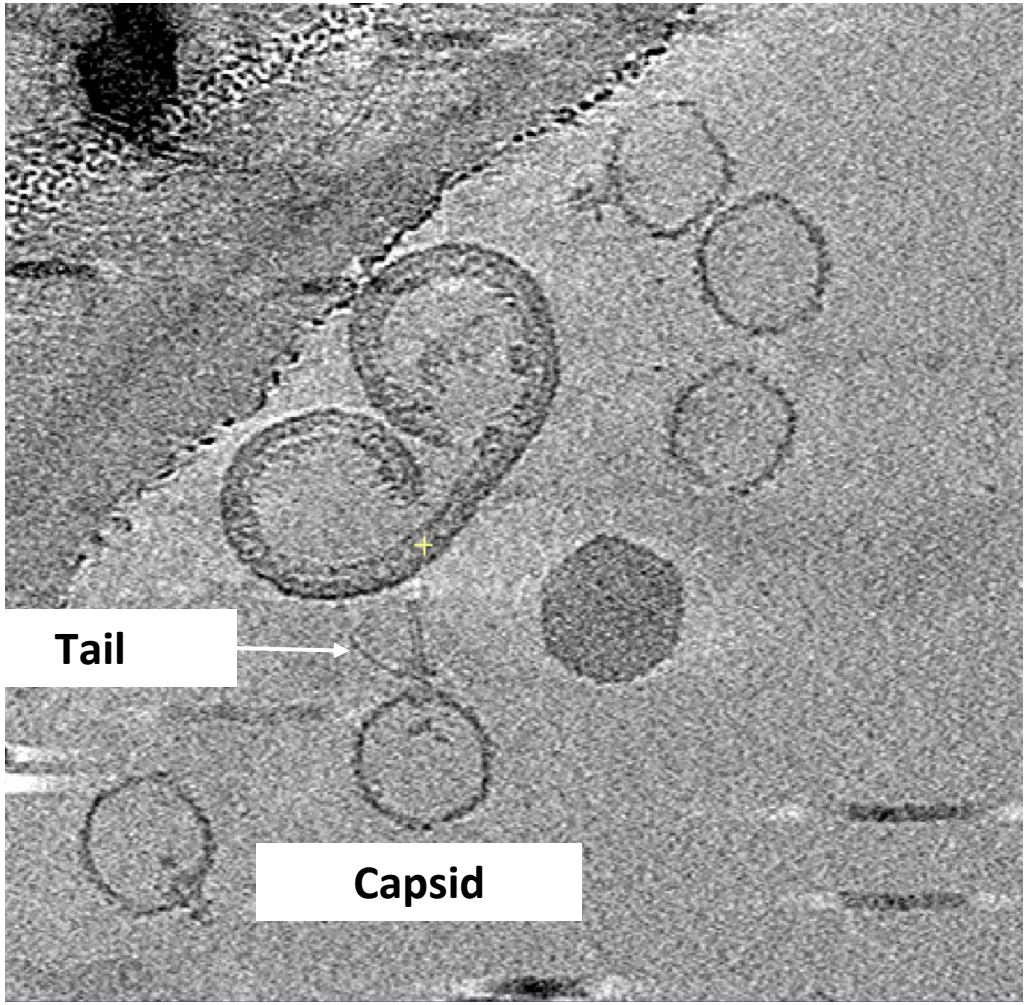




Bacteriophages imaged by electron microscopy







What is the most common biological entity on earth?

Viruses!

~ 10^{31} in total

10,000,000,000,000,000,000,000,000,000,000

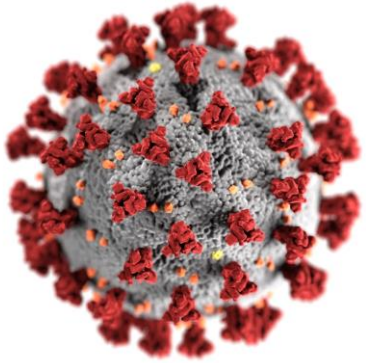
To put this into perspective, the number of viruses in just one litre of surface seawater (~10^{10}) exceeds the current number of humans on the planet (7.8×10^9). Viruses represent the smallest known biological entities in the oceans, an ecosystem containing organisms with lengths spanning nine orders of magnitude. While viruses represent 94% of the nucleic-acid-containing particles in the oceans, they only comprise a miniscule proportion of the total biomass due to their remarkably small sizes. Although the size of marine viruses varies widely, the average diameter of marine viruses is ~54 nm. To provide context, it would take ~55 million viruses of this size to cover the period at the end of this sentence. Nonetheless, the oceans contain 100 billion times as many viruses as there are sand grains on Earth (~10^{19}), and if all ~10^{30} of these marine viruses were lined up side-by-side, they would extend for 5.4×10^{22} m, which is the equivalent of 12 trillion roundtrips from Earth to Mars.

What are viruses?

An **infectious agent** that replicates only **inside the living cells** of an organism

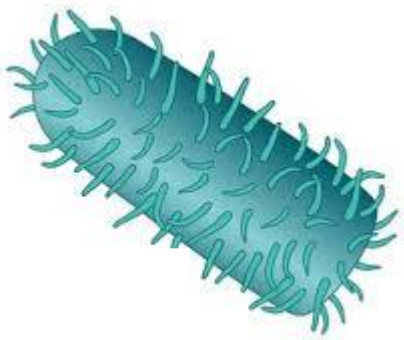
Can infect all living organisms?

Genome size:
30kb
ORFs:
50



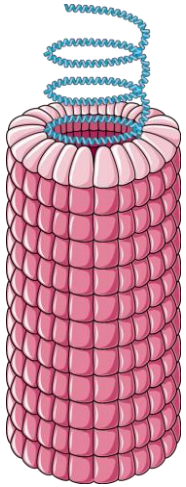
Corona

Genome size:
12kb
ORFs:
5-10



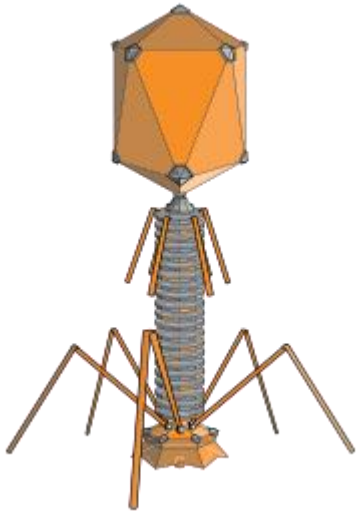
Rabies

Genome size:
6.5kb
ORFs:
4



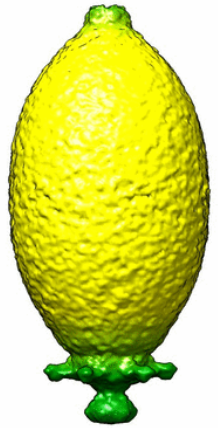
Tabacco mosaic

Genome size:
168kb
ORFs:
300



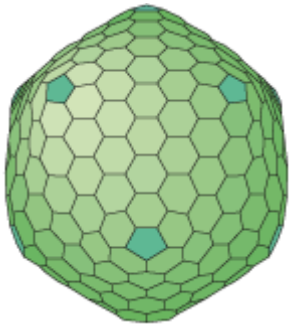
T4

Genome size:
15.5kb
ORFs:
35



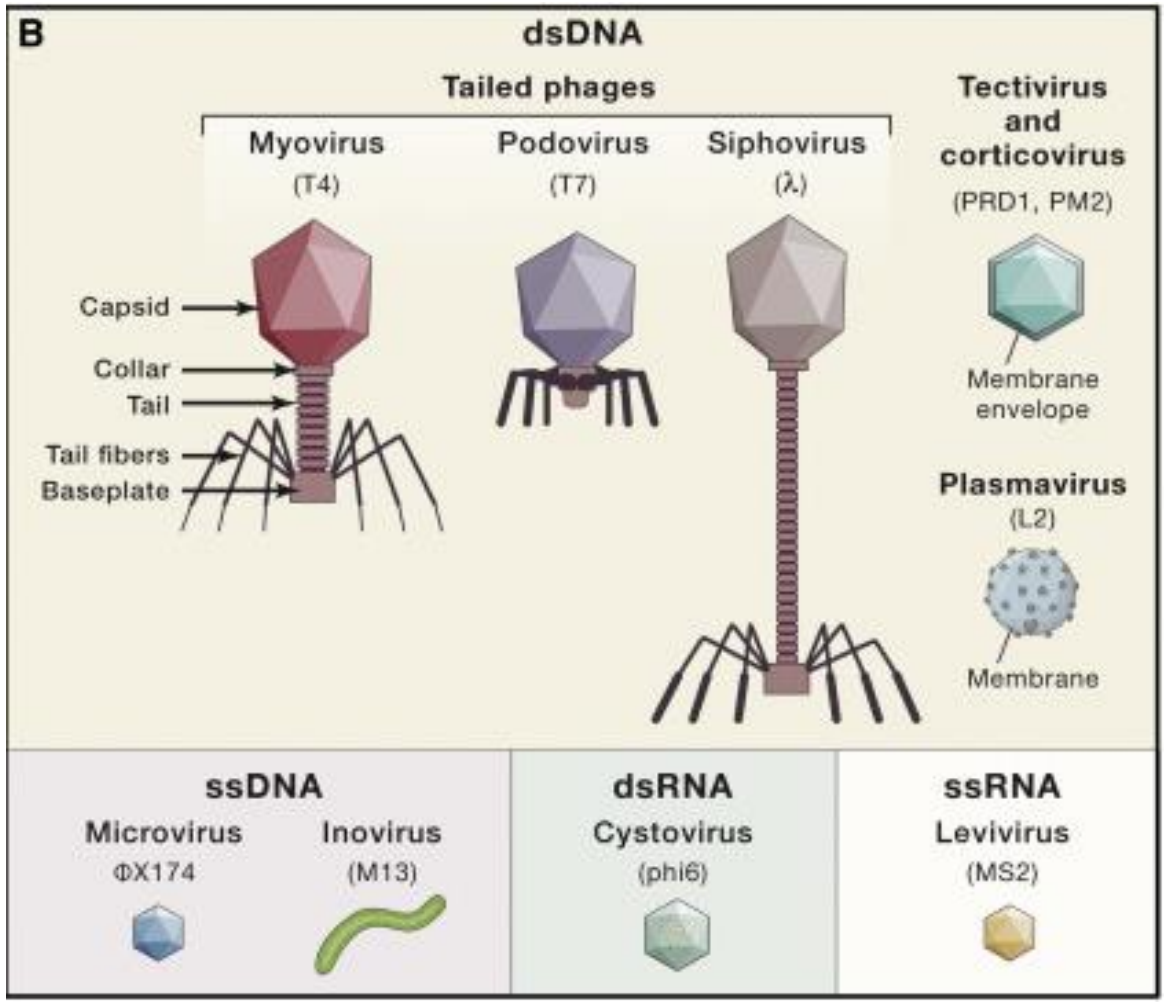
His1

Genome size:
18kb
ORFs:
21

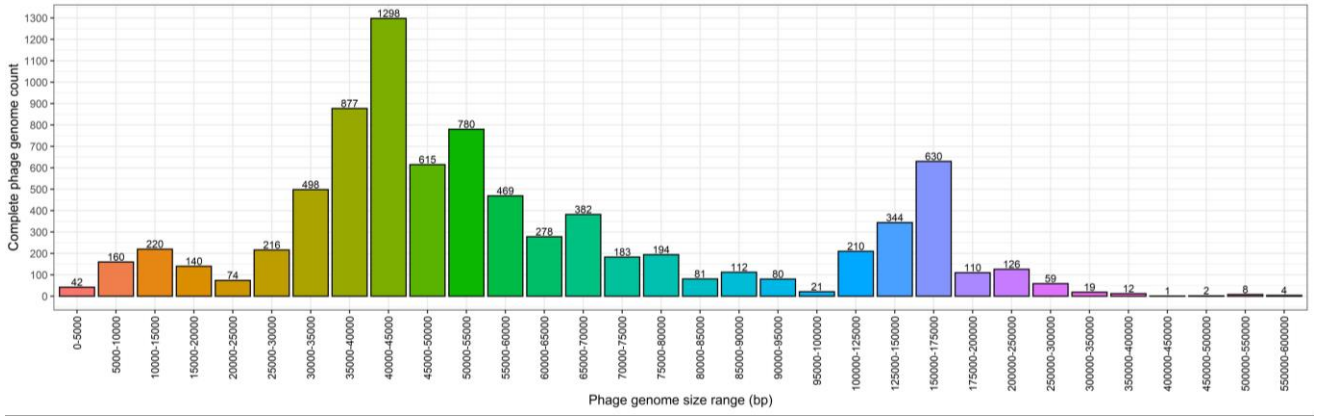


Sputnik

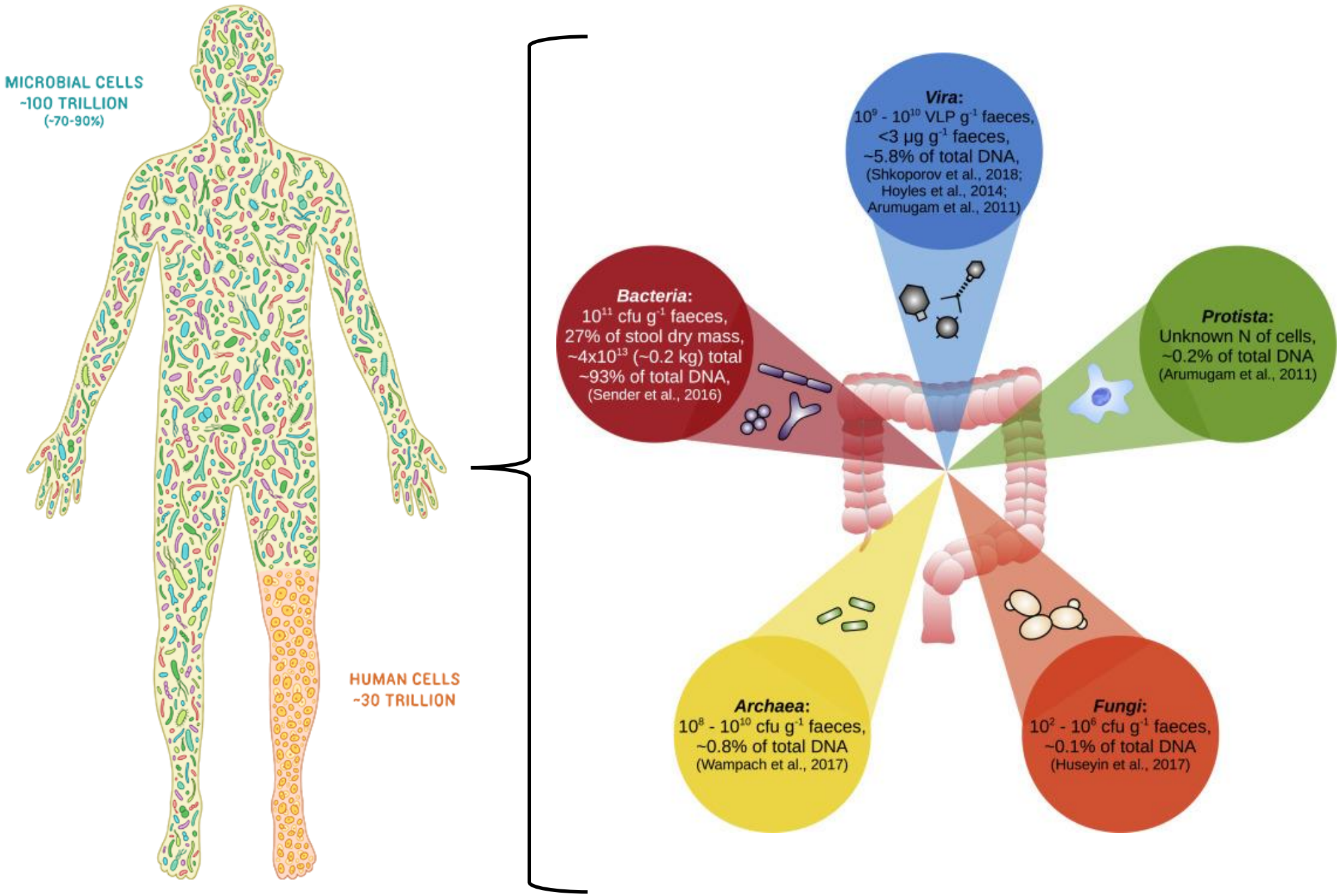
Diversity of phages



- Genome sizes range from 5kb to 600kb
- > 200kb are described as jumbo phages




MICROBIAL CELLS
~100 TRILLION
(~70-90%)
HUMAN CELLS
~30 TRILLION


MICROBIAL CELLS
~100 TRILLION
(~70-90%)
HUMAN CELLS
~30 TRILLION
Vira:
10^9 - 10^10 VLP g^-1 faeces,
<3 µg g^-1 faeces,
~5.8% of total DNA,
(Shkoporov et al., 2018;
Hoyles et al., 2014;
Arumugam et al., 2011)
Bacteria:
10^11 cfu g^-1 faeces,
27% of stool dry mass,
~4x10^13 (~0.2 kg) total
~93% of total DNA,
(Sender et al., 2016)
Protista:
Unknown N of cells,
~0.2% of total DNA
(Arumugam et al., 2011)
Archaea:
10^8 - 10^10 cfu g^-1 faeces,
~0.8% of total DNA
(Wampach et al., 2017)
Fungi:
10^2 - 10^6 cfu g^-1 faeces,
~0.1% of total DNA
(Huseyin et al., 2017)

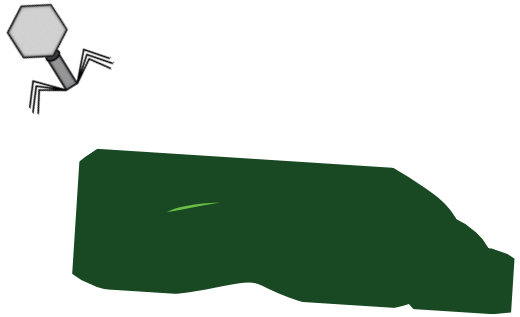
First report of phages

- Outbreak of cholera in 1892
- People that drank “holy water” from the ganges river were cured
- Water was full of bacteriophages
- How does the phage cure the people that drank from it?

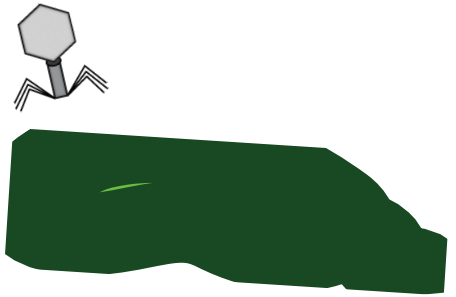


Lifecycle of a lytic bacteriophage

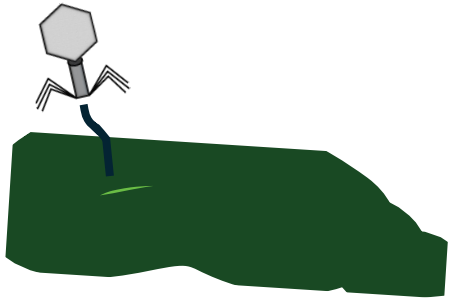
1. Recognition



2. Binding



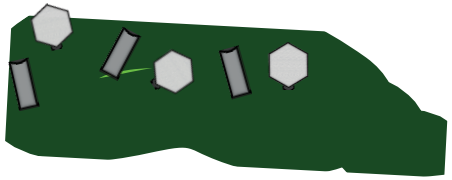
3. Injection of DNA/RNA



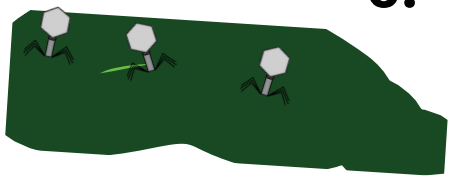
4. Replication of genetic material



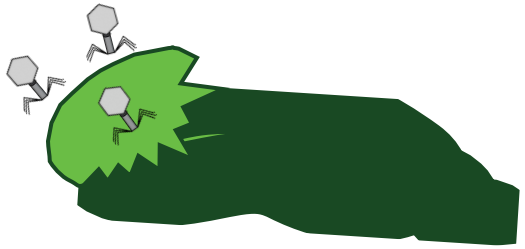
5. Production of phage proteins



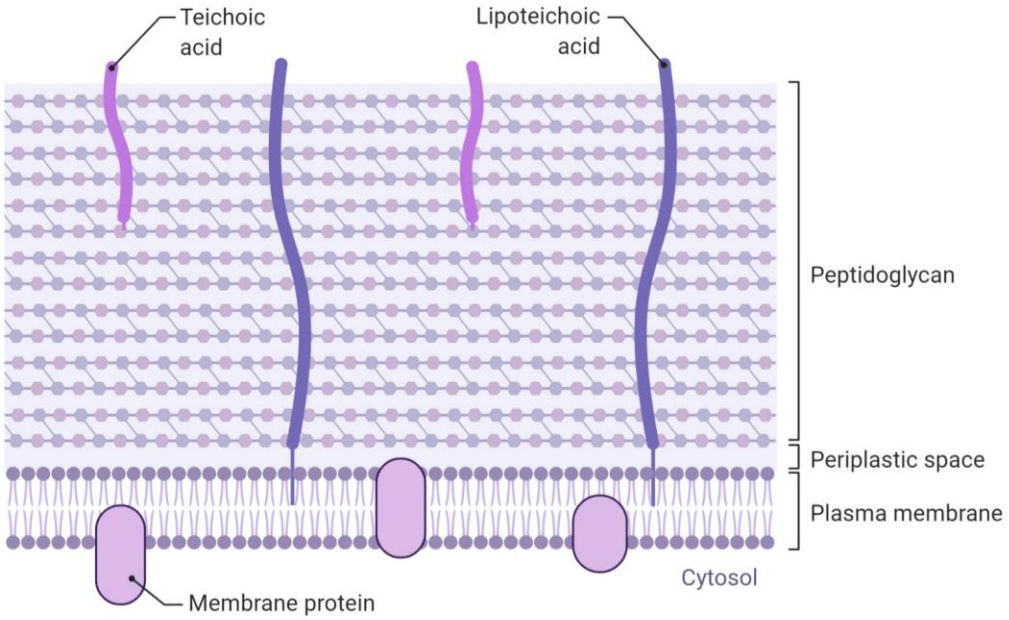
6. Assembly of new phages



7. Bacterial lysis and phage propagation



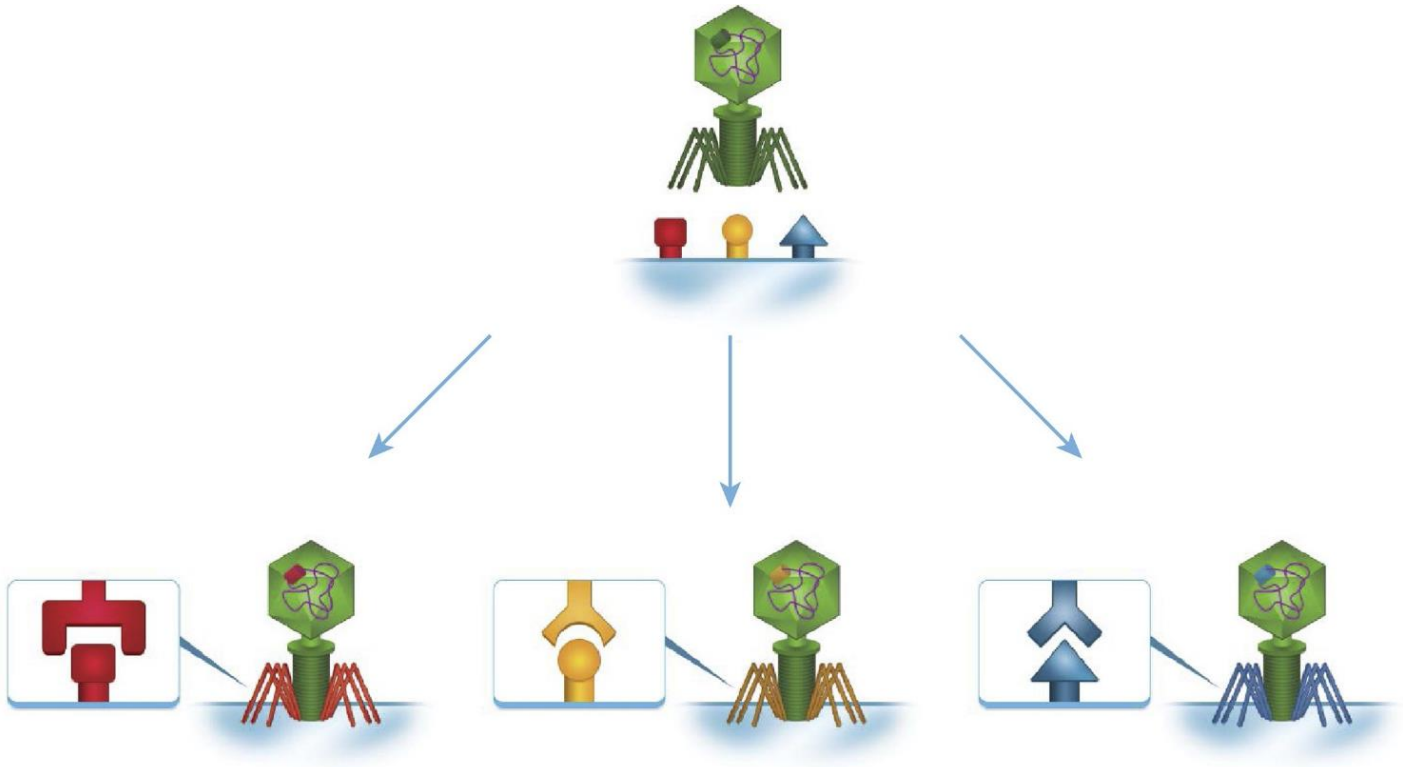
Bacteriophages attach to the bacterial cell wall



Gram-positive

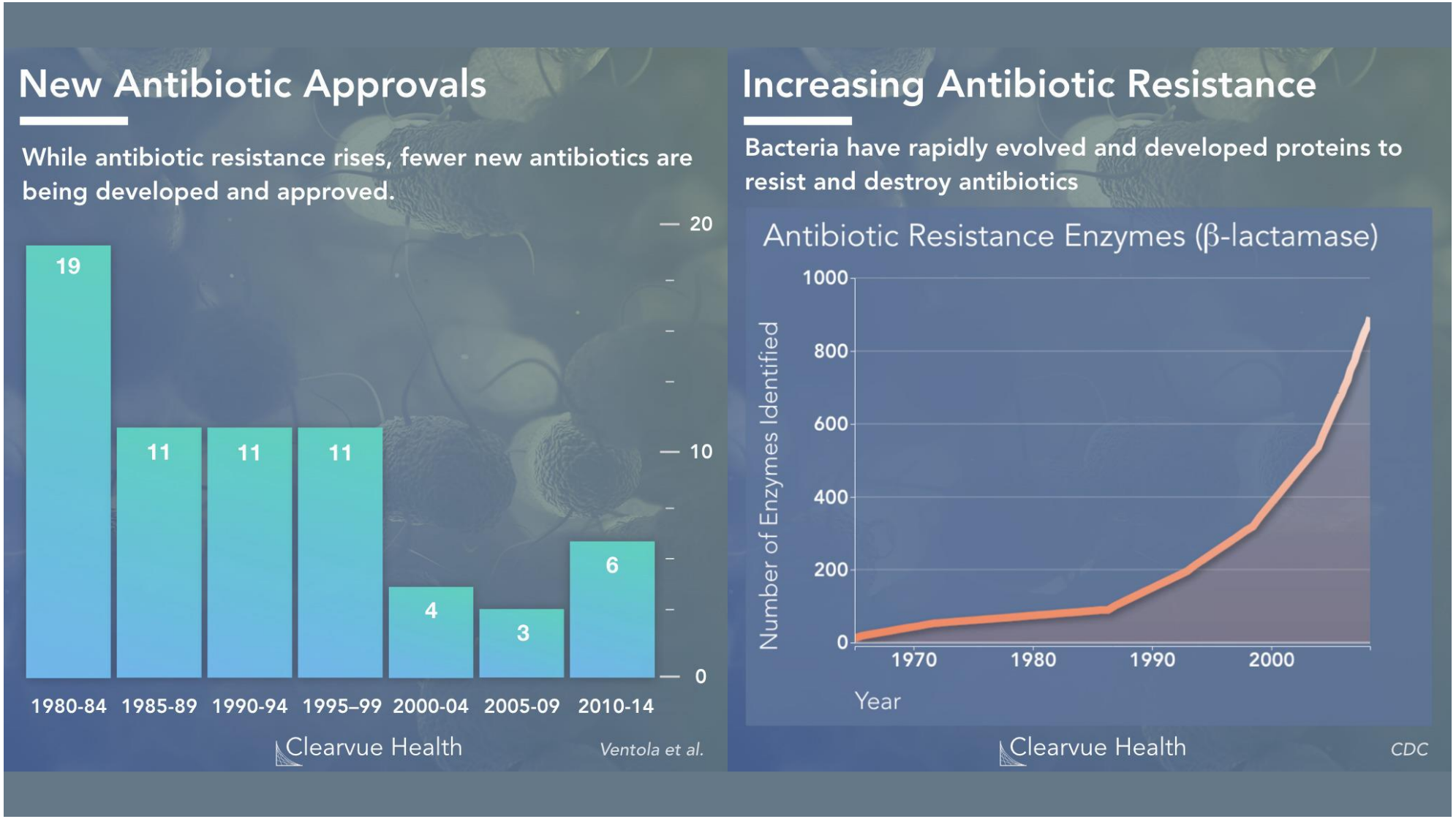
Ongenae, V., Briegel, A., & Claessen, D. (2021). Cell wall deficiency as an escape mechanism from phage infection. *Open Biology*, *11*(9), 210199.

Bacteriophages – specificity for their host



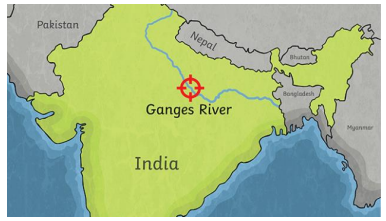
Goren et al. 2015

Could phages be the new antibiotics?



History of phage therapy

Discovery of “holy water”



Used to treat Salmonella infected chickens



First clinical trials to treat cholera

Age of antibiotics

Antibiotic resistance officially declared a crisis

1892

1917

1919

1921

1927

1928

2014

Now

D’herelle discovers anti-shigella “microbe”

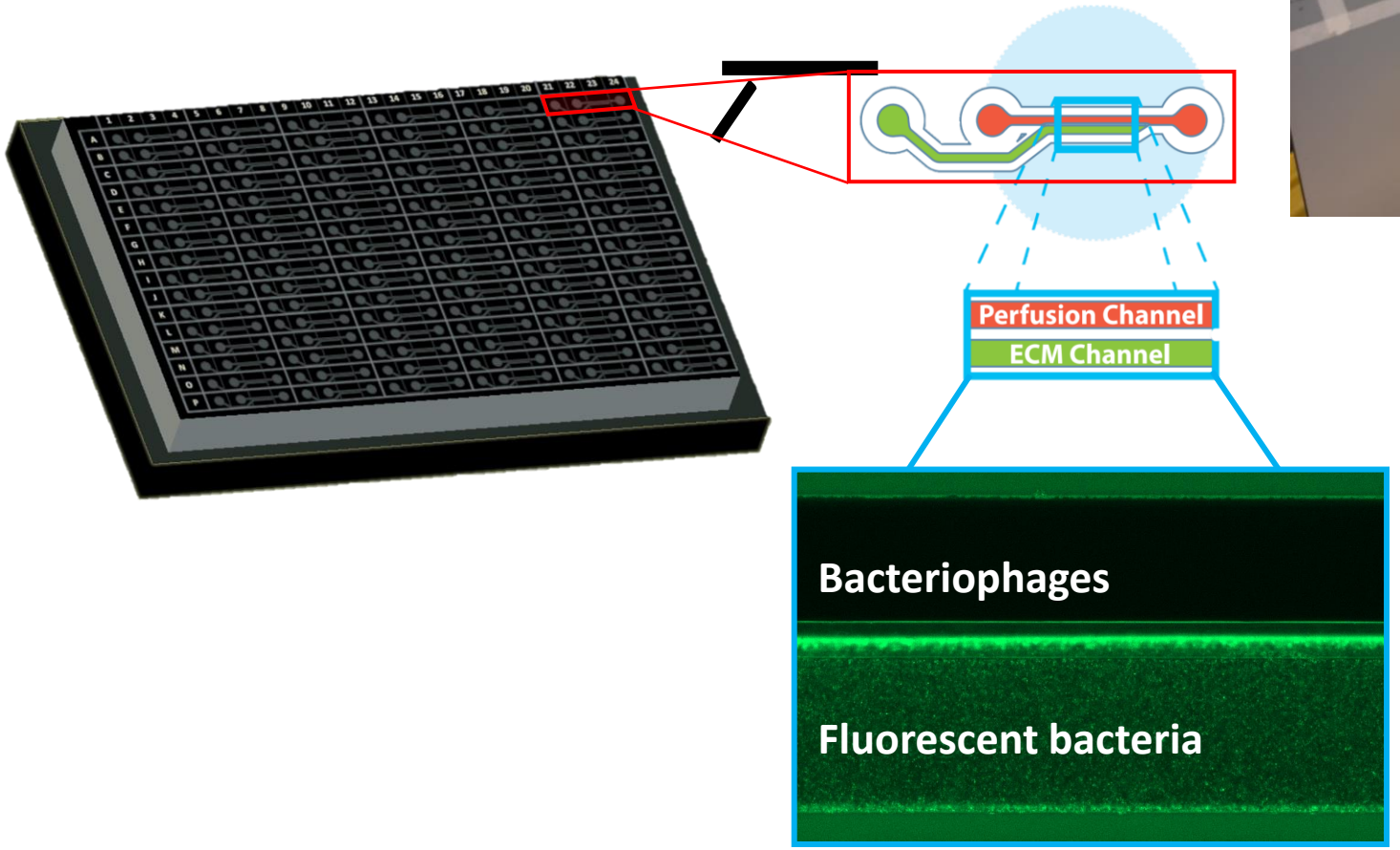
5 patients treated against shigella infection

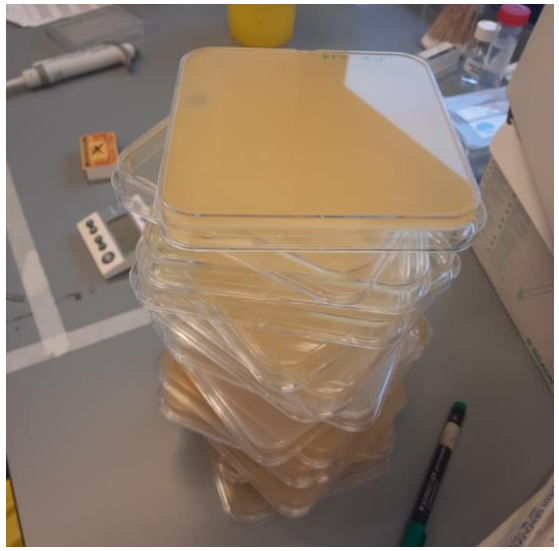
Discovery of antibiotics

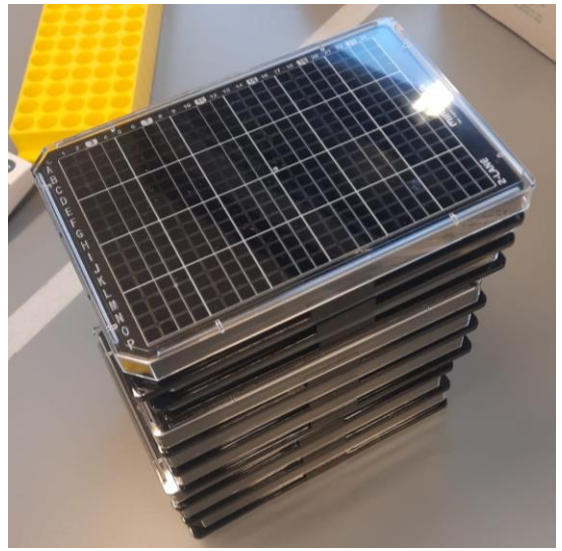
Resurgence in phage therapy and clinical trials

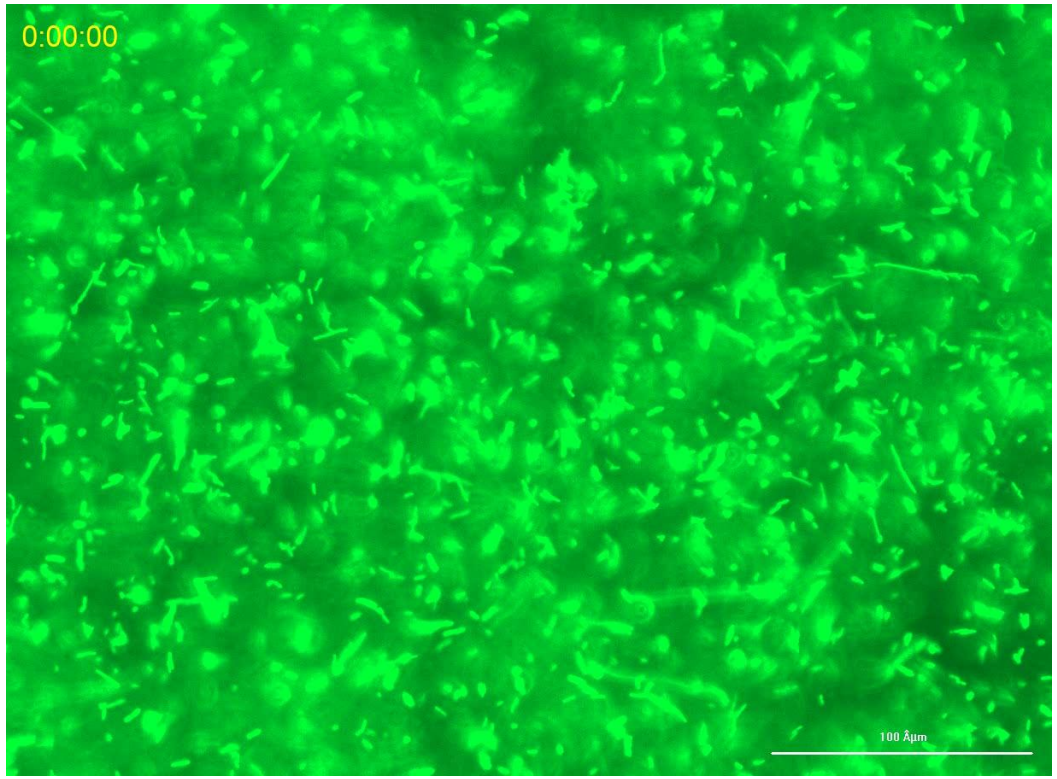
MIMETAS

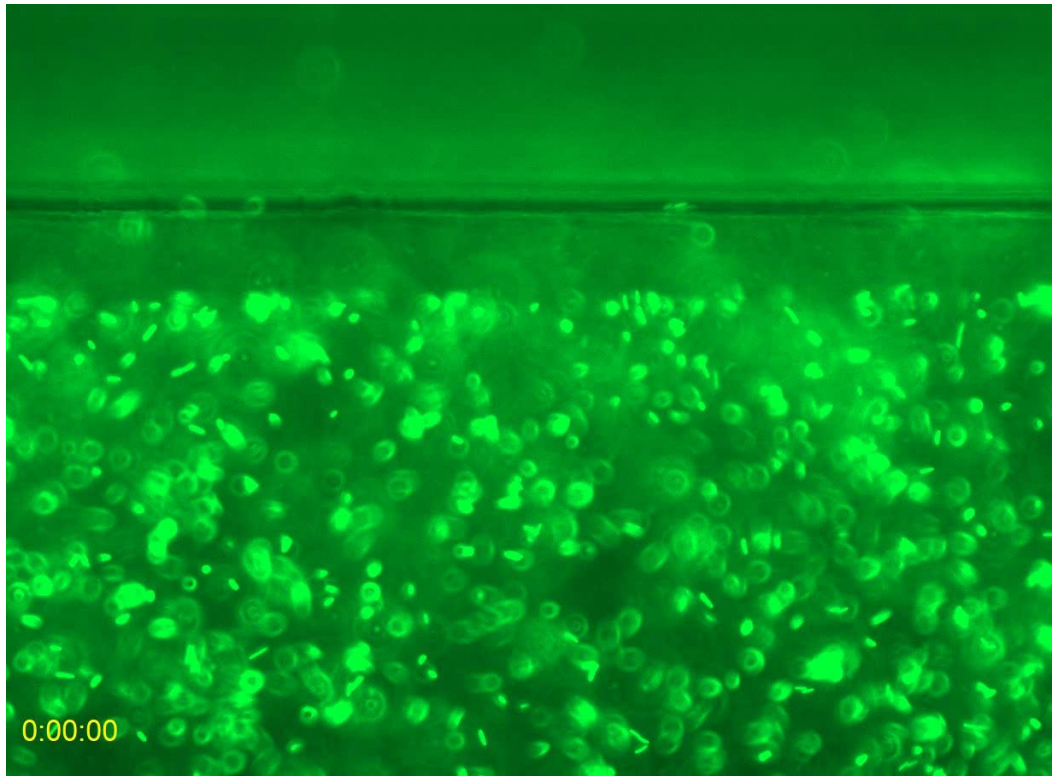
the organ-on-a-chip company

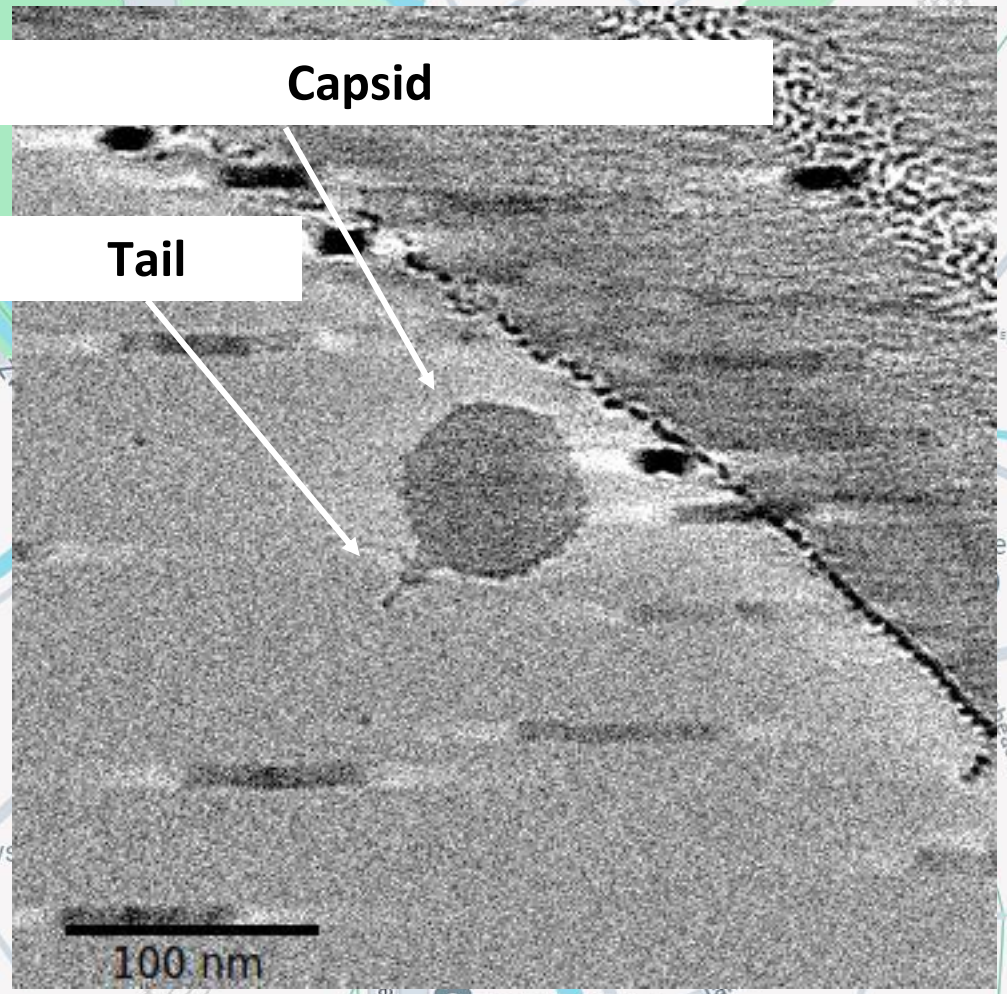


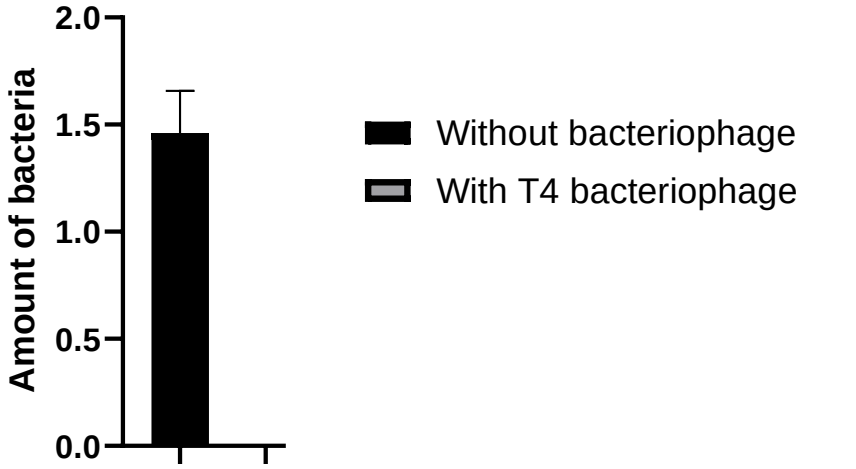


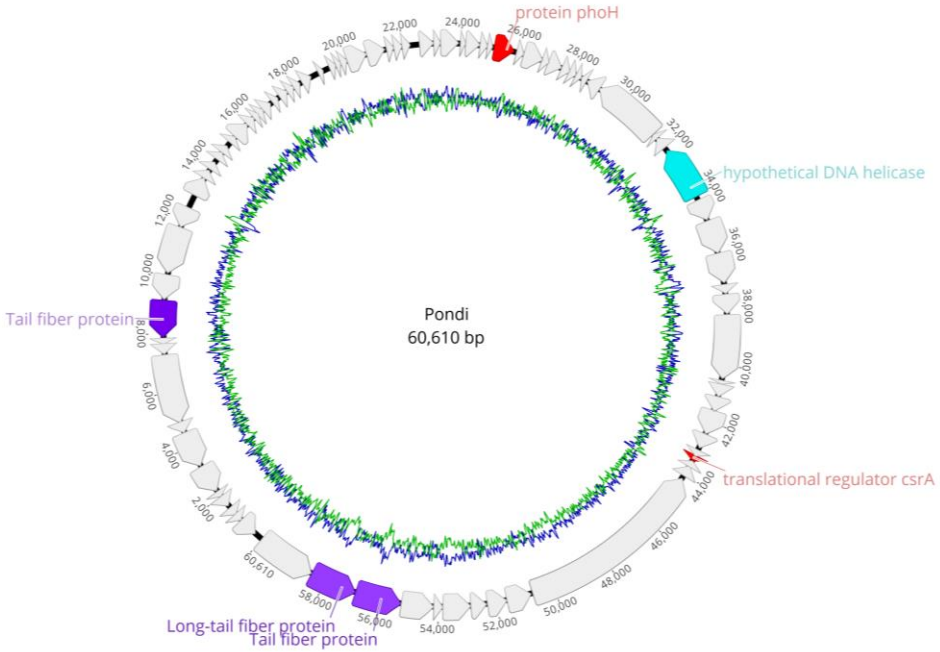



0:00:00
100 Âµm


0:00:00


Capsid
Tail
100 nm


After 24 hours of growth
Amount of bacteria
2.0
1.5
1.0
0.5
0.0
Without bacteriophage
With T4 bacteriophage


protein phoH
hypothetical DNA helicase
translational regulator csrA
Tail fiber protein
Long-tail fiber protein
Tail fiber protein
Pondi
60,610 bp

Current ICTV Taxonomy Release

Taxonomy Browser

This taxonomy browser provides access to the current virus taxonomy. This page will be updated whenever a new taxonomy release has been approved by the ICTV.

Taxonomy Search | Taxonomy Browser | Download Current Taxonomy Spreadsheet (MSL)

Unless the "Select to search across all ICTV releases" button is checked below, your search will be against the taxonomy release indicated below the search box (or below the search result set, if present). To search against the current release, refresh the page.

Search taxonomy... Search Reset

Select to search across all ICTV releases

Virus Taxonomy: 2022 Release

EC 54, Online meeting, July 2022
Email ratification March 2023 (MSL #38)
6 realms, 10 kingdoms, 17 phyla, 2 subphyla, 40 classes, 72 orders, 8 suborders, 264 families, 182 subfamilies, 2818 genera, 84 subgenera, 11273 species

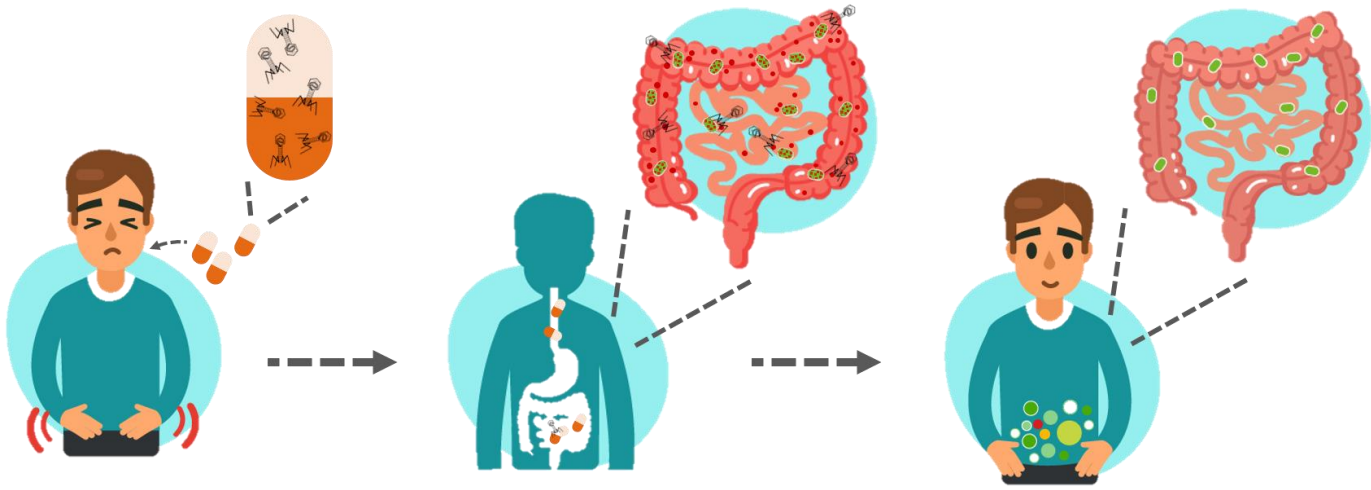
Expand ranks to show Realm Hide ranks above Realm Go

+ **Realm:** *Adnaviria*
+ **Realm:** *Duplodnaviria*
+ **Realm:** *Monodnaviria*
+ **Realm:** *Riboviria*
+ **Realm:** *Ribozyviria*
+ **Realm:** *Varidnaviria*
+ **Class:** *Naldaviricetes*
+ **Family:** *Alphasatellitidae*
+ **Family:** *Ampullaviridae*

Could phages be the new antibiotics? -> **Phage therapy**

Phages are:

- Drugs that amplify themselves
- Non-harmful to humans
- Phages are cleared by the body
- Fewer side-effects
- Can be highly specific (but host range varies markedly)
  -> keeps microbiome intact



Een behandeling bij de kliniek of vanuit huis?



Een behandeling met bacteriofagen in Nederland vanuit huis

Een behandeling thuis kan plaats vinden bij een veelvoorkomende bacteriële infectie.

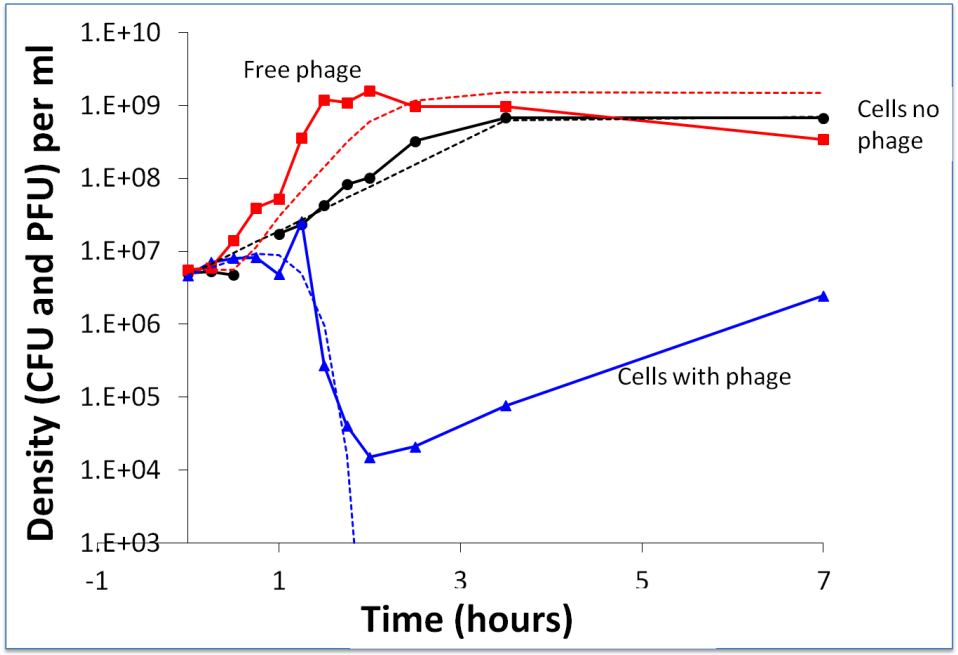
De bacteriofagen die nodig zijn worden opgestuurd en afgeleverd aan huis. Bacteriofagen in Nederland zijn dus mogelijk. Uiteraard is onderzoek middels een E-Swab een vereiste.

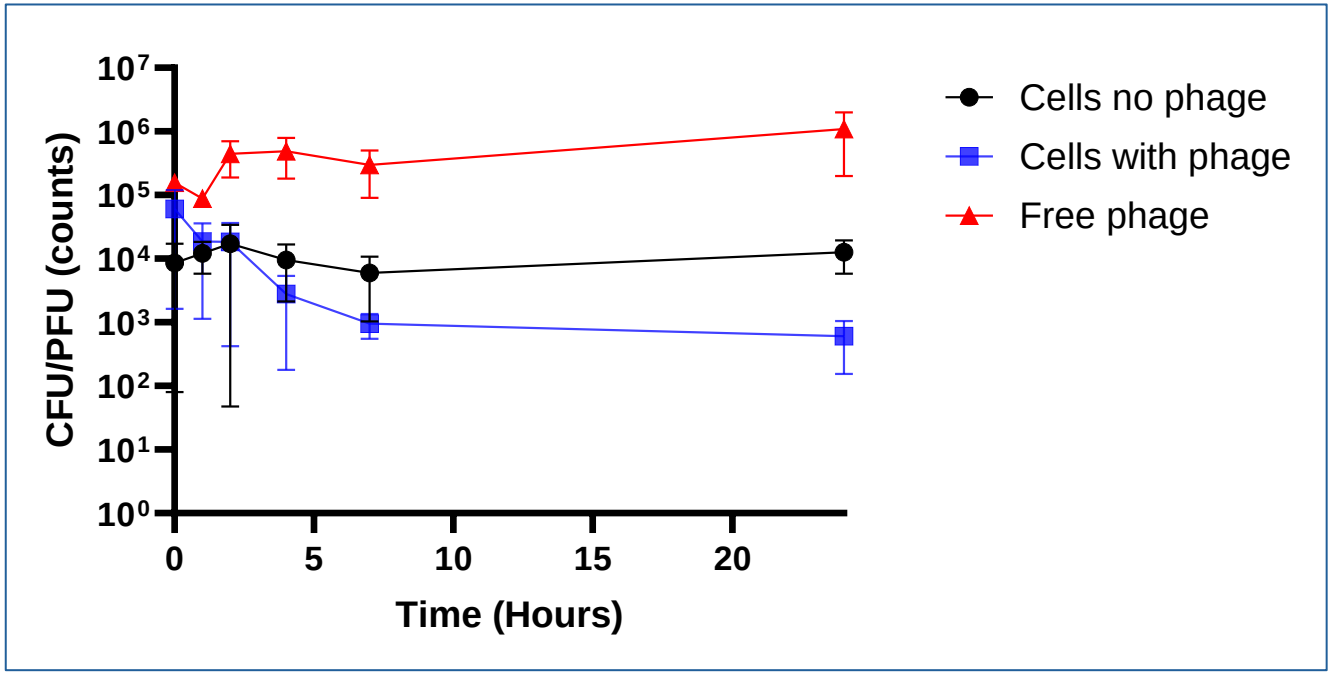
Opstarten van een behandeling in de Eliava kliniek

Het opstarten van een behandeling met bacteriofagen bij de Eliava kliniek zelf ziet er als volgt uit:

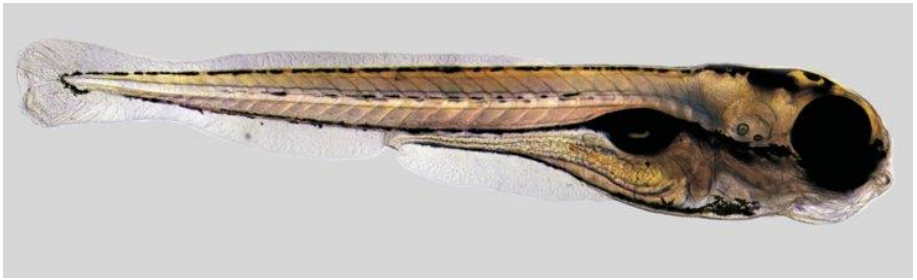
- de reis (indien nodig)
- het verblijf (incl. transfer en evt. tolk)
- bezoek aan de kliniek, gelijk starten met de fagen en de onderzoeken
- 14 dagen in Georgië zelf, 10 werkdagen in behandeling. (rechtstreekse vlucht)
- Eenmaal in Nederland gaat de behandeling ongeveer 1-3 maanden verder (Afhankelijk van de aandoening)
- Nazorg van Lydia zolang het nodig is

Evolution of resistance

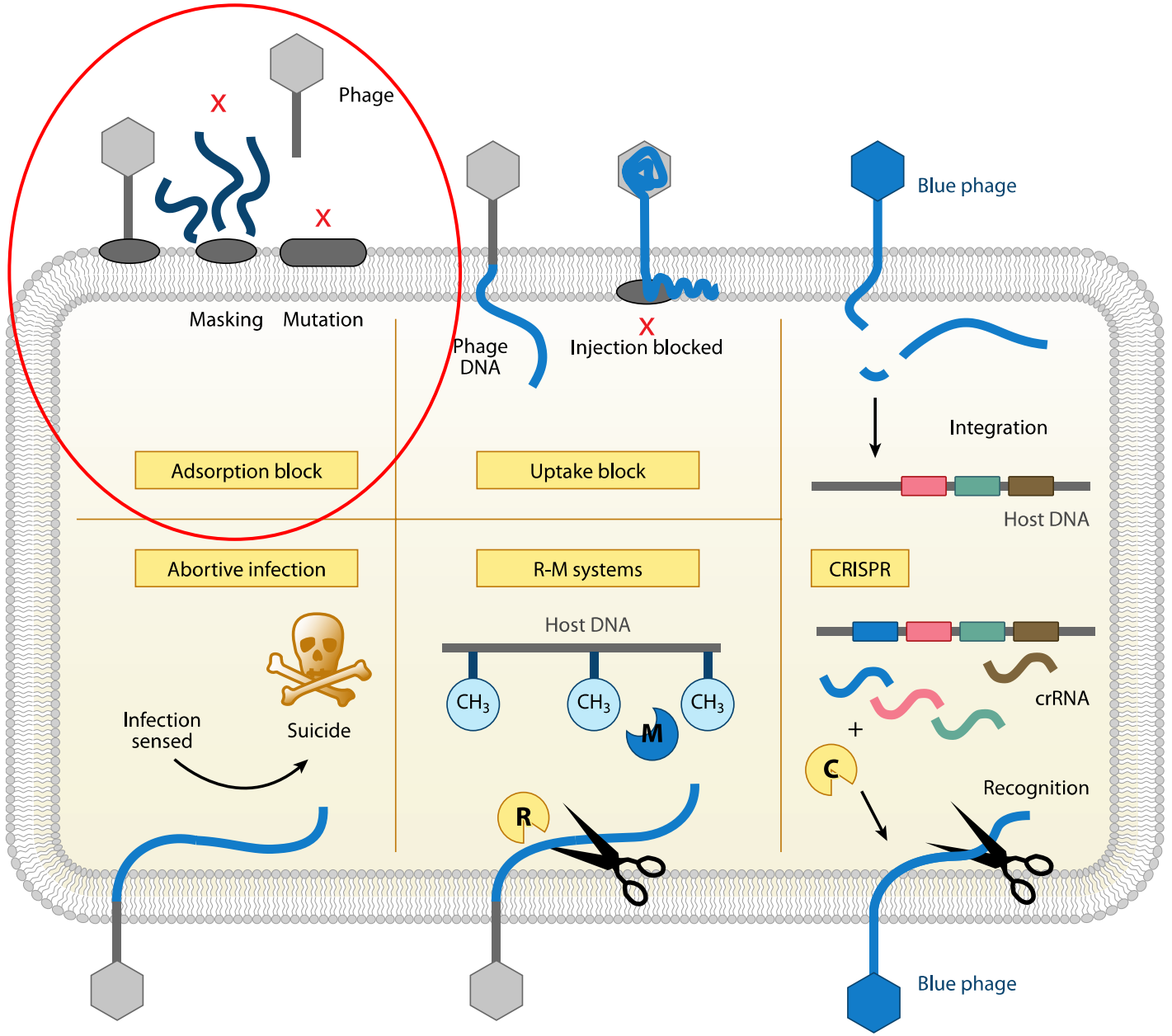




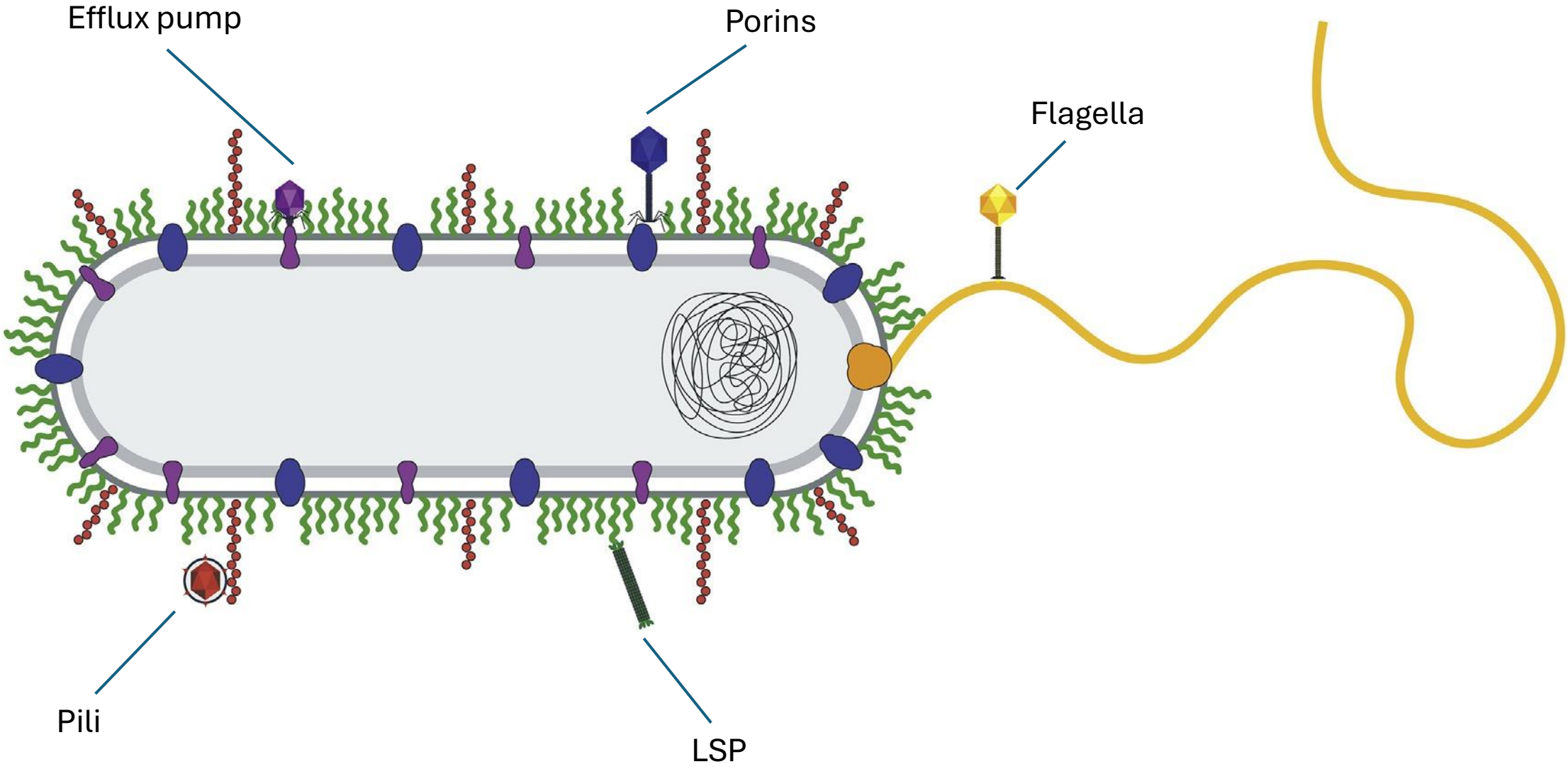
Note: dynamics of bacteriophage predation and evolved resistance can vary wildly in a host



Overview of defense mechanisms



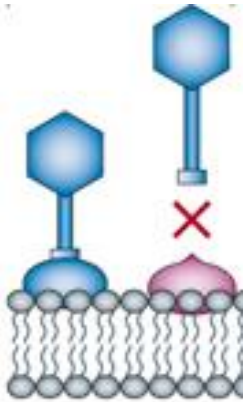
What does a phage bind to?



Preventing phage adsorption

Blocking phage receptors

Production of extracellular matrix - biofilm



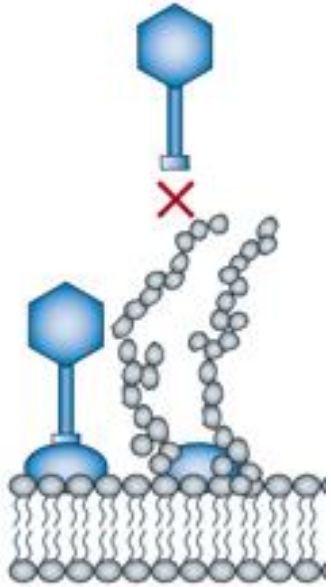
host mutates receptor

Preventing phage adsorption

Blocking phage receptors

Production of extracellular matrix - biofilm

Biofilm Consortia of cells embedded in an extracellular matrix



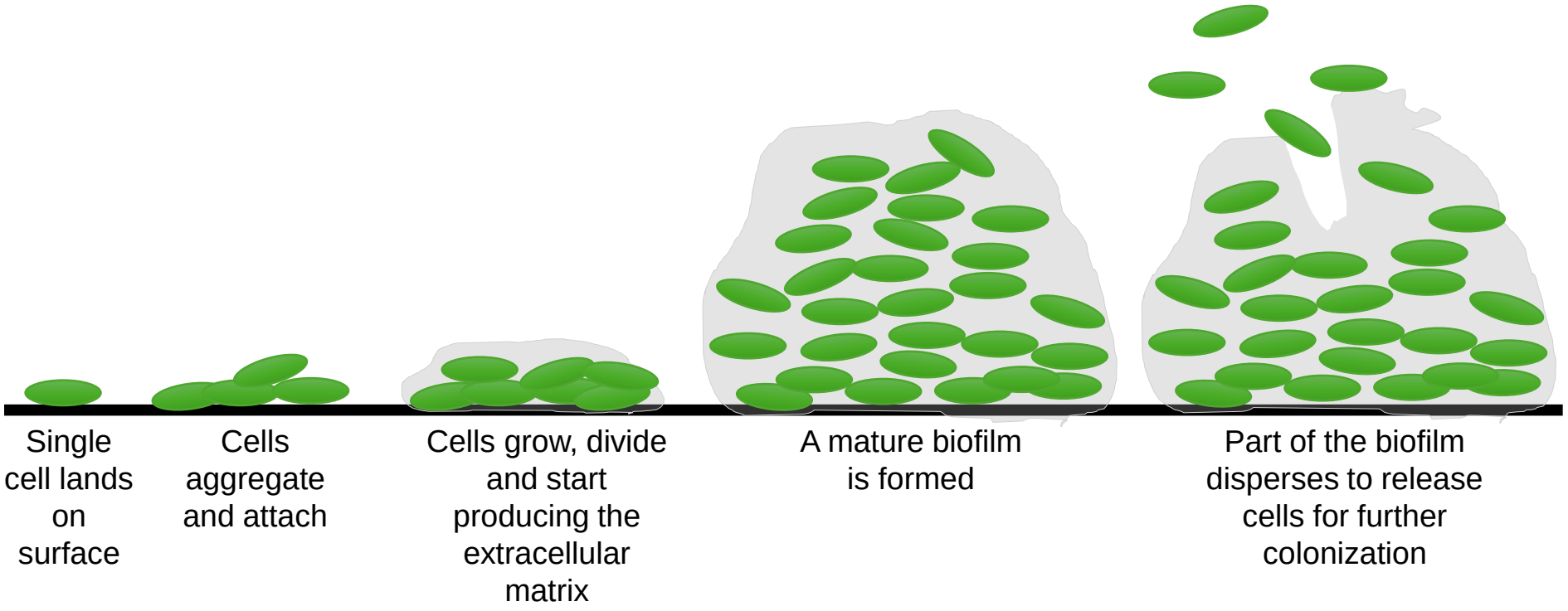
host produces EPS
(extracellular polysaccharides)

Preventing phage adsorption

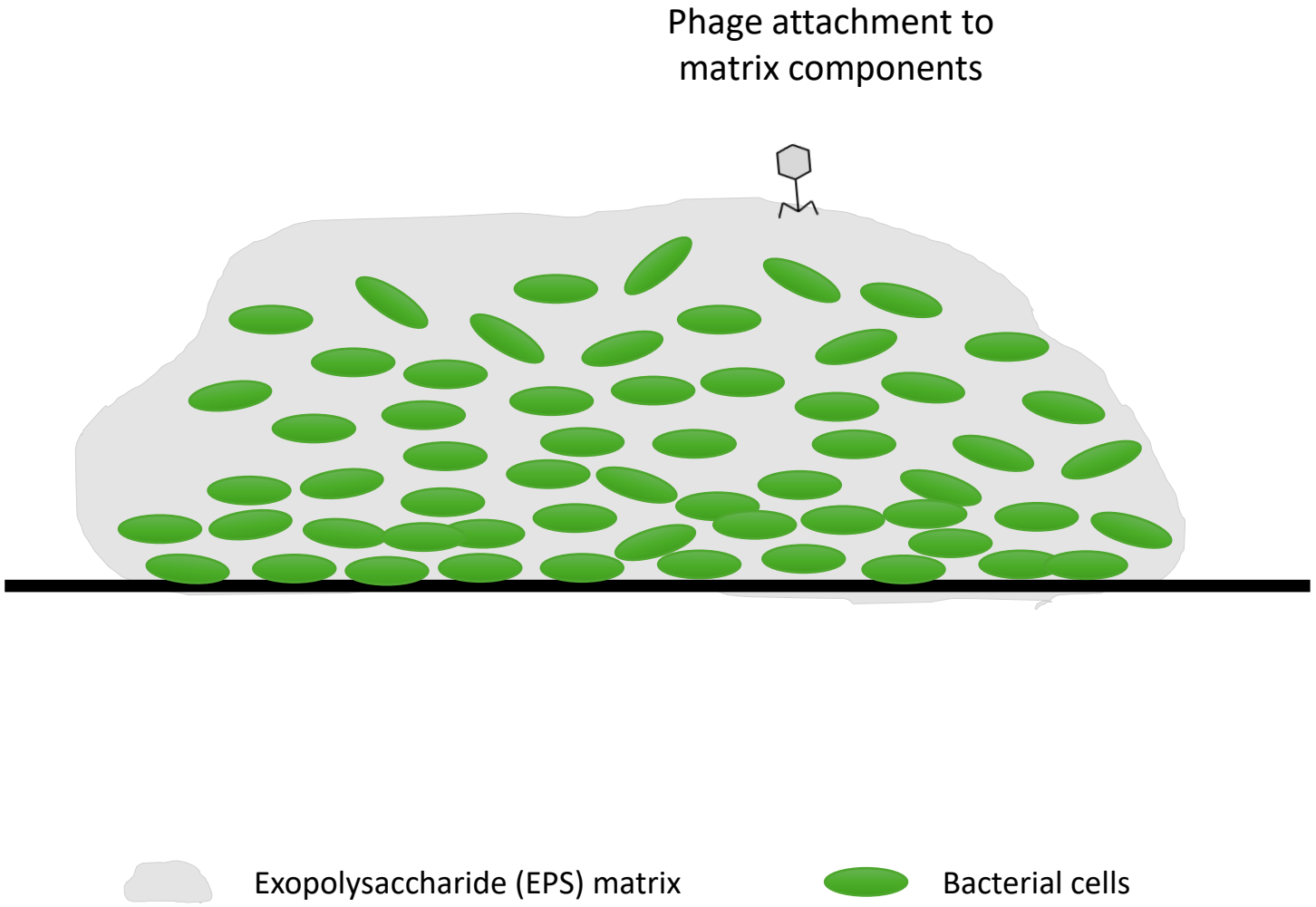
Blocking phage receptors

Production of extracellular matrix - biofilm

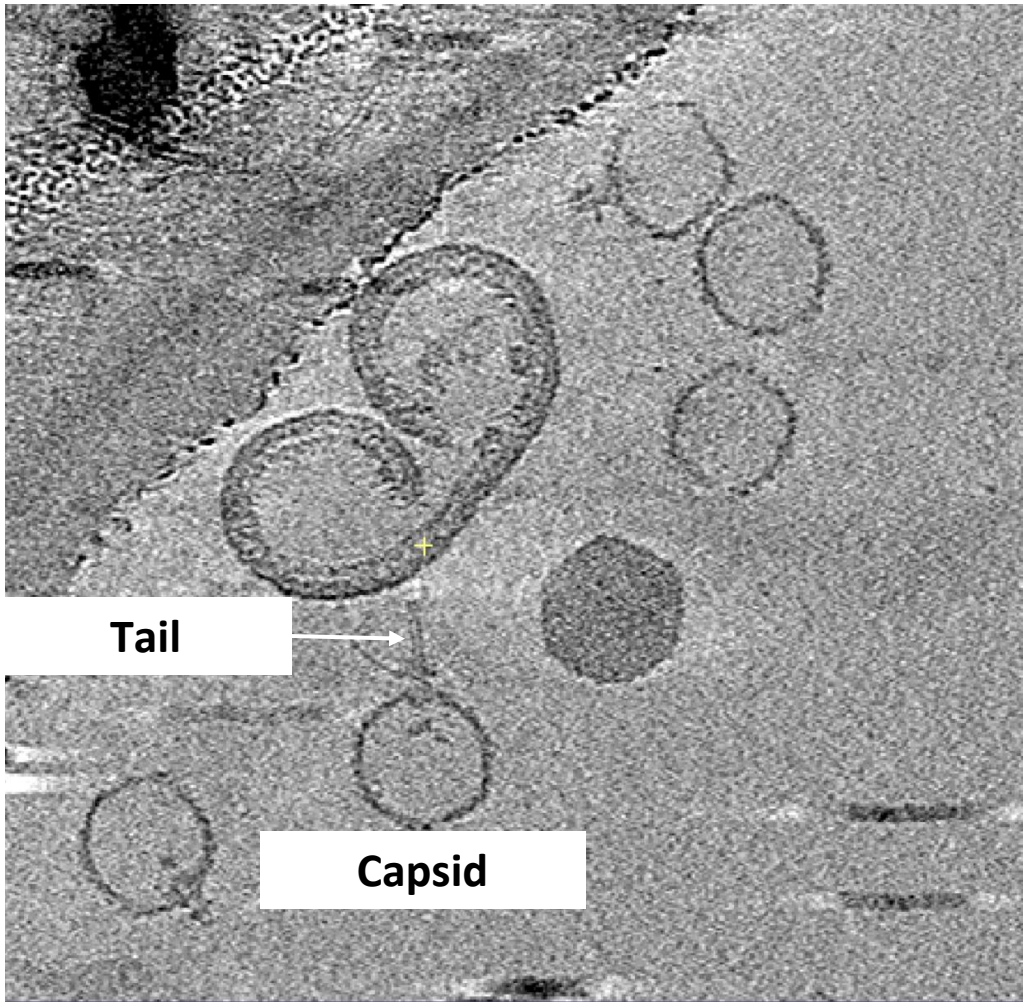
Biofilm Consortia of cells embedded in an extracellular matrix

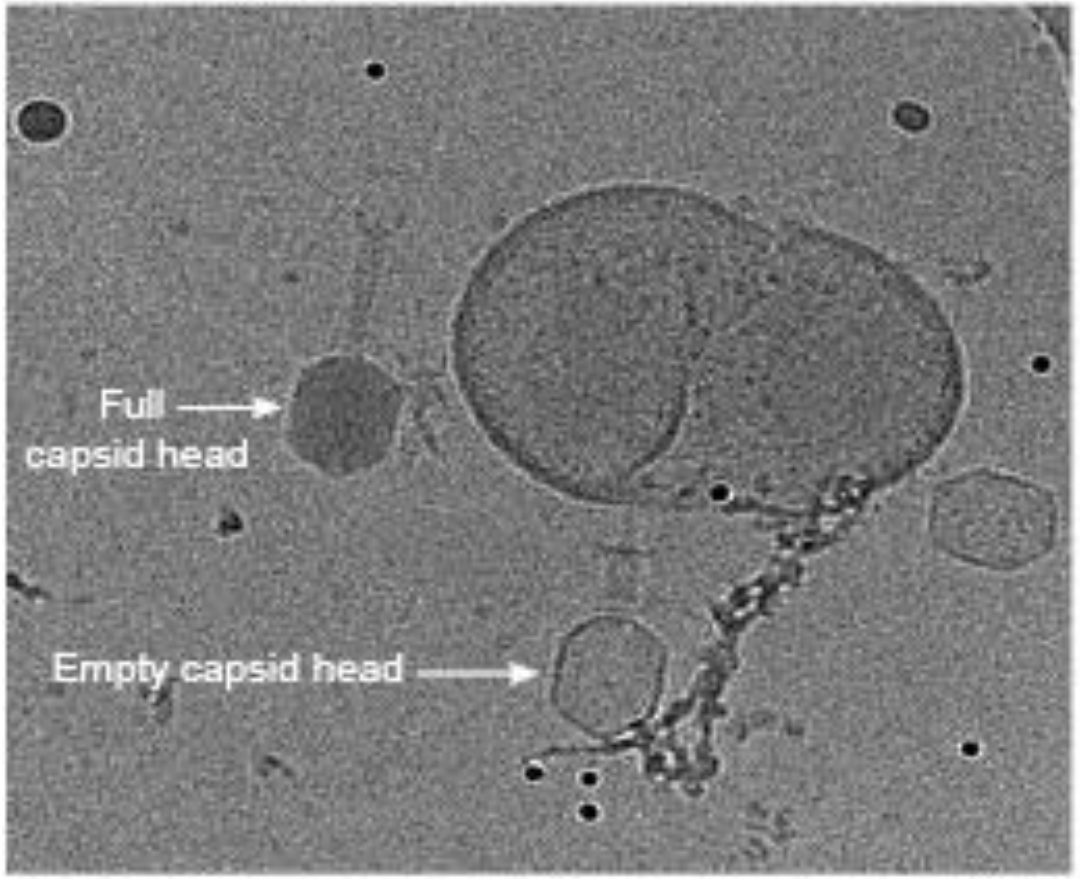


Preventing phage adsorption - Biofilm



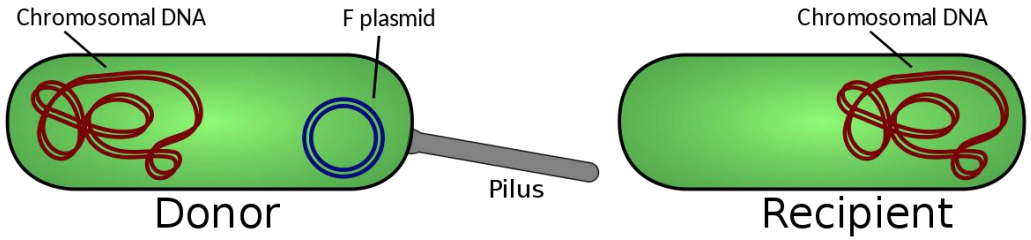
OMV– outer membrane vesicles used a decoys



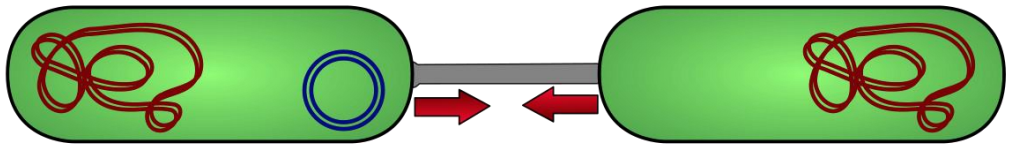


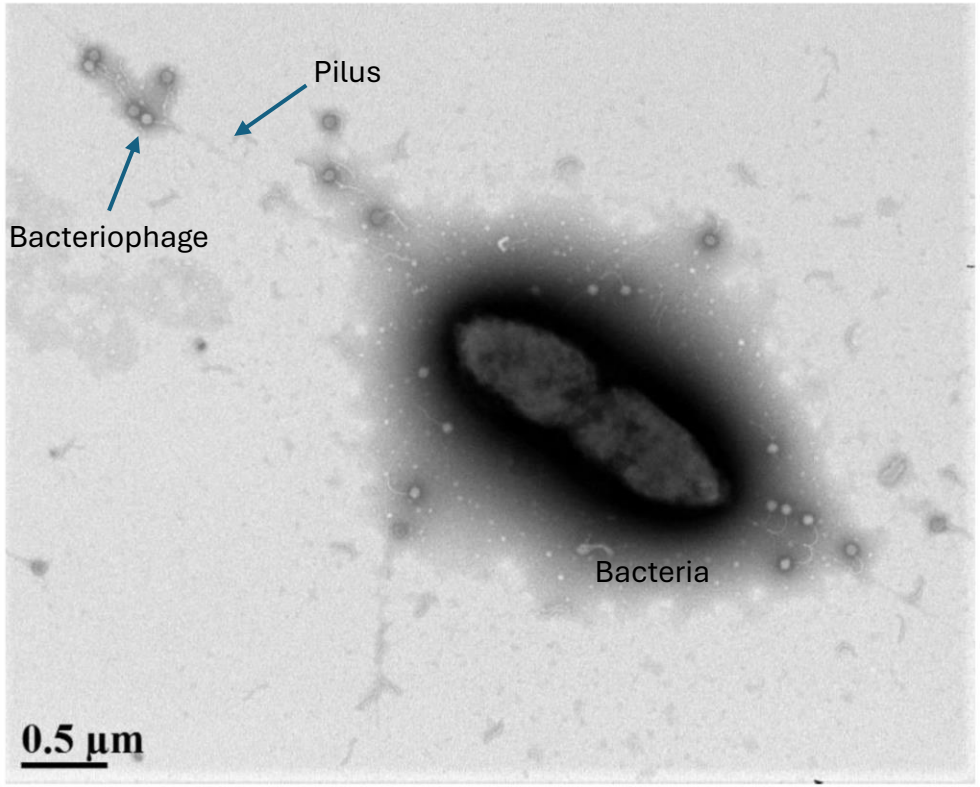
Pili attaching phage

1.



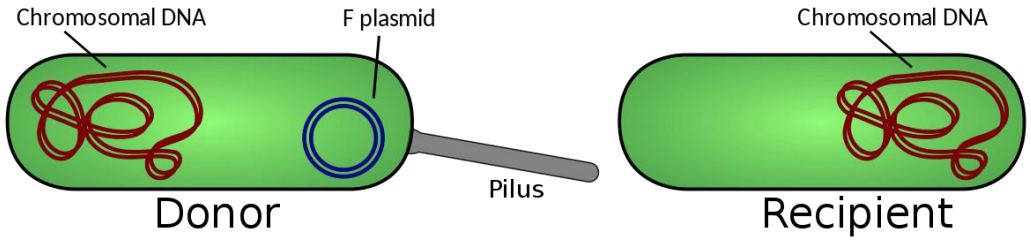
2.



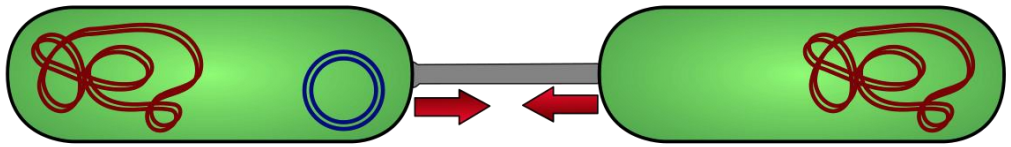


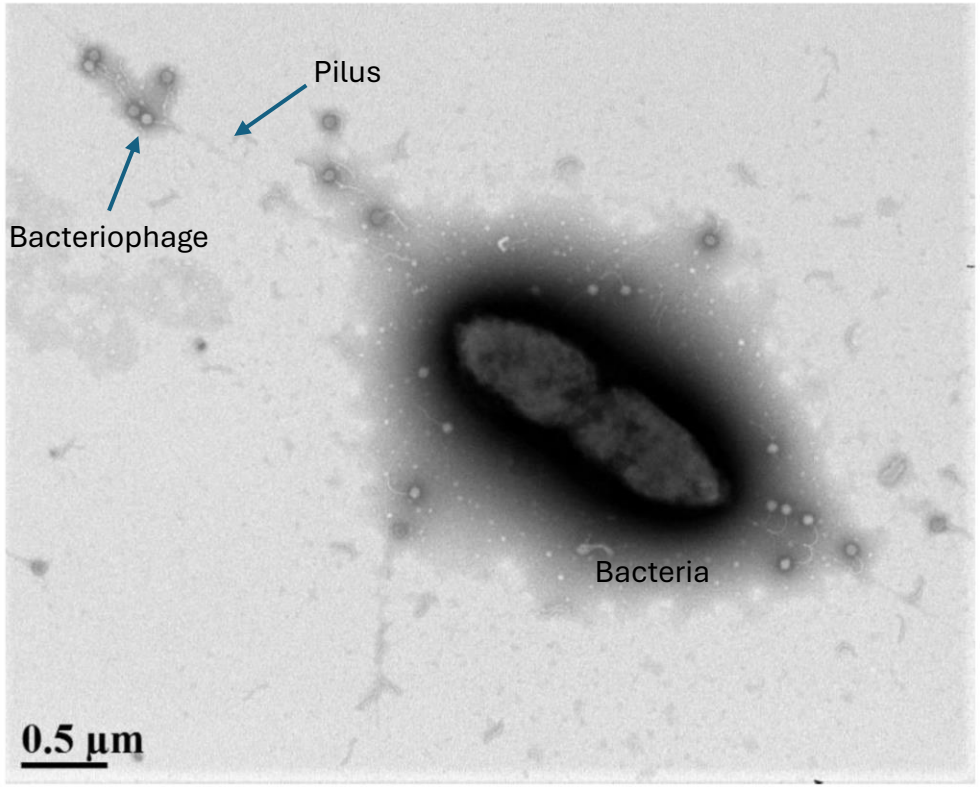
Pili attaching phage

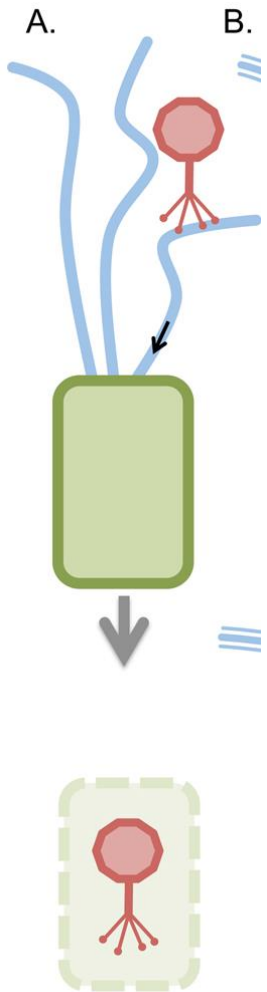
1.



2.

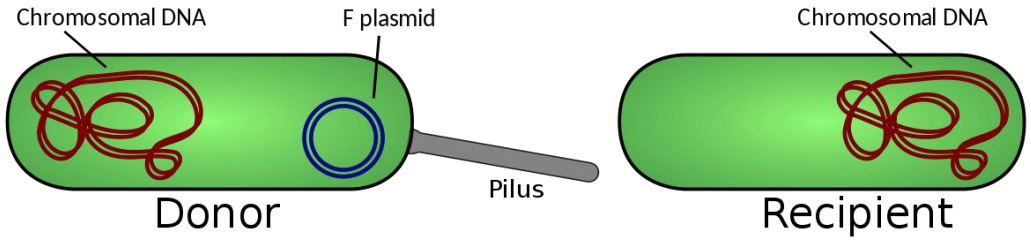




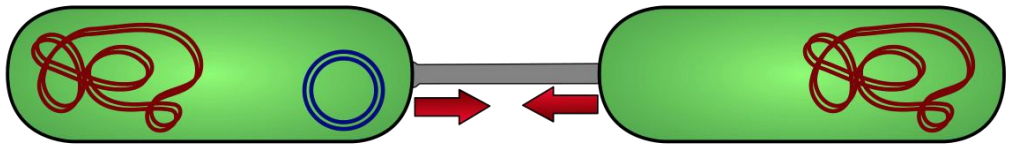


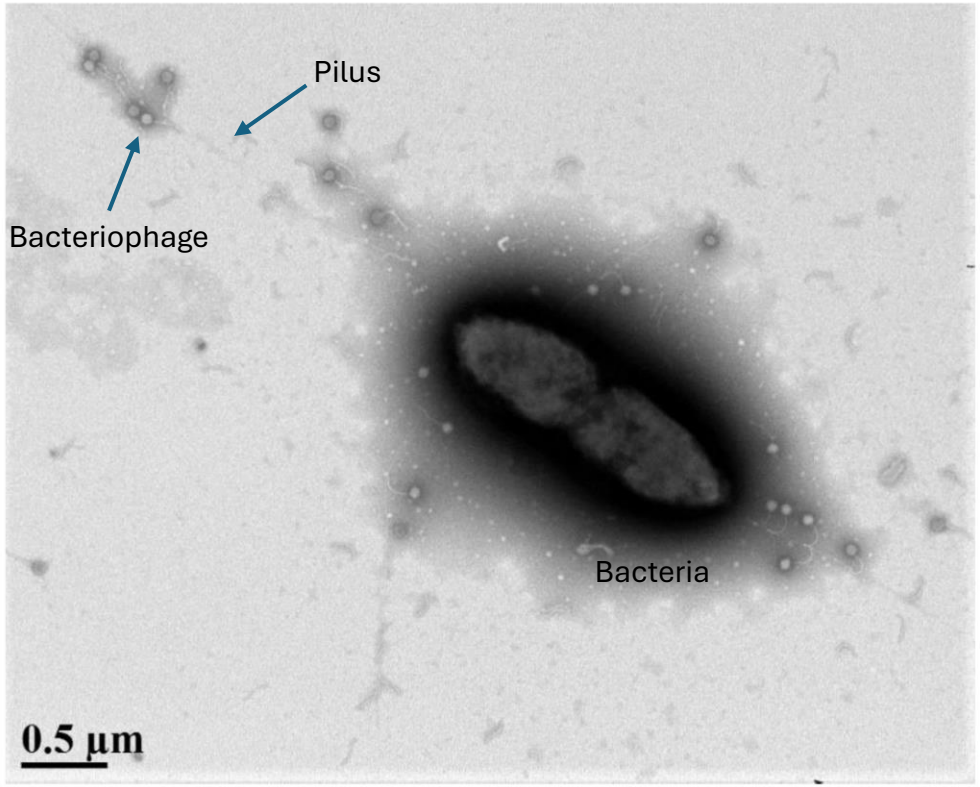
Pili attaching phage

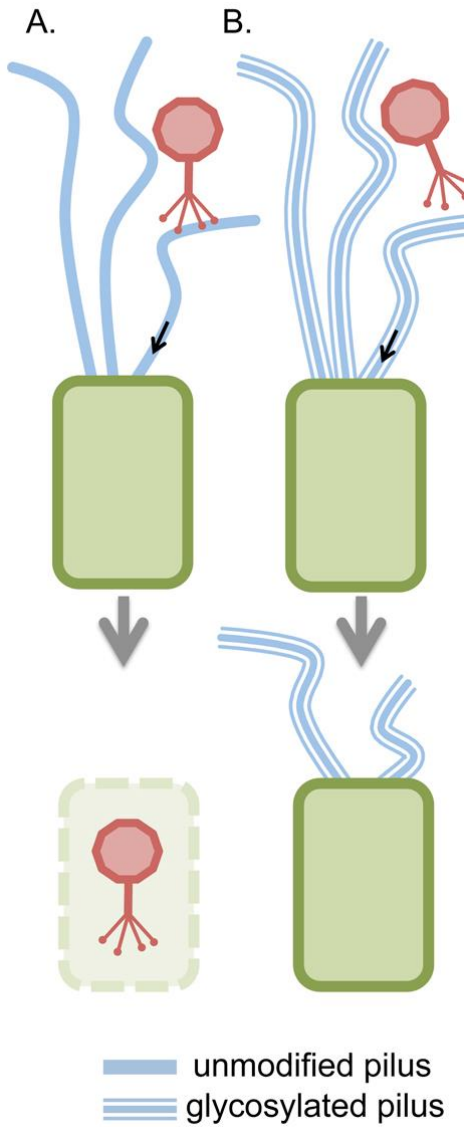
1.



2.





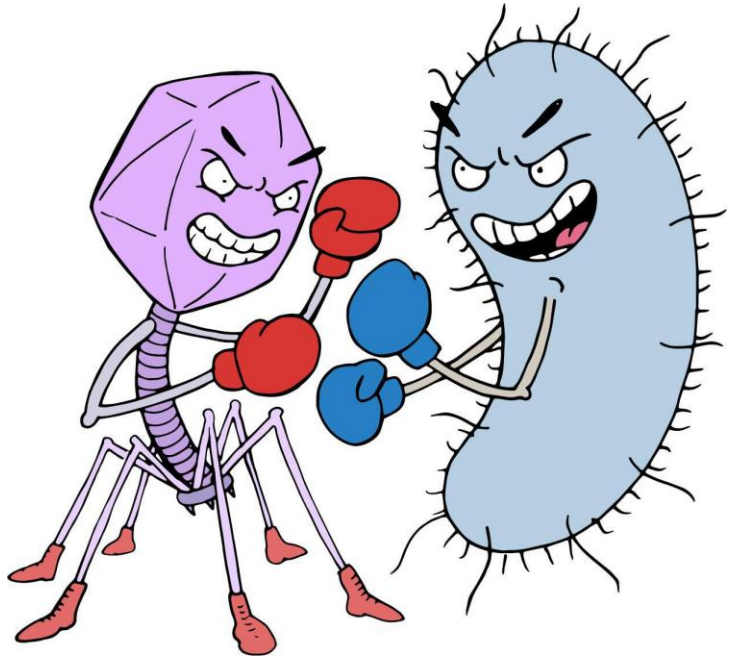


If resistance evolves so easily, why aren’t all bacteria resistant? (and why are there still phages?)

Bacteria have evolved numerous redundant strategies to resist phage infection

And at the same time, phage have evolved strategies to overcome bacterial defense

This is called coevolution.

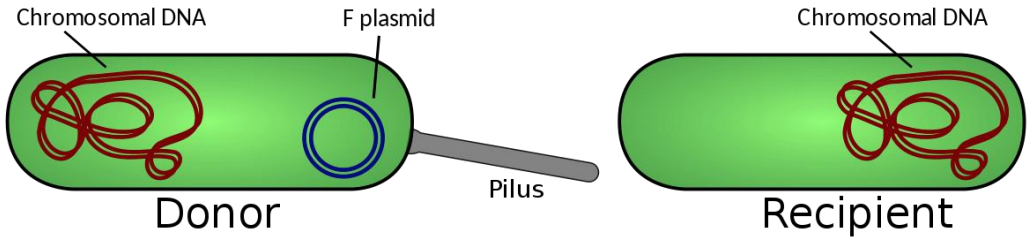


Pili attaching phage

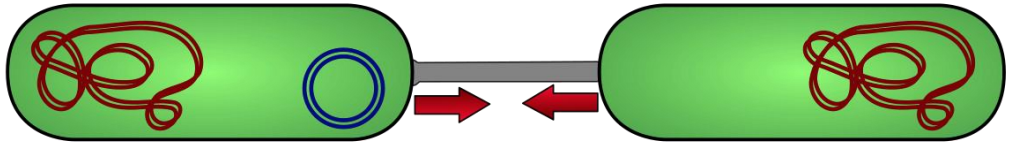
Degrades modification

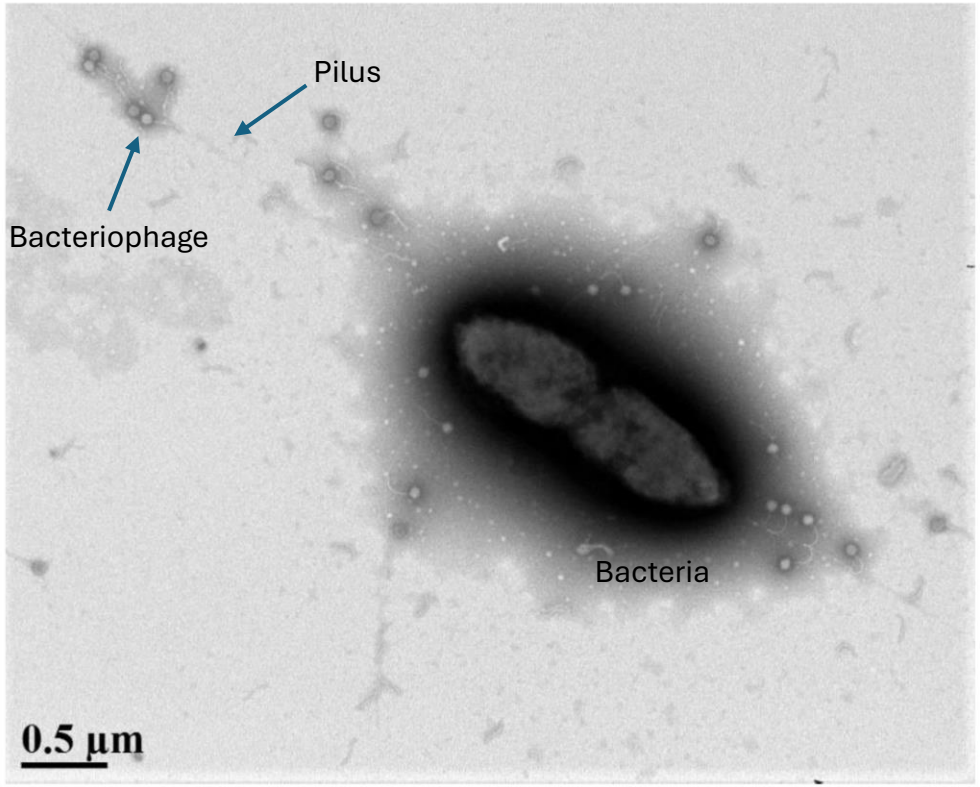
Prefers modified pilus

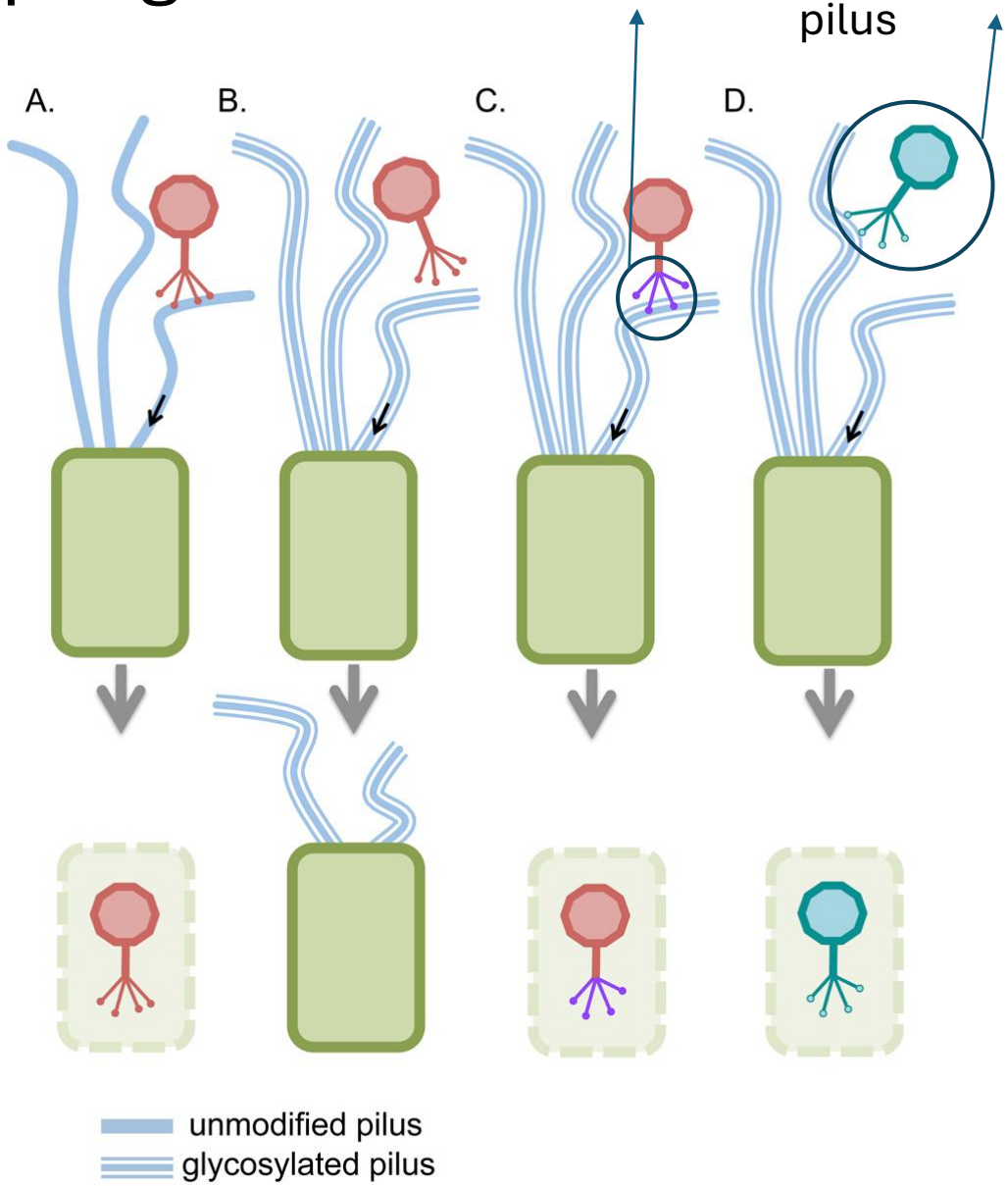
1.



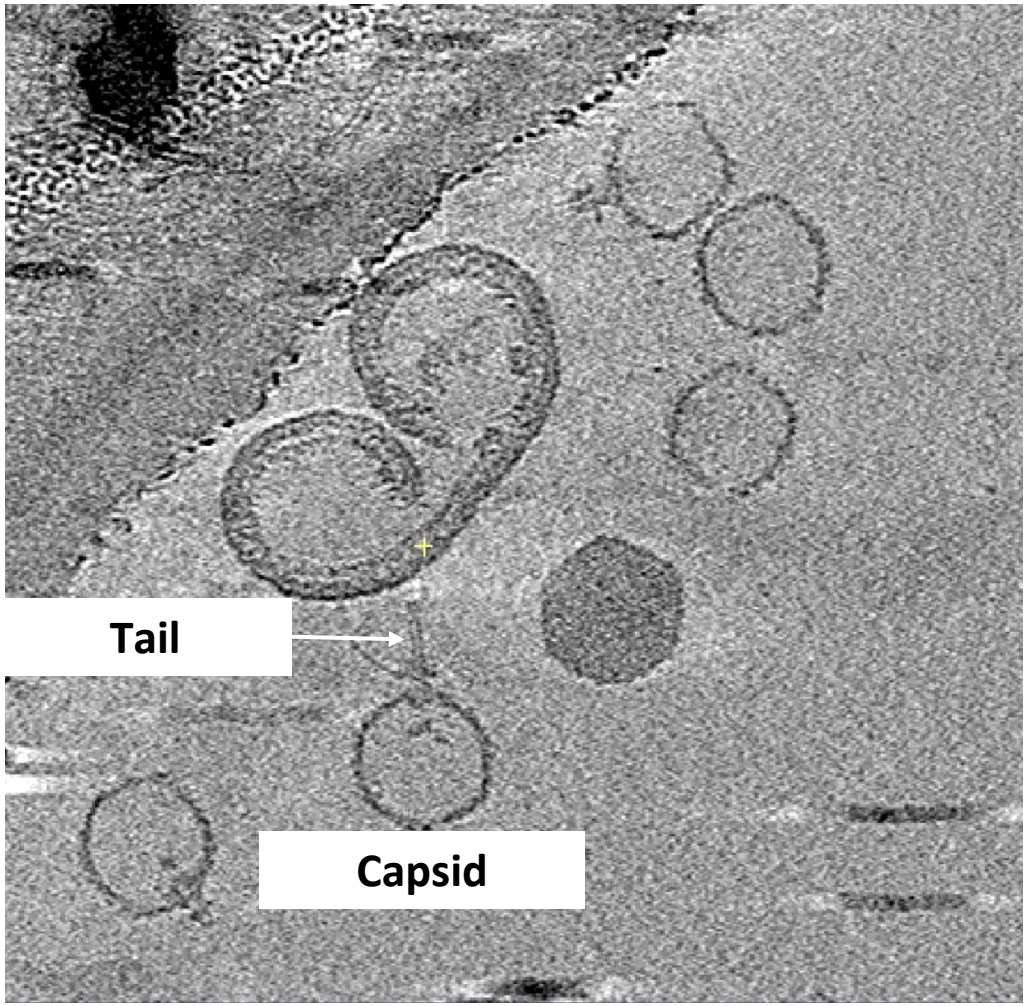
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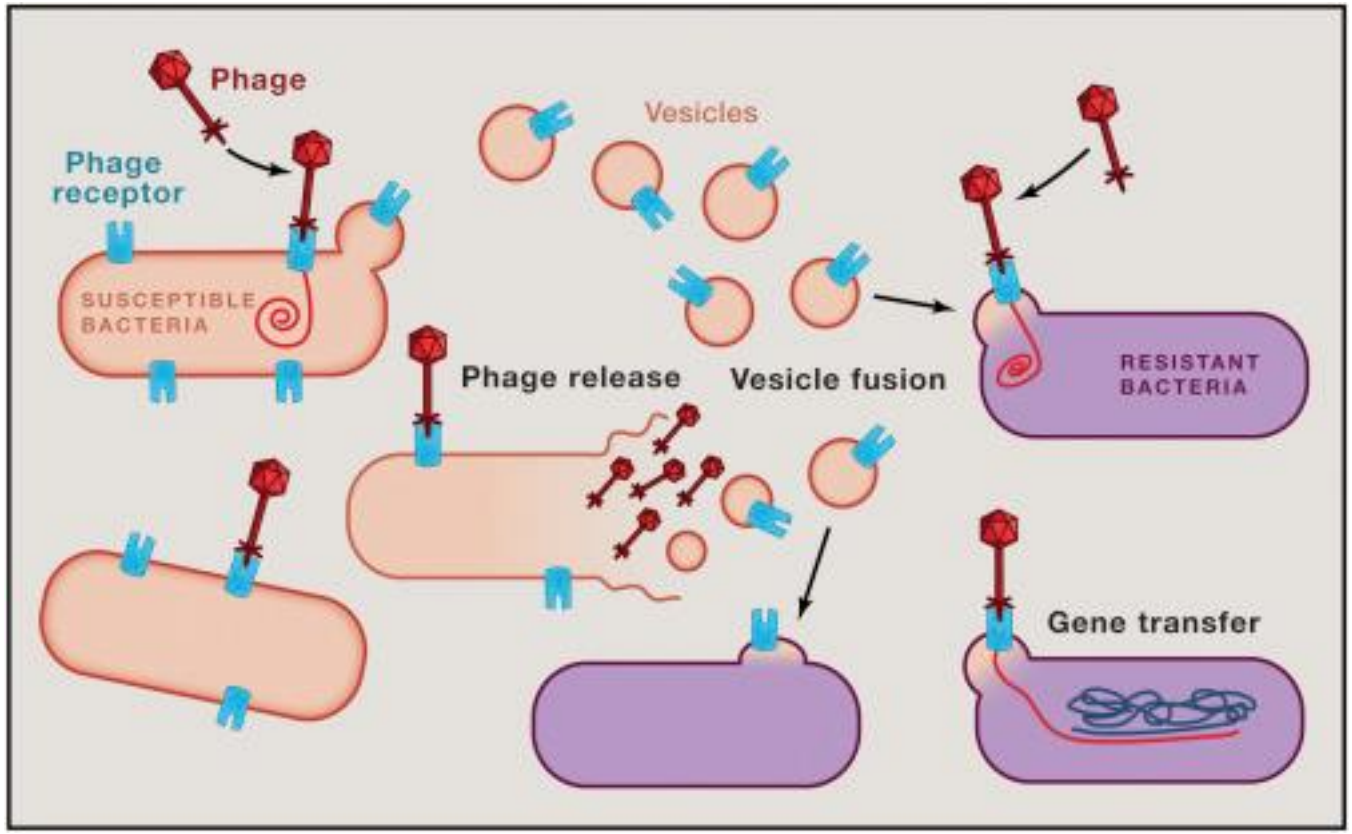






OMV– outer membrane vesicles used a decoys

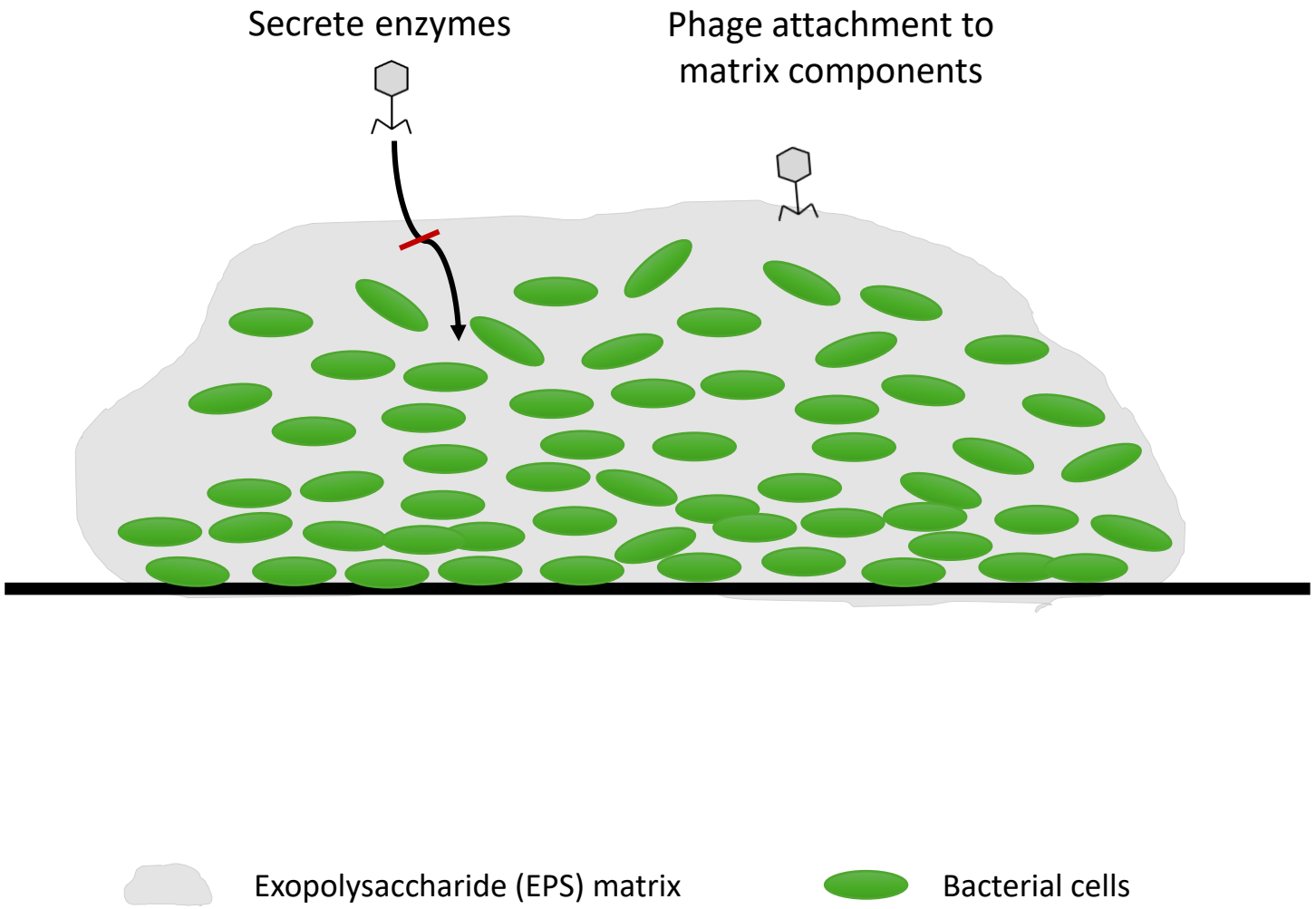




Vesicles Spread Susceptibility to Phages

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[1]Department of Molecular Genetics, Weizmann Institute of Science, 76100 Rehovot, Israel
*Correspondence: rotem.sorek@weizmann.ac.il
http://dx.doi.org/10.1016/j.cell.2016.12.035

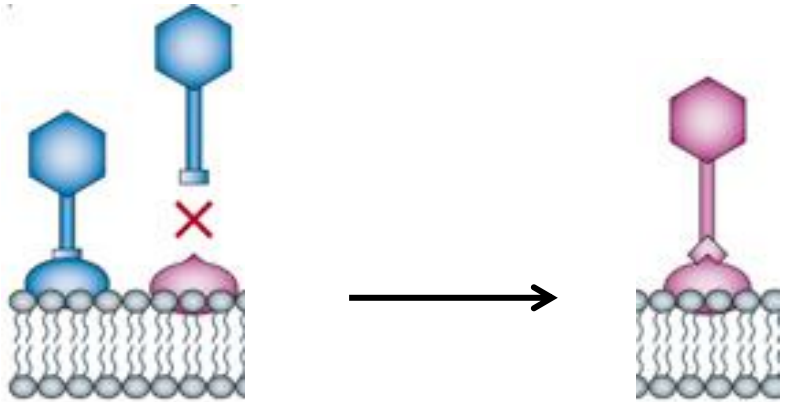
Preventing phage adsorption - Biofilm



Preventing phage adsorption

Blocking phage receptors

Production of extracellular matrix - biofilm

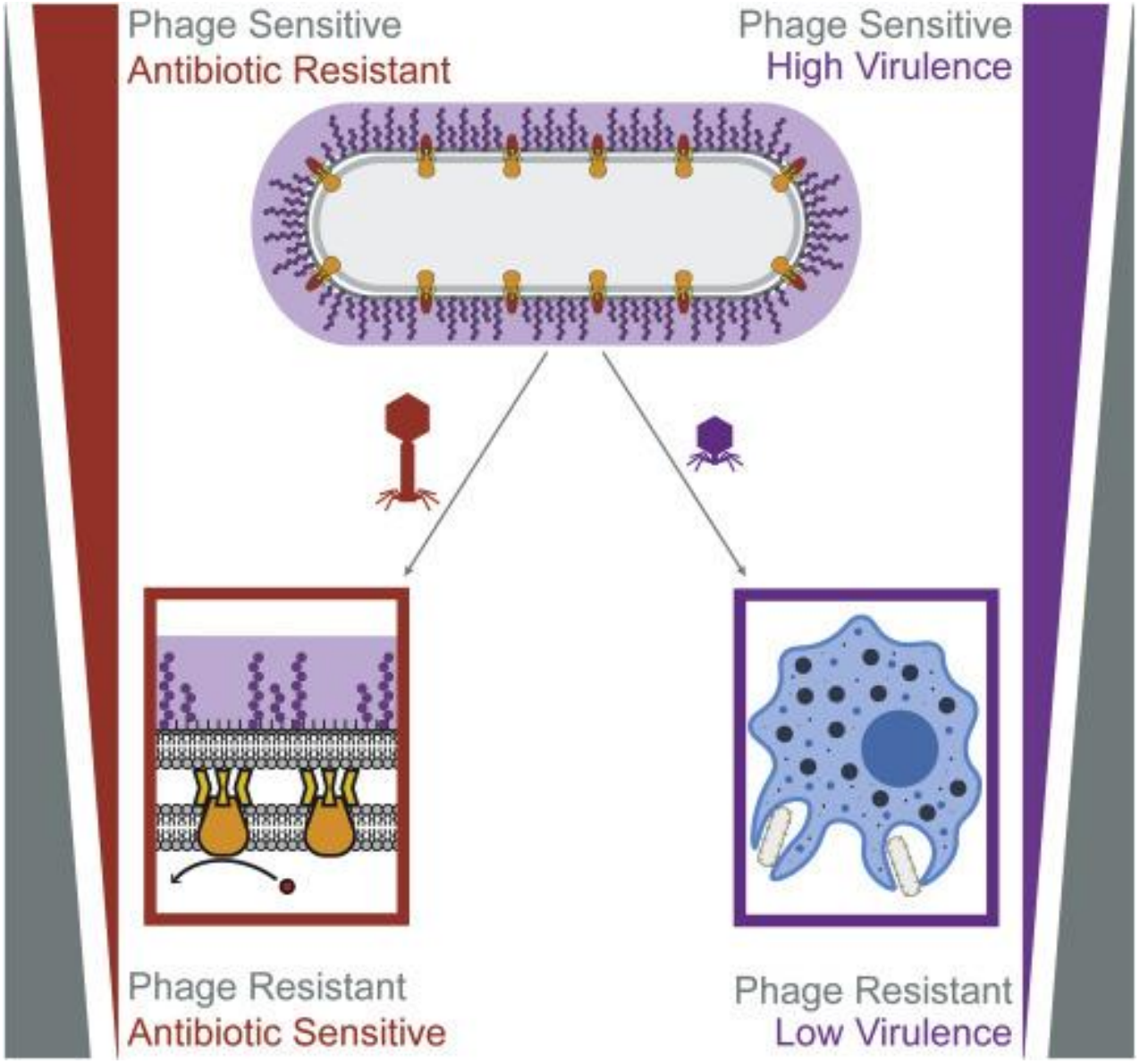


host mutates receptor

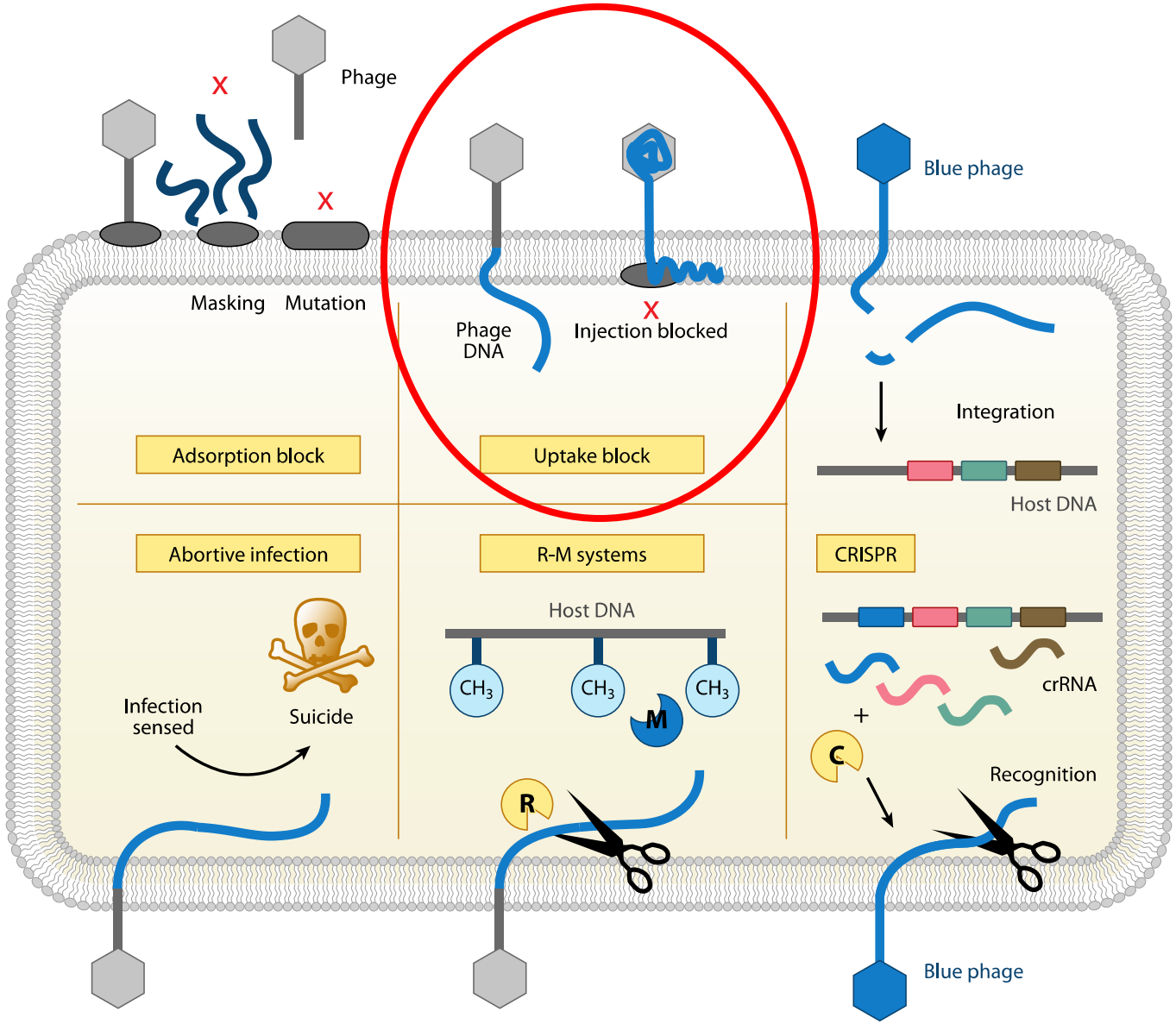
virus recognizes mutated receptor

Second generation phage ‘Adapted phage’

Resistance comes at a cost



- Phages that bind to efflux pumps will increase susceptibility to antibiotics
- This would make combination therapy more suitable (antibiotics + phages)
- Phages that bind to structural virulence factors will decrease virulence
  - Lipopolysaccharide (O1 antigen)
  - Capsular antigen


Phage
X
X
Masking
Mutation
Phage DNA
X
Injection blocked
Blue phage
Integration
Host DNA
Adsorption block
Uptake block
Abortive infection
R-M systems
CRISPR
Host DNA
CH3
CH3
CH3
M
Infection sensed
Suicide
crRNA
+
C
Recognition
R
Blue phage

When phages infect they block the entrance of DNA

-- called SIE systems (super-infection exclusion)

-- “a preexisting viral infection prevents a secondary infection with the same or a closely related virus”

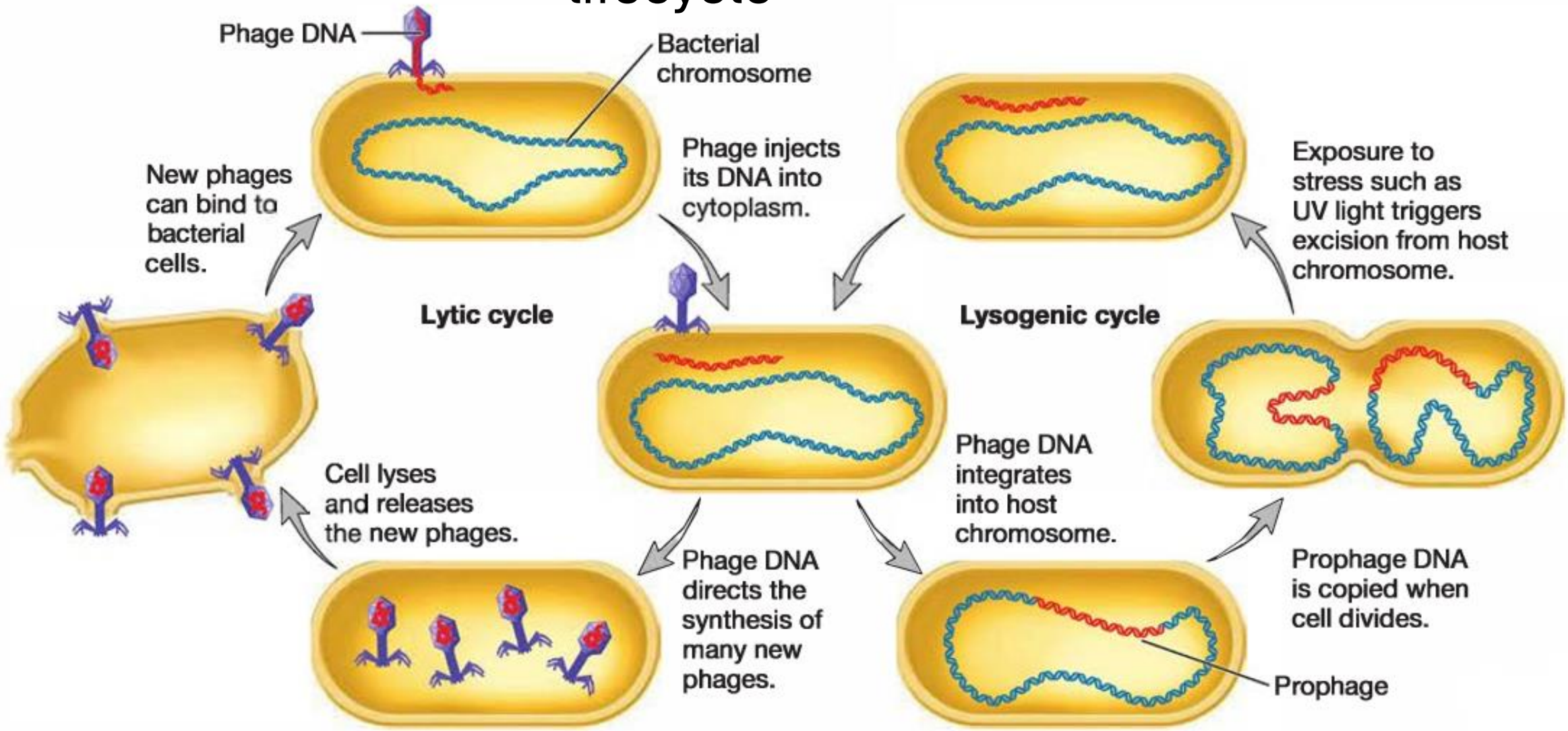
When phages infect they block the entrance of DNA

-- called SIE systems (super-infection exclusion)

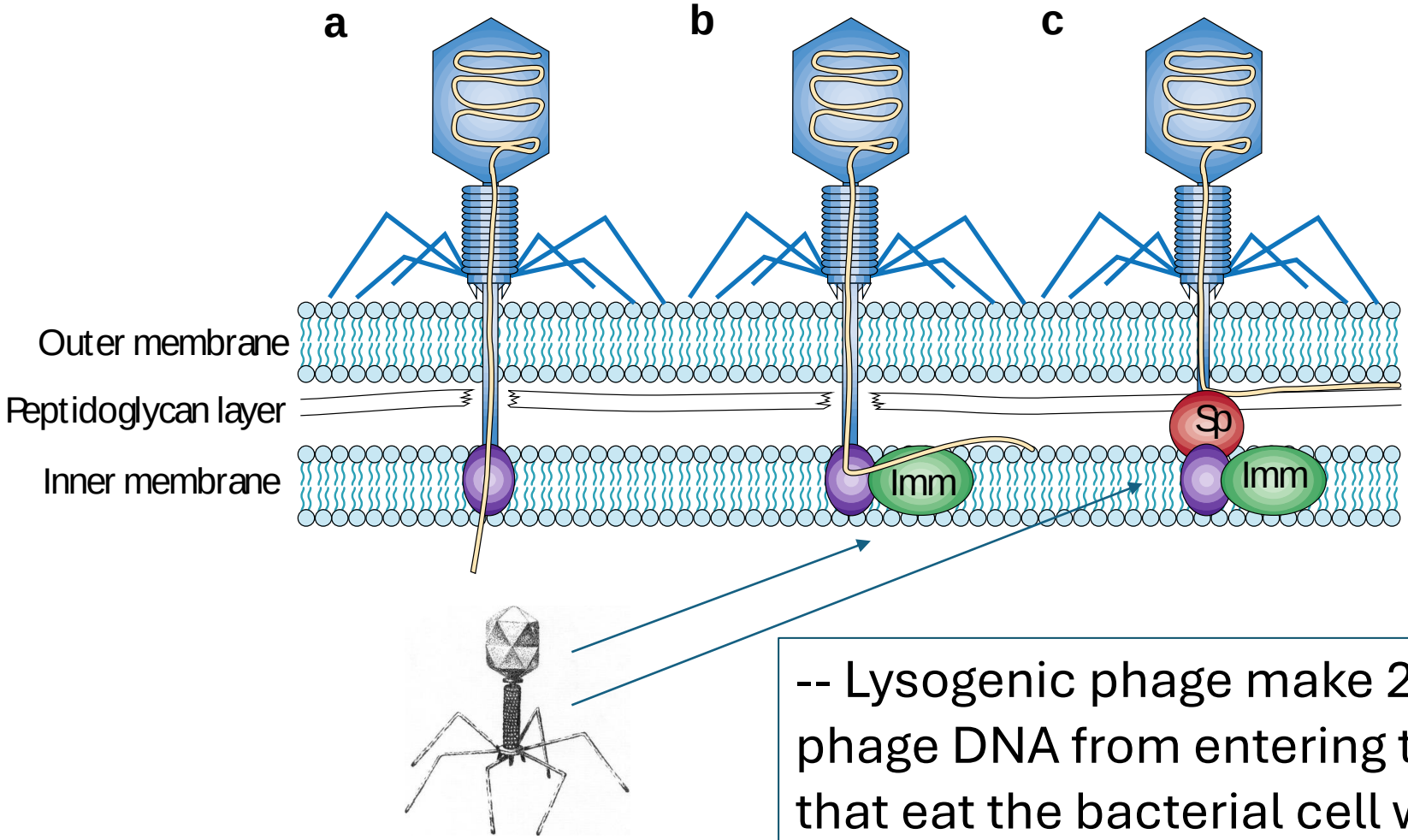
-- “a preexisting viral infection prevents a secondary infection with the same or a closely related virus”

-- But when is a bacteria already infected?

Lysogenic lifecycle



When phages infect they block the entrance of DNA



-- Lysogenic phage make 2 proteins that prevent phage DNA from entering the cell (blocks enzymes that eat the bacterial cell wall)

NOTE: works best with “homotypic” phage

Example of SIE

Control

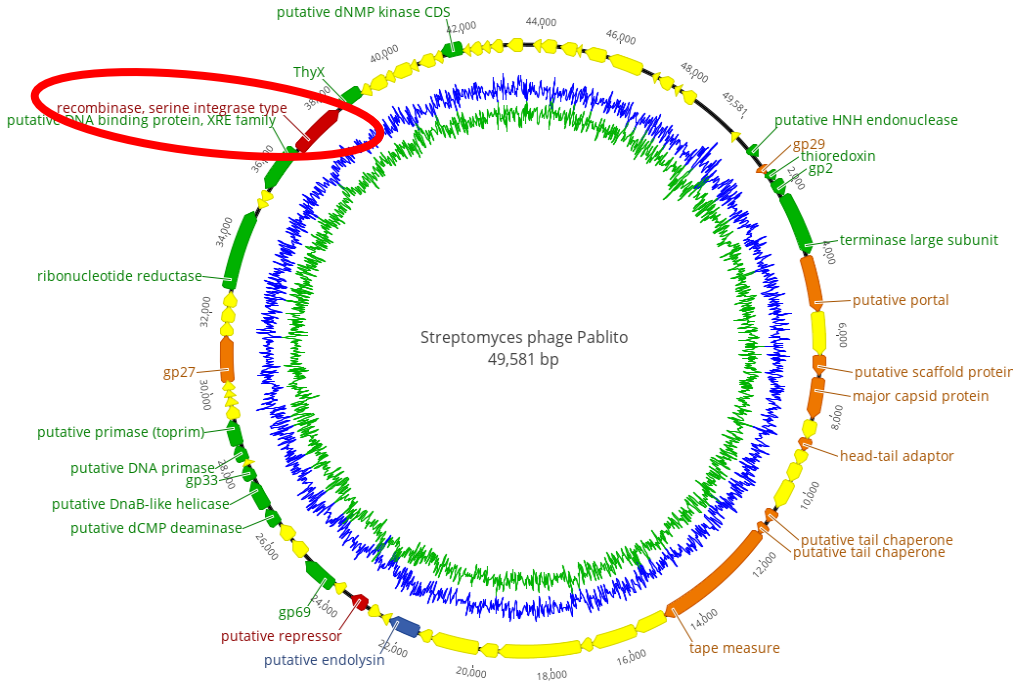




Lysogenic phage

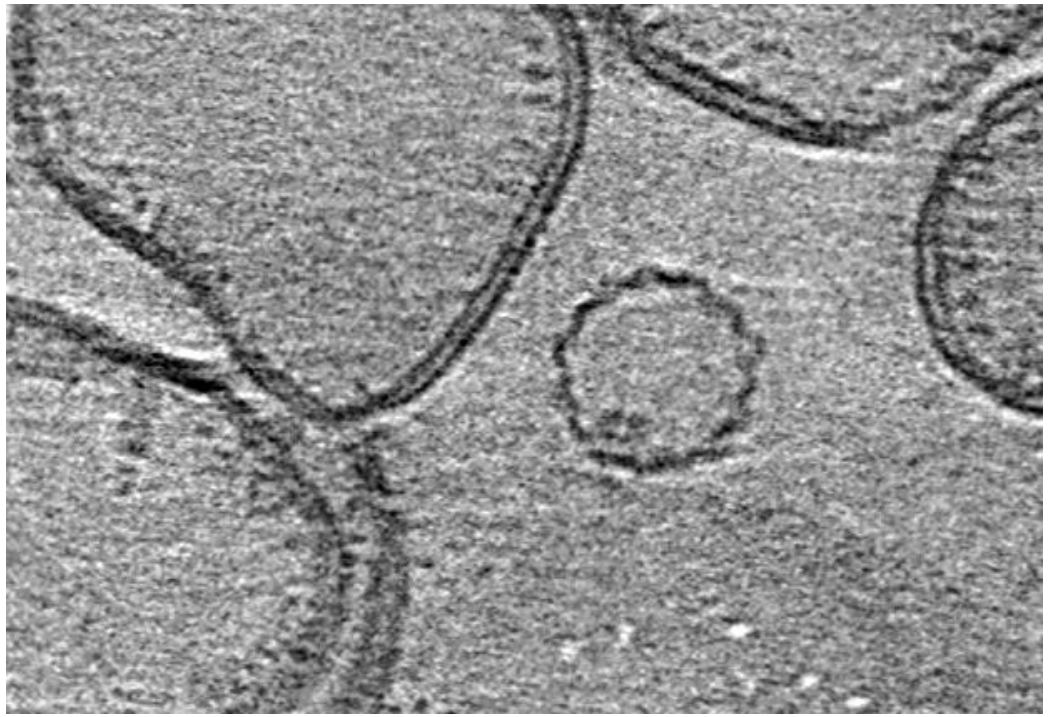






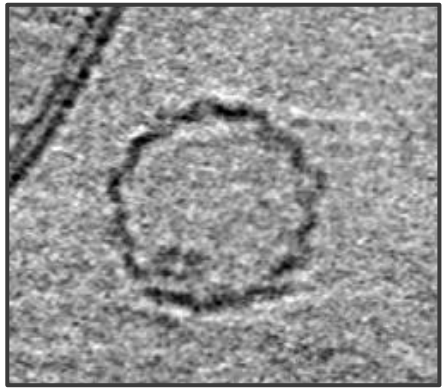
- Phage needs an integrase in the genome to be lysogenic
- The bacteria is resistant to the phage because phage Pablito is already integrated

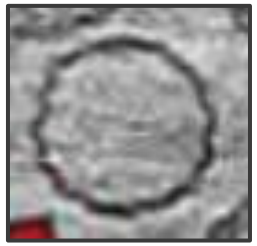
Other lytic phages can induce lysogenic phages

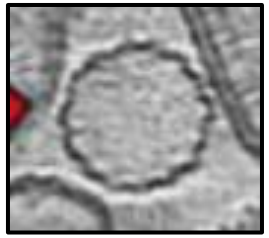




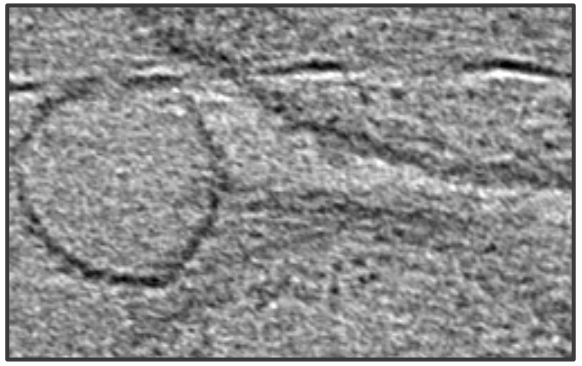
Lytic phage MS2

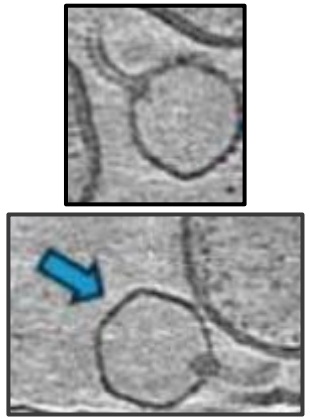




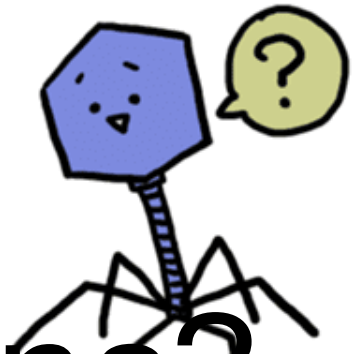


Lysogenic phage λ



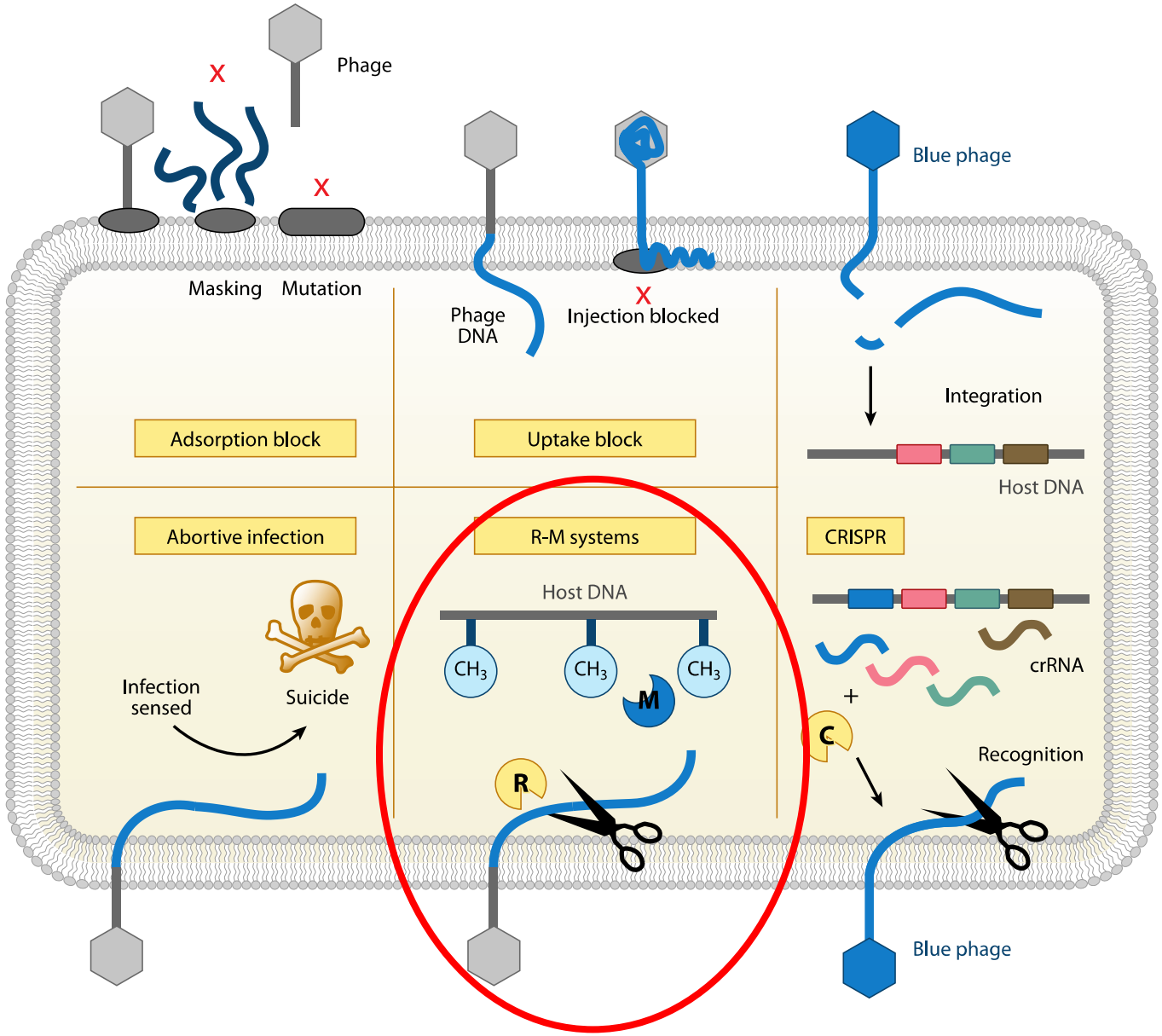


Questions?
Part-1



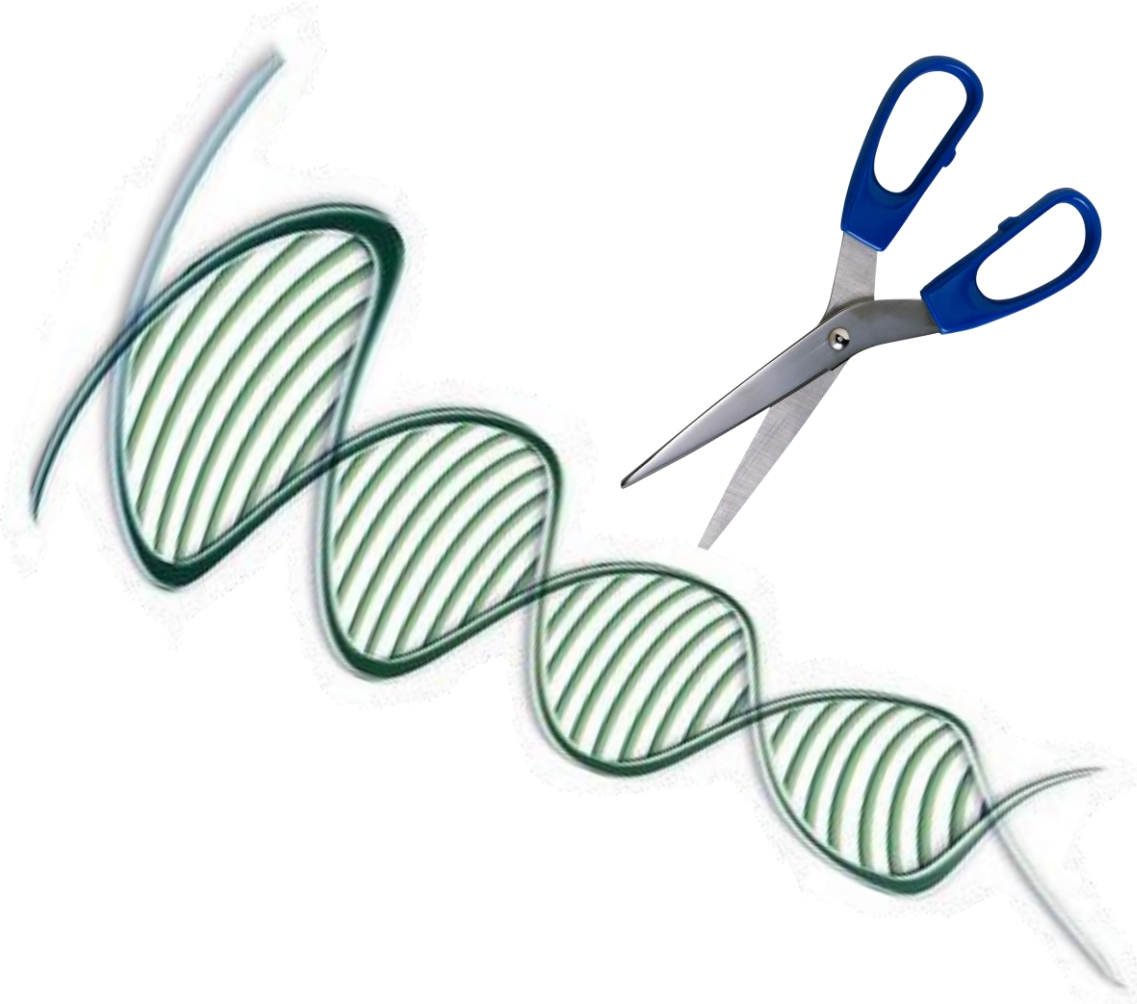
Break




Phage
X
X
Masking
Mutation
Phage DNA
X
Injection blocked
Blue phage
Integration
Host DNA
Adsorption block
Uptake block
Abortive infection
R-M systems
CRISPR
Host DNA
CH3
CH3
CH3
M
R
Infection sensed
Suicide
crRNA
+
C
Recognition
Blue phage

Restriction - modification

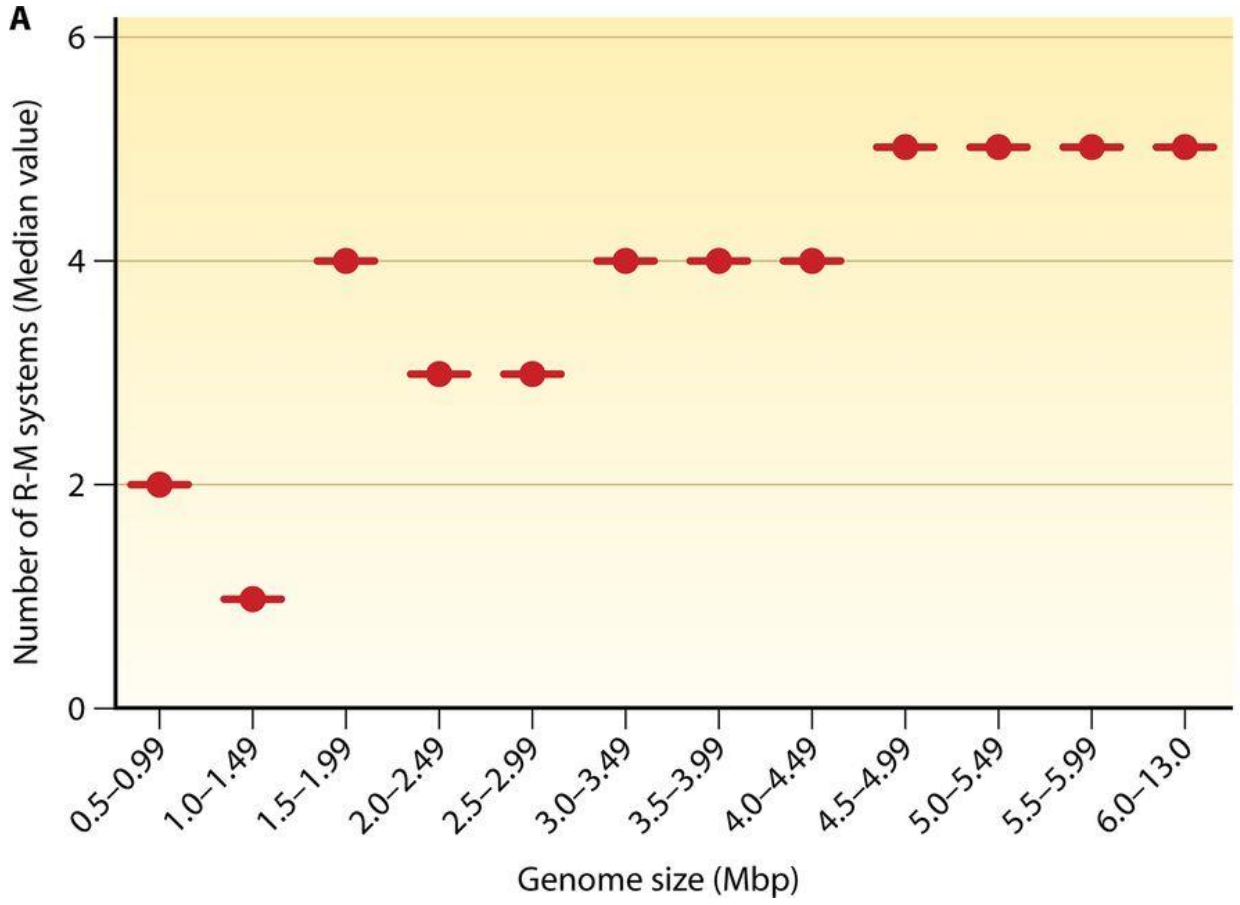
Site-specific restriction endonucleases are used to identify DNA molecules



R-M is extremely widespread!!

-- ~90% of the genomes contain at least one R-M system and ~80% contain multiple R-M systems

-- more than 19,000 known systems



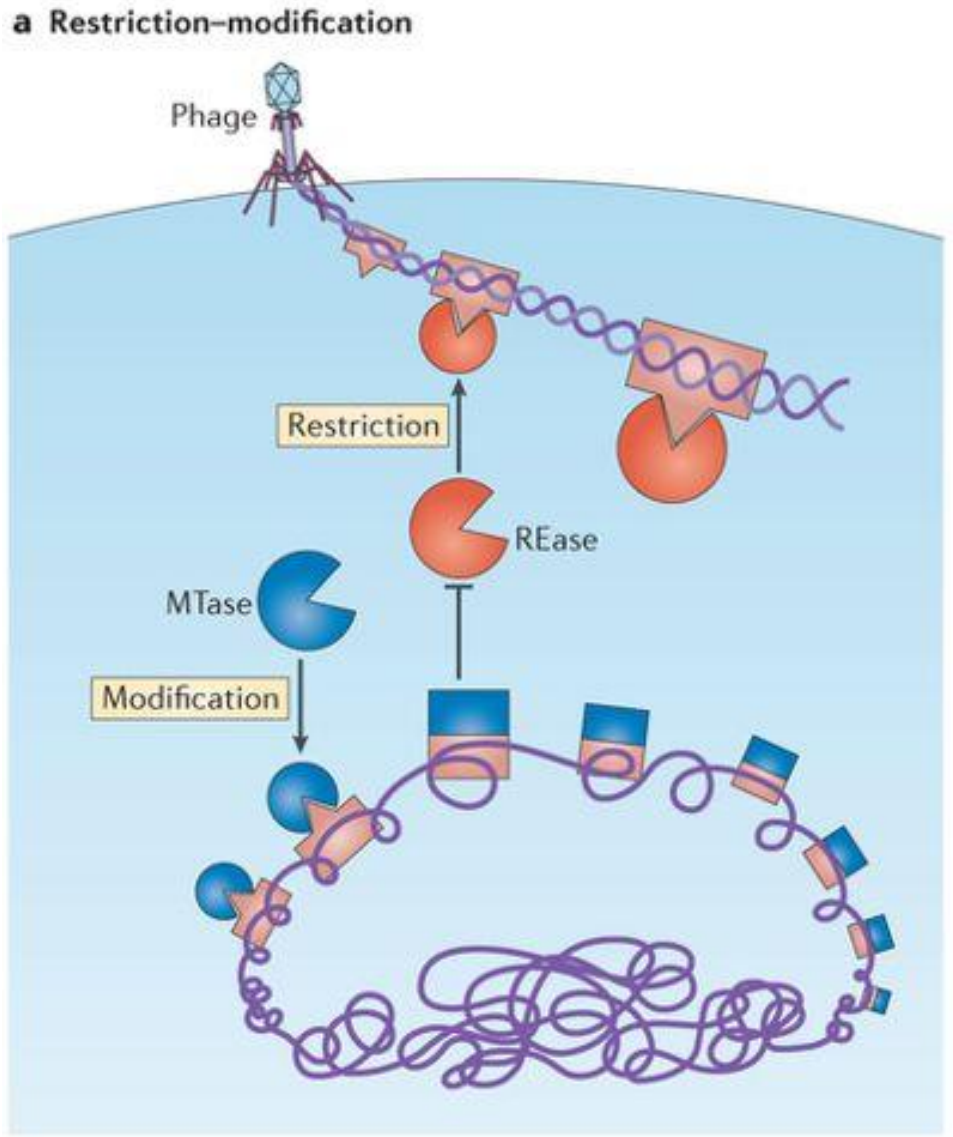
Restriction endonucleases are part of a bacterium's innate defense against invaders

Restriction-modification systems allow the bacterium to distinguish self from non-self DNA

Restriction: bacterial endonucleases cleave both strands of foreign DNA at ***specific*** recognition sites

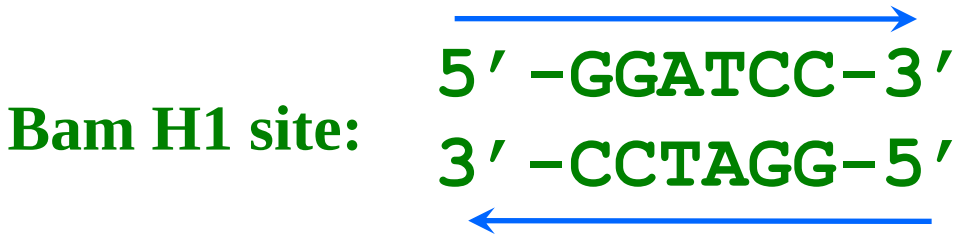
Modification: bacteria protect their own DNA by adding a methyl group to the recognition sites in their own DNA

Modification ensures self/non-self discrimination

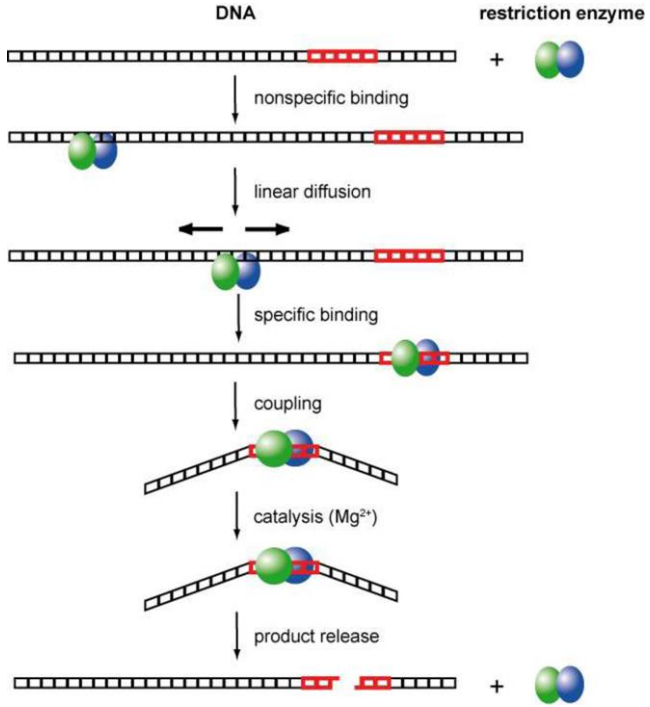


Restriction Endonucleases

Recognition sites have symmetry (palindromic)



Enzyme Activity



Scanning

GGACGCTAGCTGAT**GAATTC**GCATCGGATCCGAATCCGCTCTTTCAA
CCTGCGATCGACTA**CTTAAG**CGTAGCCTAGGCTTAGGCGAGAAAGTT

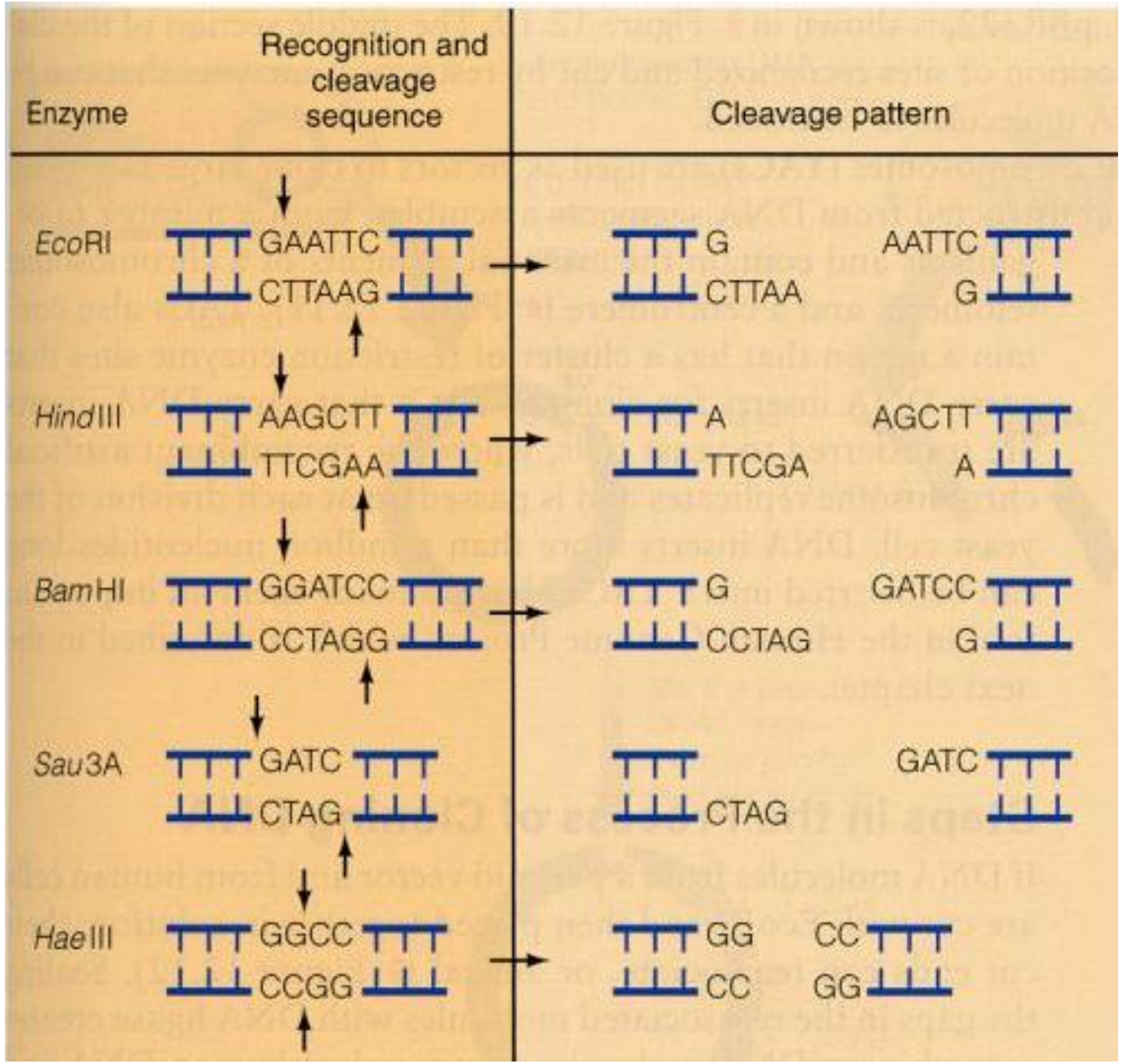
Recognition Sequence

GGACGCTAGCTGAT**GAATTC**GCATCGGATCCGAATCCGCTCTTTCAA
CCTGCGATCGACTA**CTTAAG**CGTAGCCTAGGCTTAGGCGAGAAAGTT

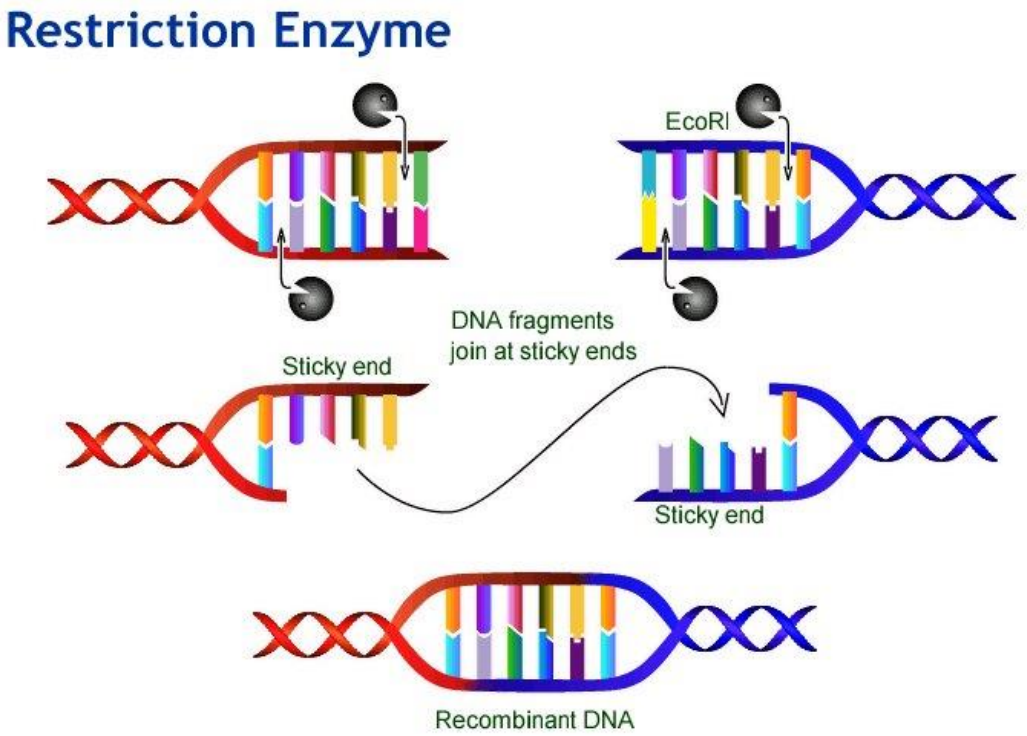
Cleavage

GGACGCTAGCTGAT**G**
CCTGCGATCGACTA**CTTAA**

AATTCGCATCGGATCCGAATCCGCTCTTTCAA
GCGTAGCCTAGGCTTAGGCGAGAAAGTT


Enzyme
Recognition and cleavage sequence
Cleavage pattern
EcoRI
GAATTC
CTTAAG
G
CTTAA
AATTC
G
HindIII
AAGCTT
TTCGAA
A
TTCGA
AGCTT
A
BamHI
GGATCC
CCTAGG
G
CCTAG
GATCC
G
Sau3A
GATC
CTAG
CTAG
GATC
HaeIII
GGCC
CCGG
GG
CC
CC
GG

DNA recombinant technology



The frequency by which a Class II restriction enzyme will cut DNA is mainly a function of the length of the sequence it is sensitive to. For instance:



*Bsu*RI has a recognition sequence of 4 base-pairs. Out of sheer probability, we see that with 4 positions, and each position having potentially 4 different values, there are 4^4 = 256 different possibilities for any given 4-base long strand. Therefore, theoretically (assuming completely random DNA), this enzyme will cut 1 in 256 4-base-pair long sites.



*Eco*RI, on the other hand, recognizes a sequence of 6 base-pairs. 4^6 = 4096 possible combinations with this length, and so *Eco*RI will cut 1 in 4096 6-base-pair long sites.



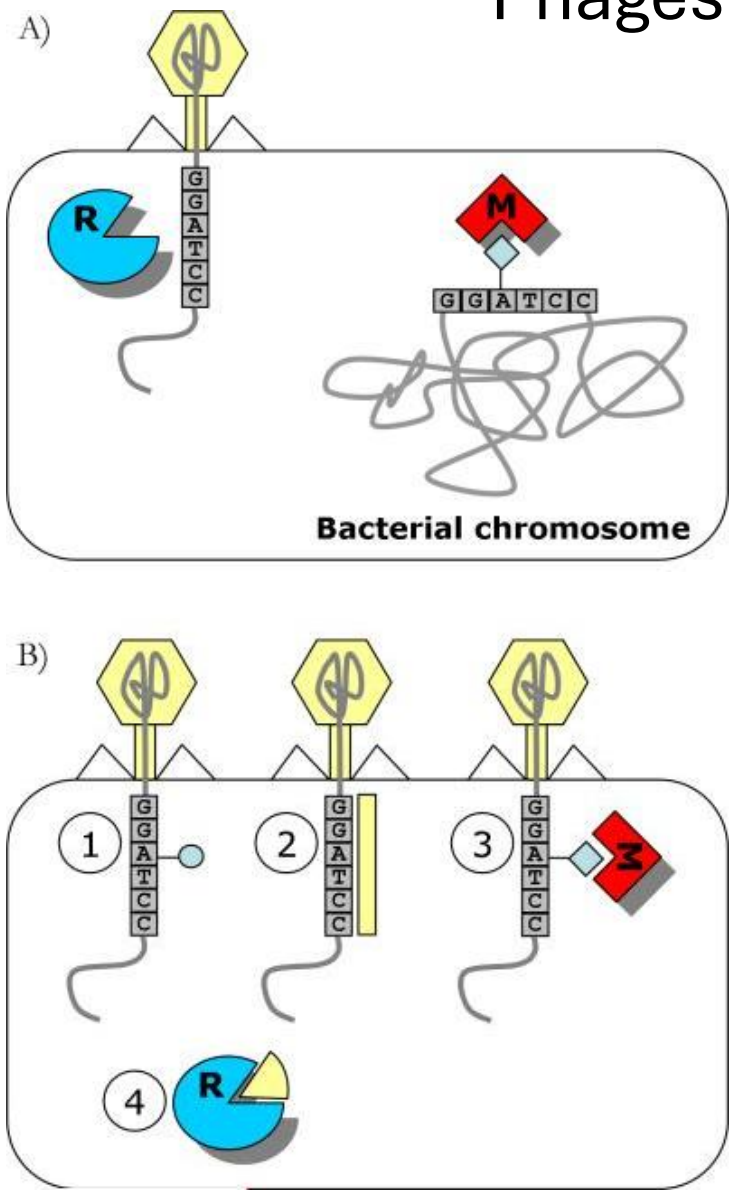
*Not*I has a recognition sequence of 8 base-pairs. 4^8 = 65536 possible combinations, and so *Not*I will theoretically cut one out of 65536 8-base-pair long sites.

Different restriction sites vary in size

-- size determines the expected frequency of the restriction site in the genome

-- more sites / phage genome = more efficient restriction

Phages can also protect themselves!



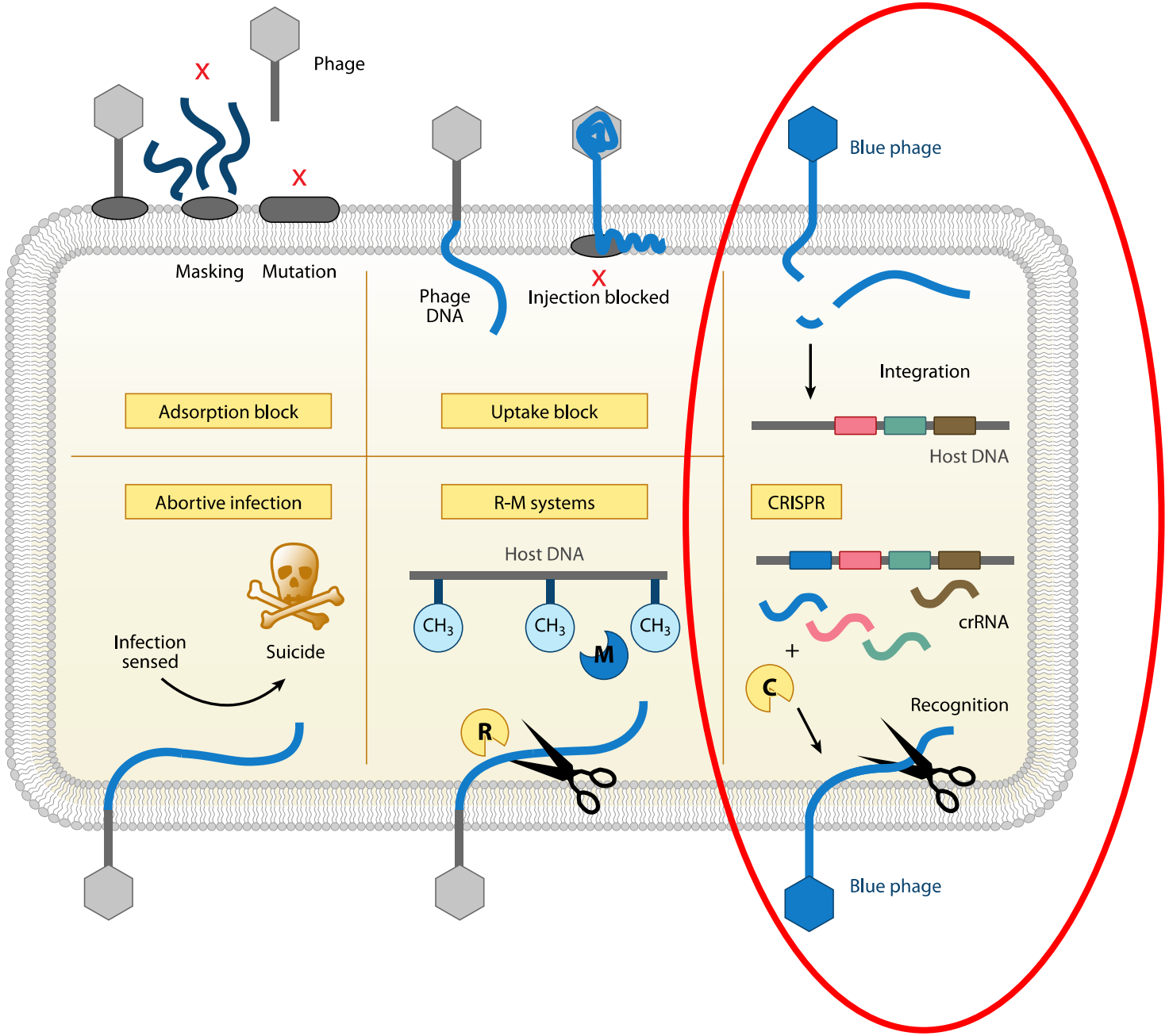
<u>Coevolution</u>

1) New bases to avoid restriction (methylation requires certain base pairs, phage can change these)

2) Phage can hide restriction sites with specific masking proteins

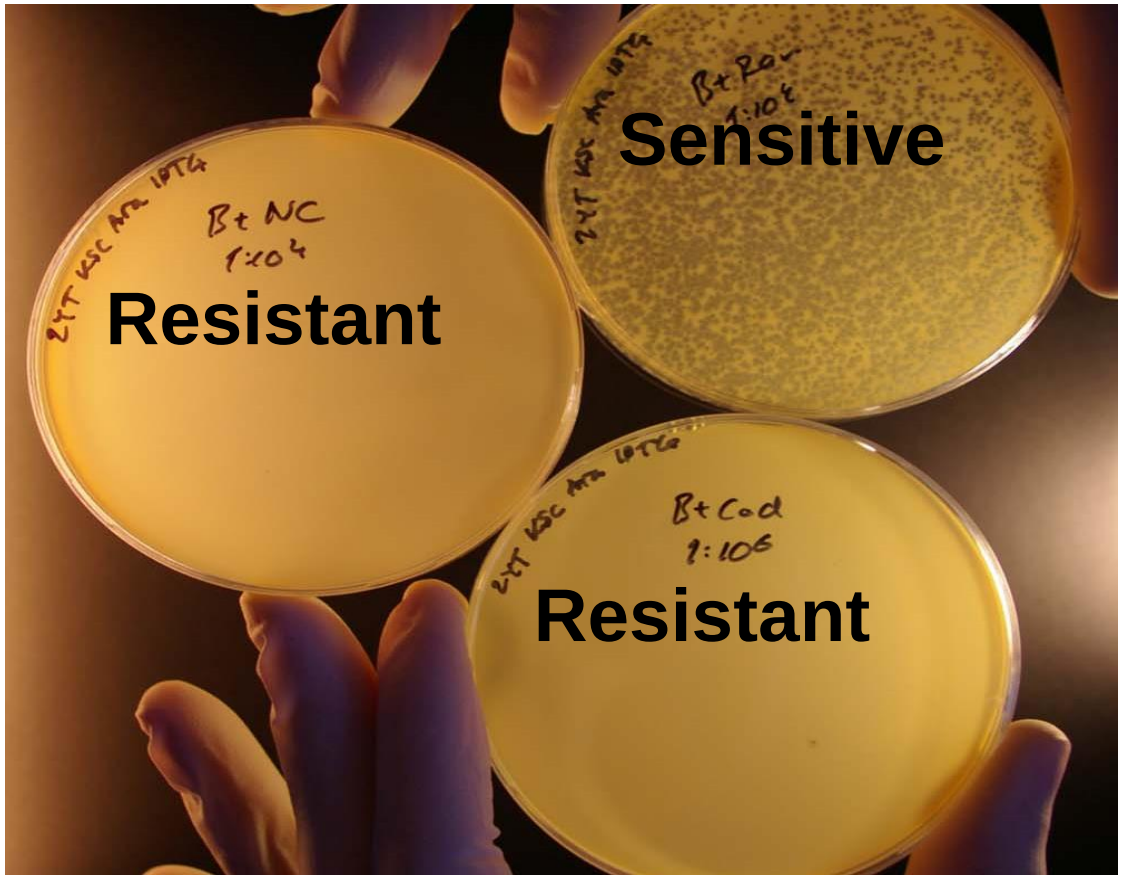
3) Phage can stimulate methylases OR express their own methylases

4) Phage can express proteins that block restriction

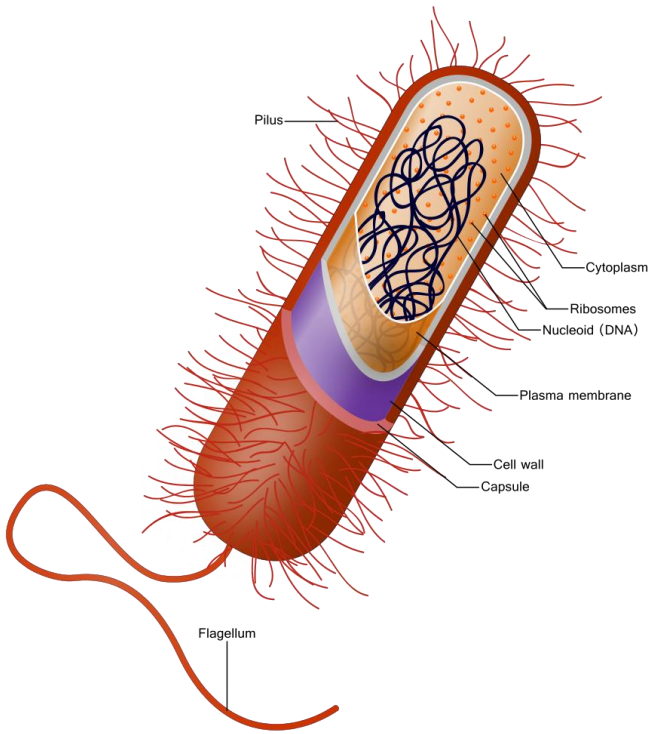

Phage
X
X
Masking
Mutation
Phage DNA
X
Injection blocked
Blue phage
Integration
Host DNA
Adsorption block
Uptake block
Abortive infection
R-M systems
CRISPR
Host DNA
CH3
CH3
CH3
M
Infection sensed
Suicide
crRNA
+
C
Recognition
R
Blue phage

CRISPR: clustered regularly interspaced short palindromic repeat

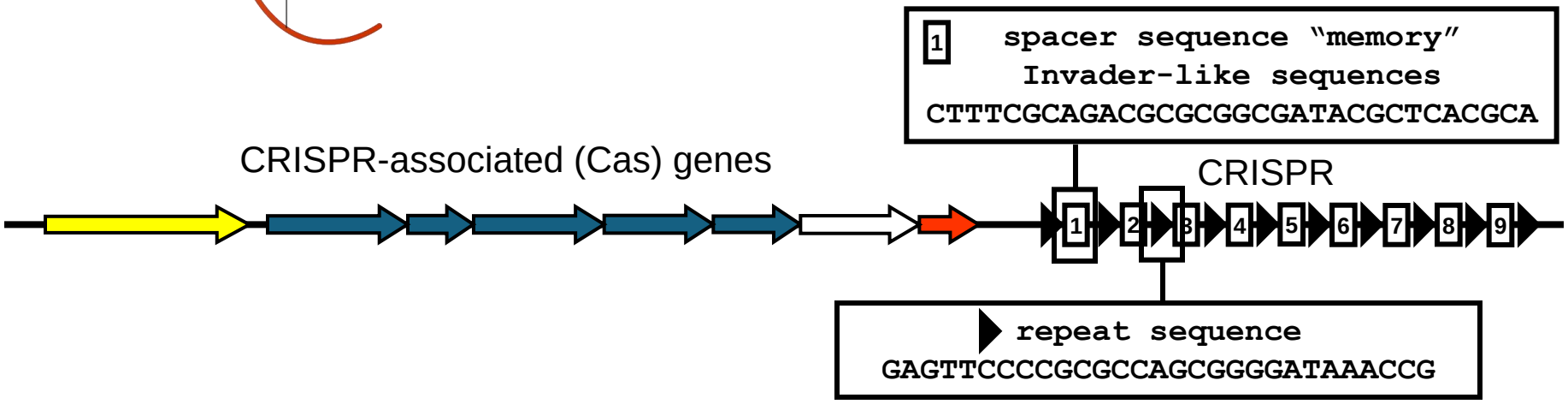
-- present in ~50% of bacteria and more than 90% of archaea
-- immune memory in microbes
-- acquired during the lifetime of a cell (like our immune system)
AND persists through reproduction to daughter cells



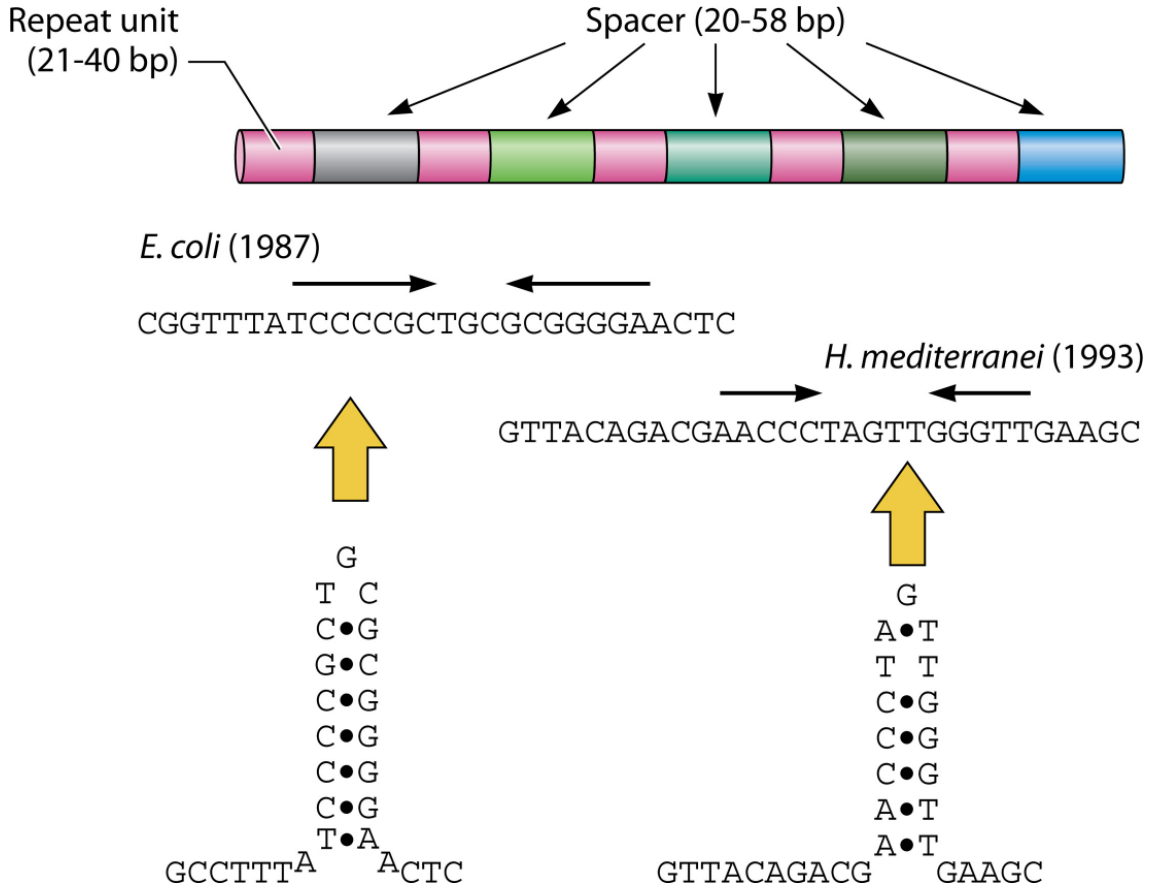
CRISPR: an immune system in Bacteria



Cluster of **R**egularly **I**nterspaced
Short **P**alindromic **R**epeats



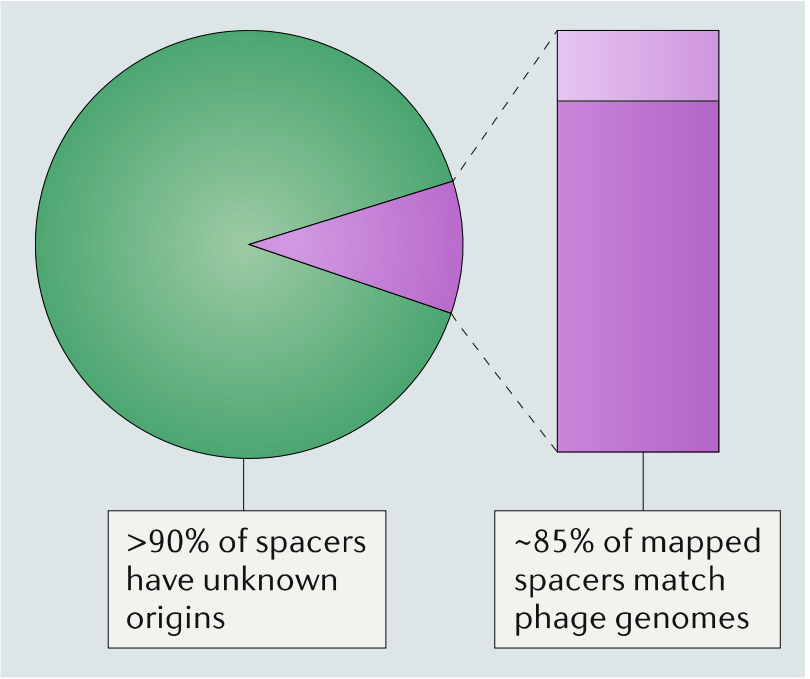
CRISPR: clustered regularly interspaced short palindromic repeat

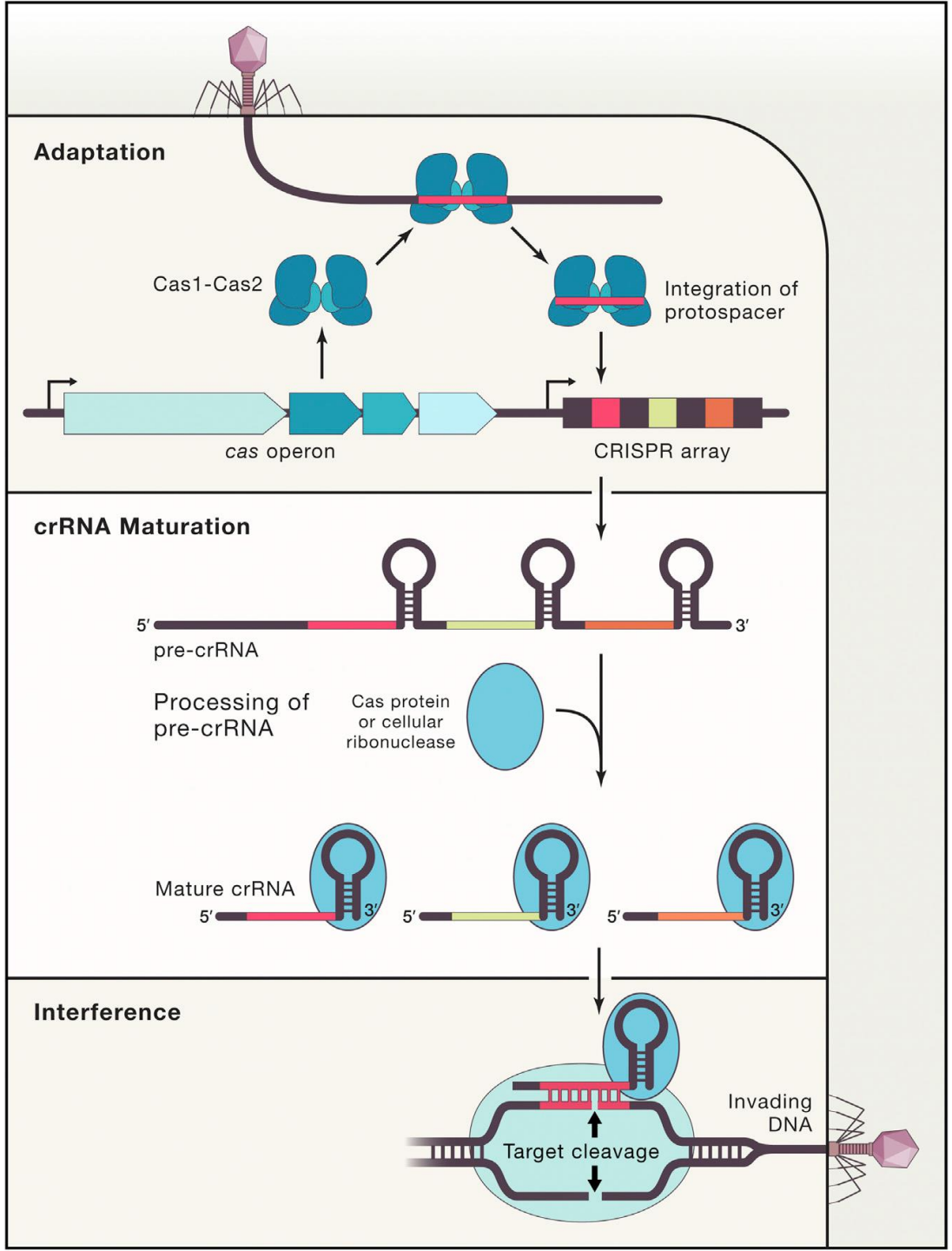


-- Repeats are species specific

-- spacers match phage sequences

-- this match was the first hint to the function of CRISPR



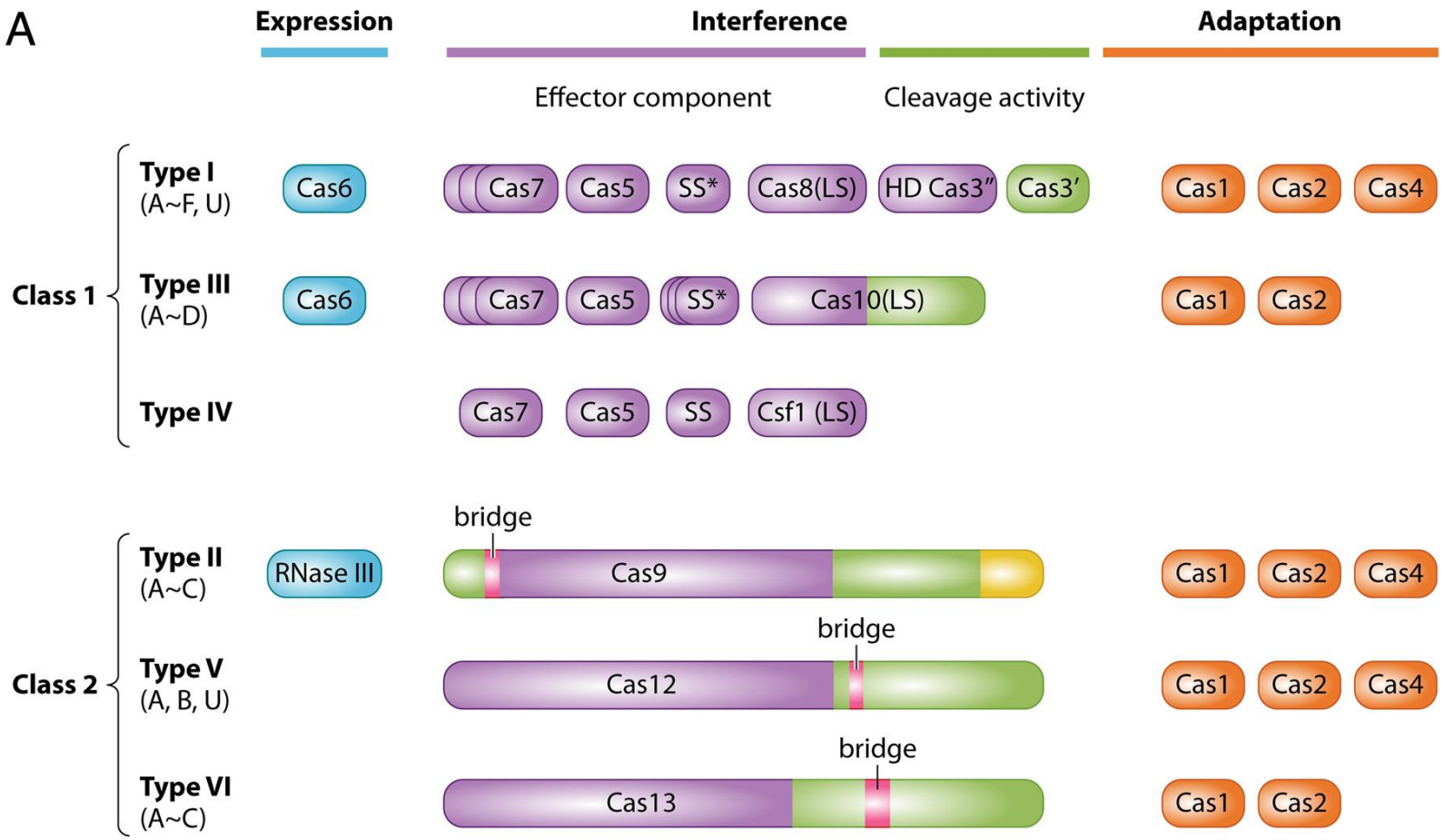


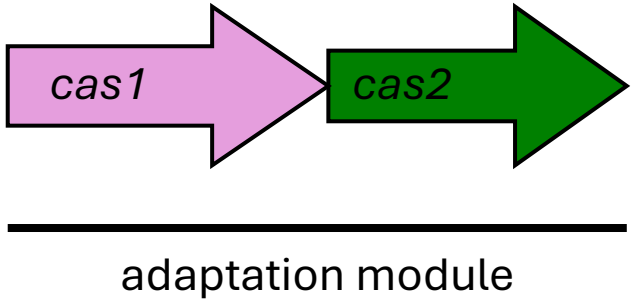
During adaptation, the Cas1-Cas2 complex selects a part of the foreign DNA and integrates it into the host's CRISPR array.

In the next stage (crRNA maturation), the CRISPR array is transcribed into a long pre crRNA that is further processed by Cas proteins

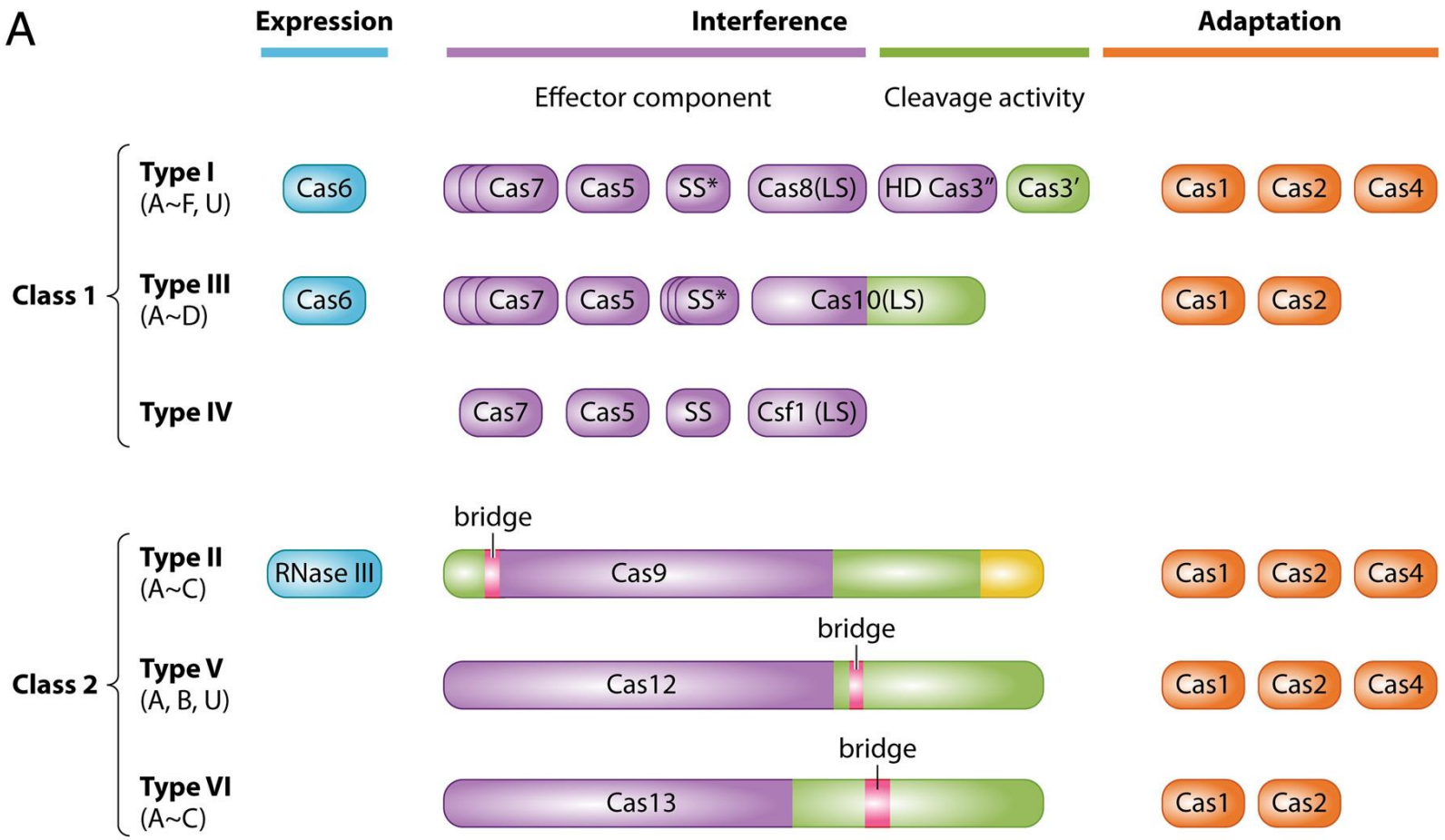
In the interference stage, the mature crRNAs guide Cas nucleases to the cognate foreign DNA. The Cas proteins cleave the foreign nucleic acid upon binding of the crRNA to the target sequence.

CRISPR diversity





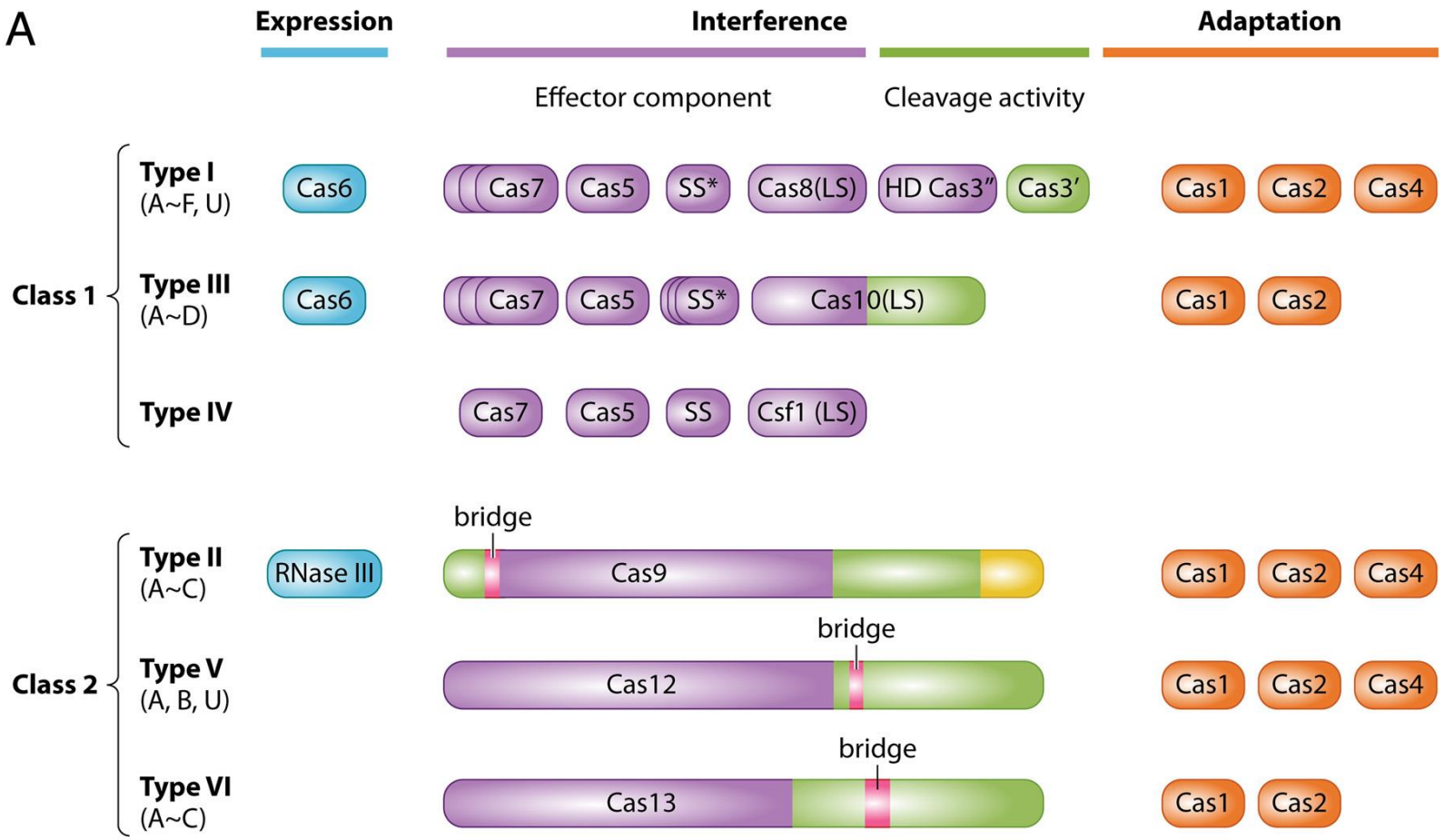
CRISPR diversity



CLASS 1 CRISPR-CAS SYSTEMS

- Lots of Cas genes
- Highly variable architecture
- Abundant in both bacteria and archaea

CRISPR diversity



CLASS 1 CRISPR-CAS SYSTEMS

- Lots of Cas genes
- Highly variable architecture
- Abundant in both bacteria and archaea

CLASS 2 CRISPR-CAS SYSTEMS

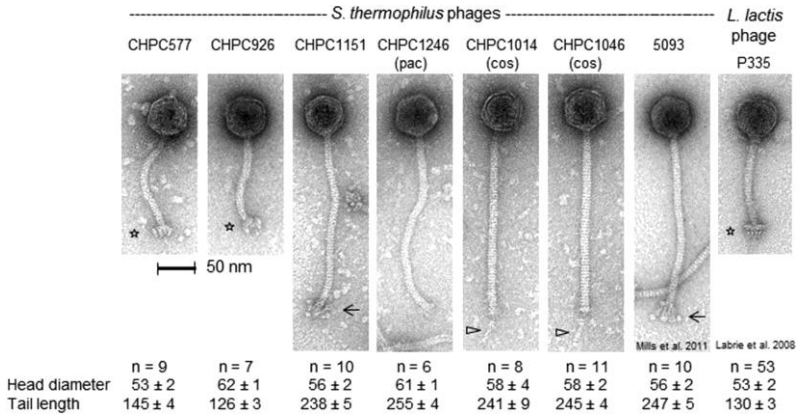
- Simpler & less variable
- Architecture better suited for genome editing
- Almost exclusively in bacteria

Adaptive immunity

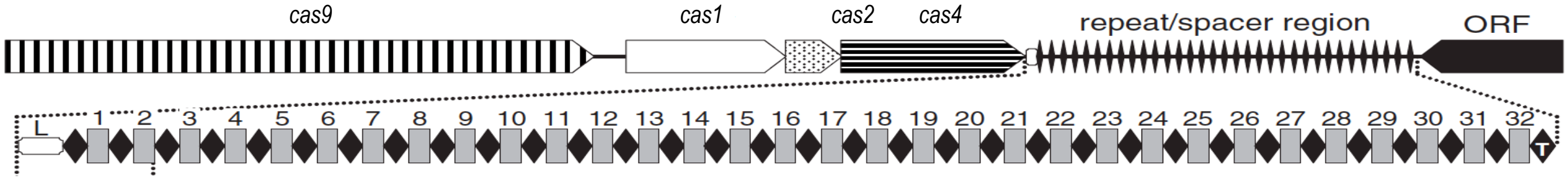


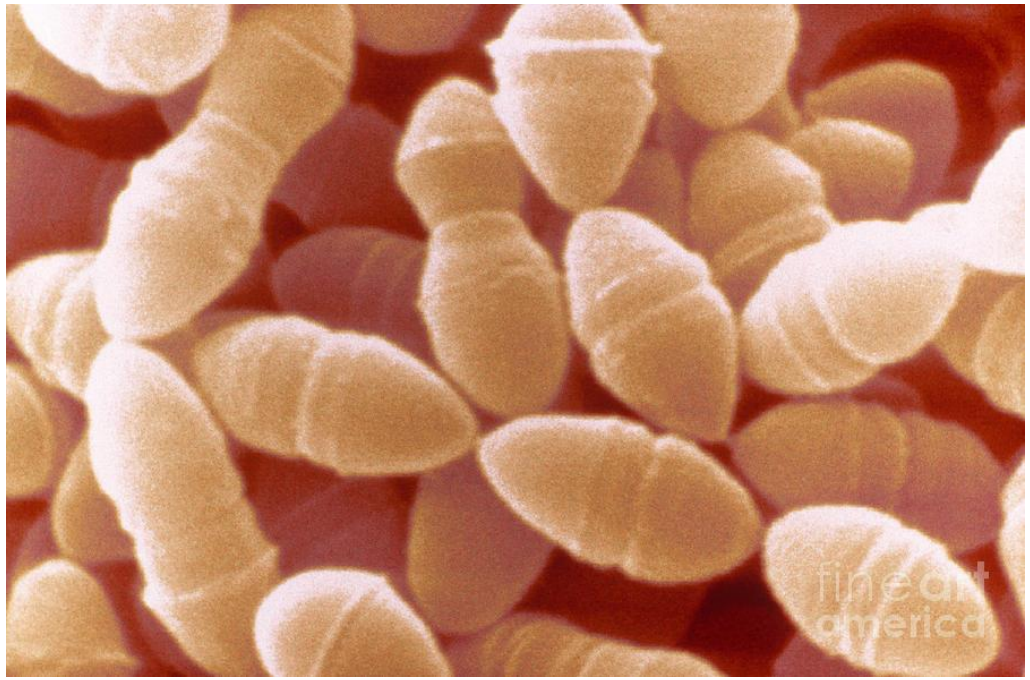
Streptococcus thermophilus



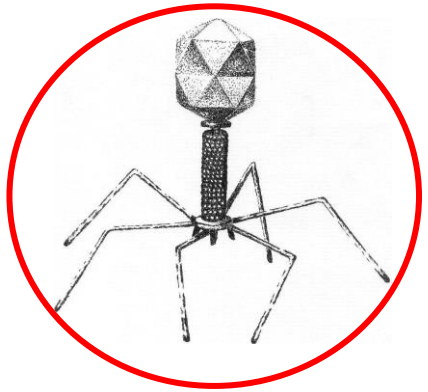


Adaptation – acquisition of a novel phage-derived sequence in the CRISPR

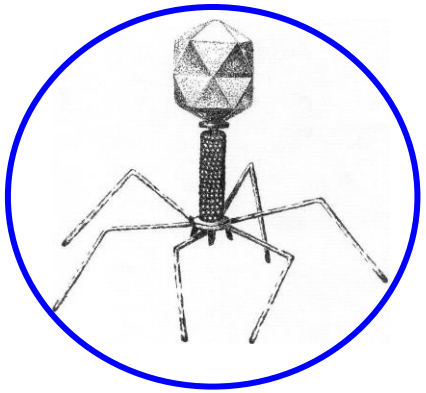




Phage Φ858

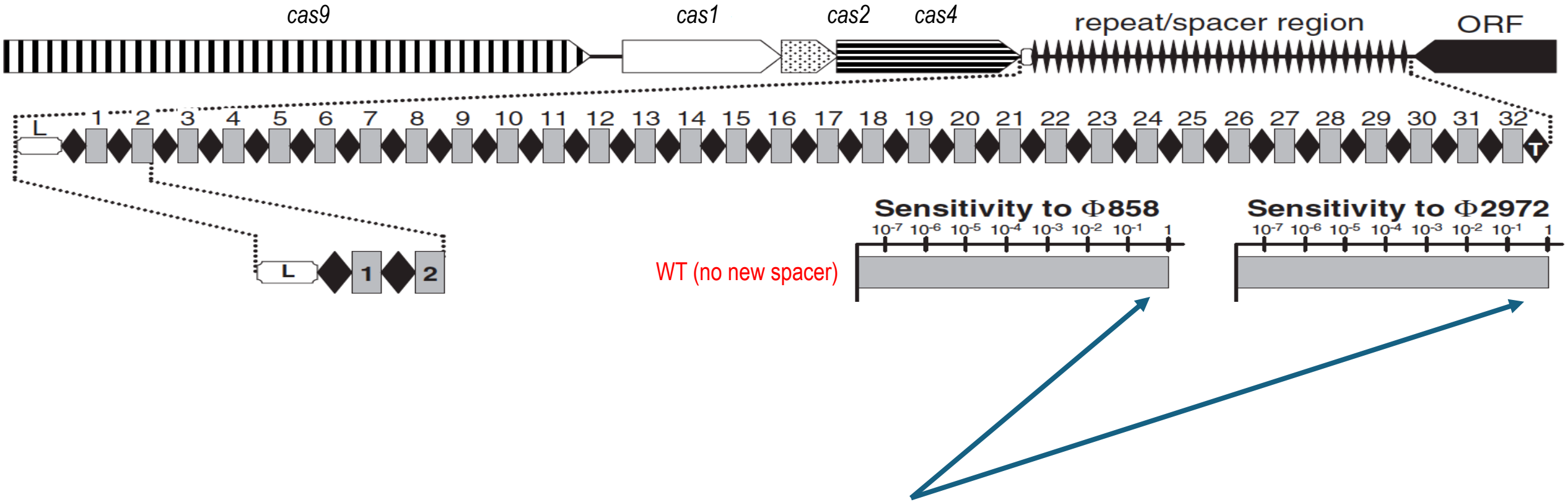


Phage Φ2972



Barrangou et al Science 2008

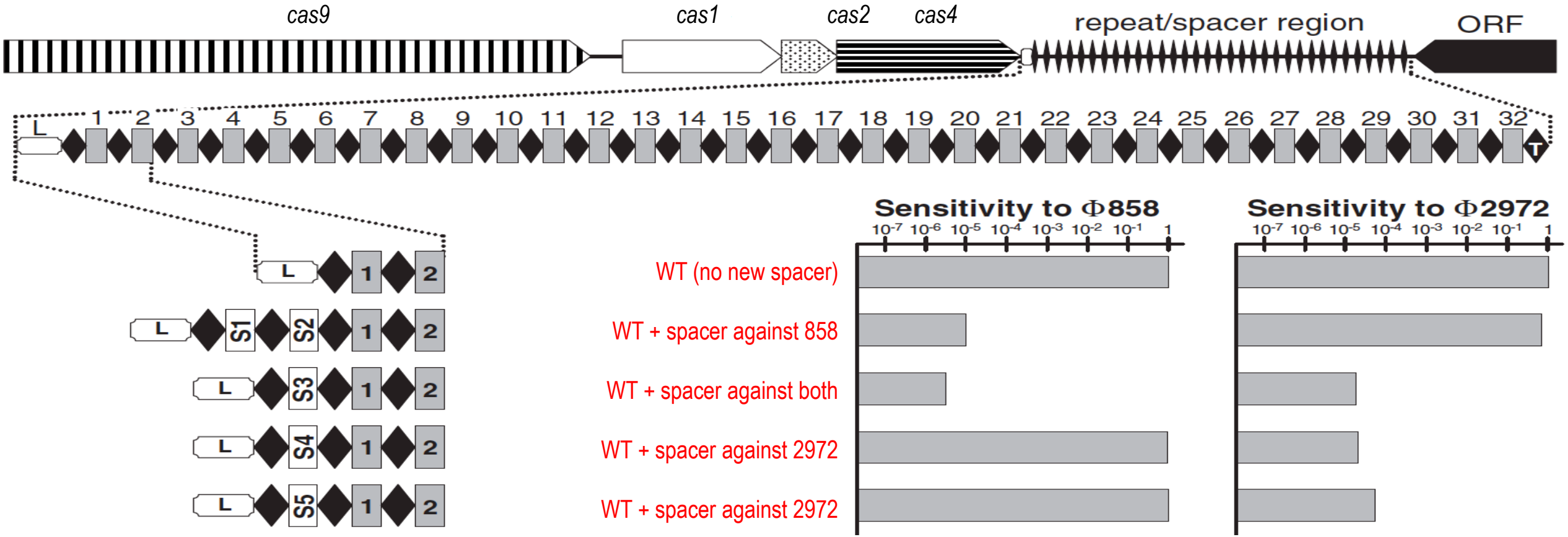
<u>Adaptation</u> – acquisition of a novel phage-derived sequence in the CRISPR



-- in the absence of new spacers, sensitivity to phage is 100%

Barrangou et al Science 2008

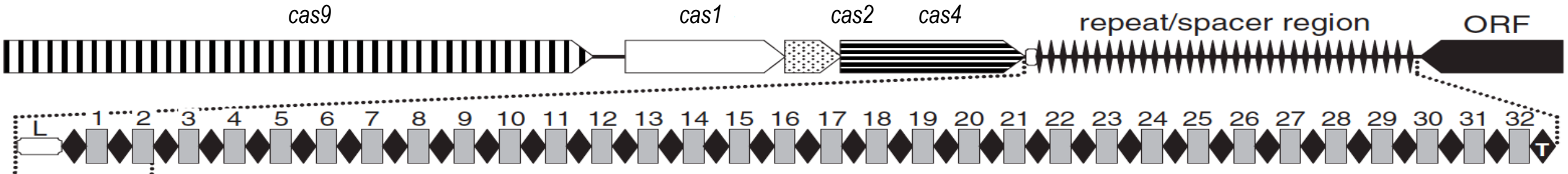
Adaptation – acquisition of a novel phage-derived sequence in the CRISPR

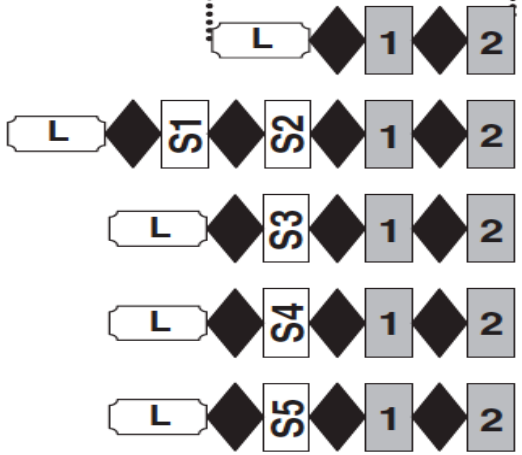


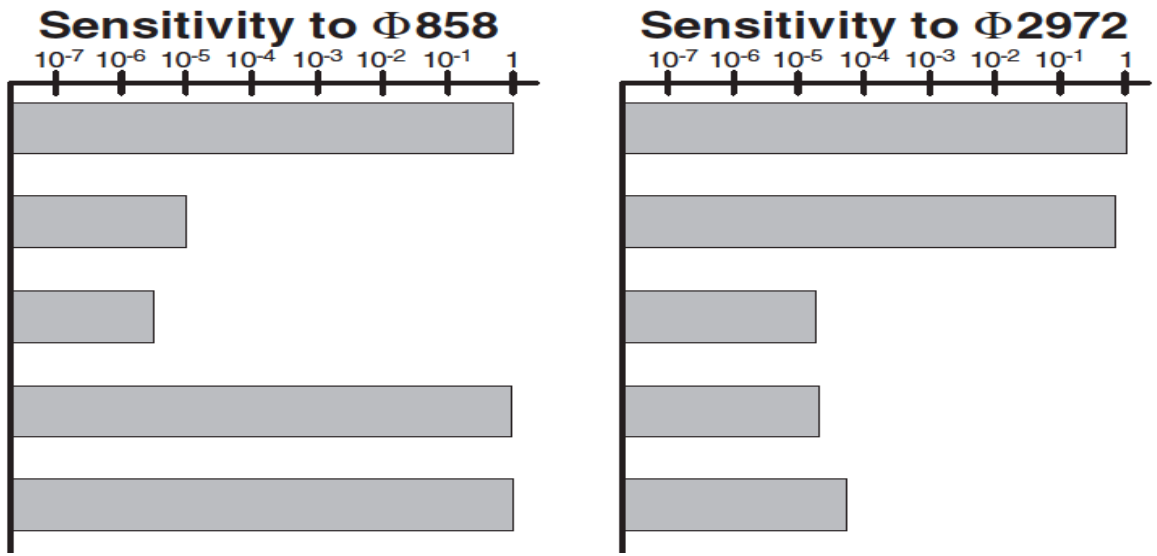
--- new spacers matched sequences in one or both phages

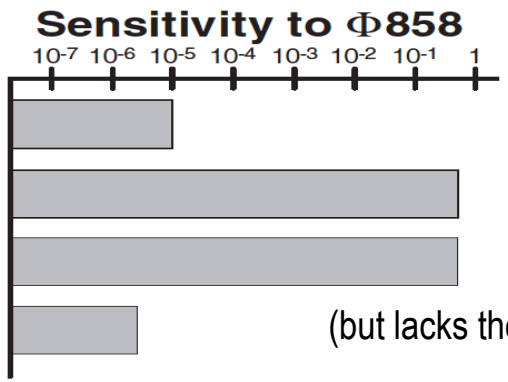
Barrangou et al Science 2008

Adaptation – acquisition of a novel phage-derived sequence in the CRISPR









WT + spacer against 858

WT + spacer against 858 ΔCRISPR

WT + spacer against 858 ΔCas9

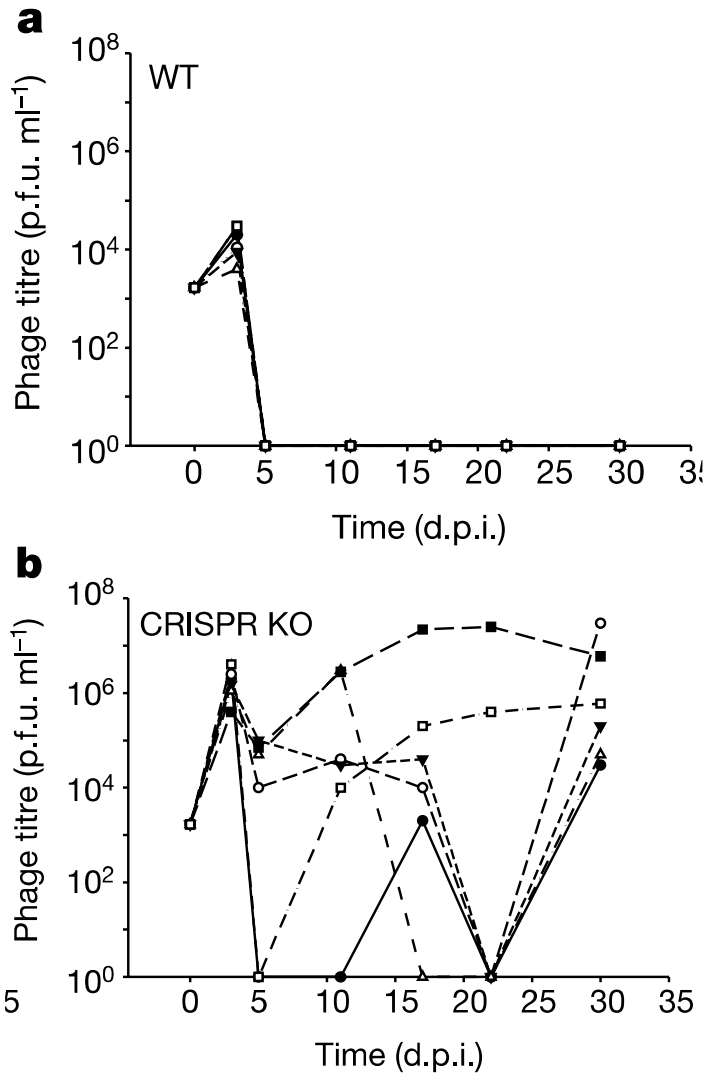
WT + spacer against 858 ΔCas4

(but lacks the ability to acquire spacers)

Barrangou et al Science 2008

The diversity-generating benefits of a prokaryotic adaptive immune system

Stineke van Houte[1], Alice K. E. Ekroth[1], Jenny M. Broniewski[1], Hélène Chabas[1,2], Ben Ashby[3,4], Joseph Bondy-Denomy[5], Sylvain Gandon[2], Mike Boots[3,4], Steve Paterson[6], Angus Buckling[1] & Edze R. Westra[1]

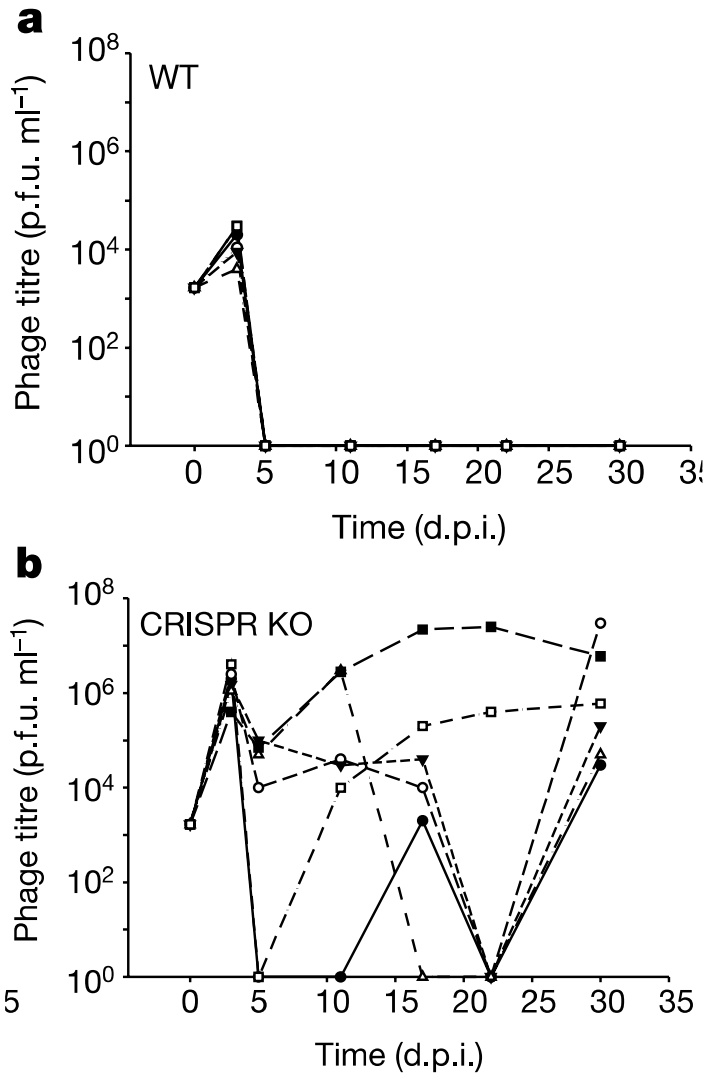


-- with CRISPR the phage go extinct

-- without CRISPR the phage persist

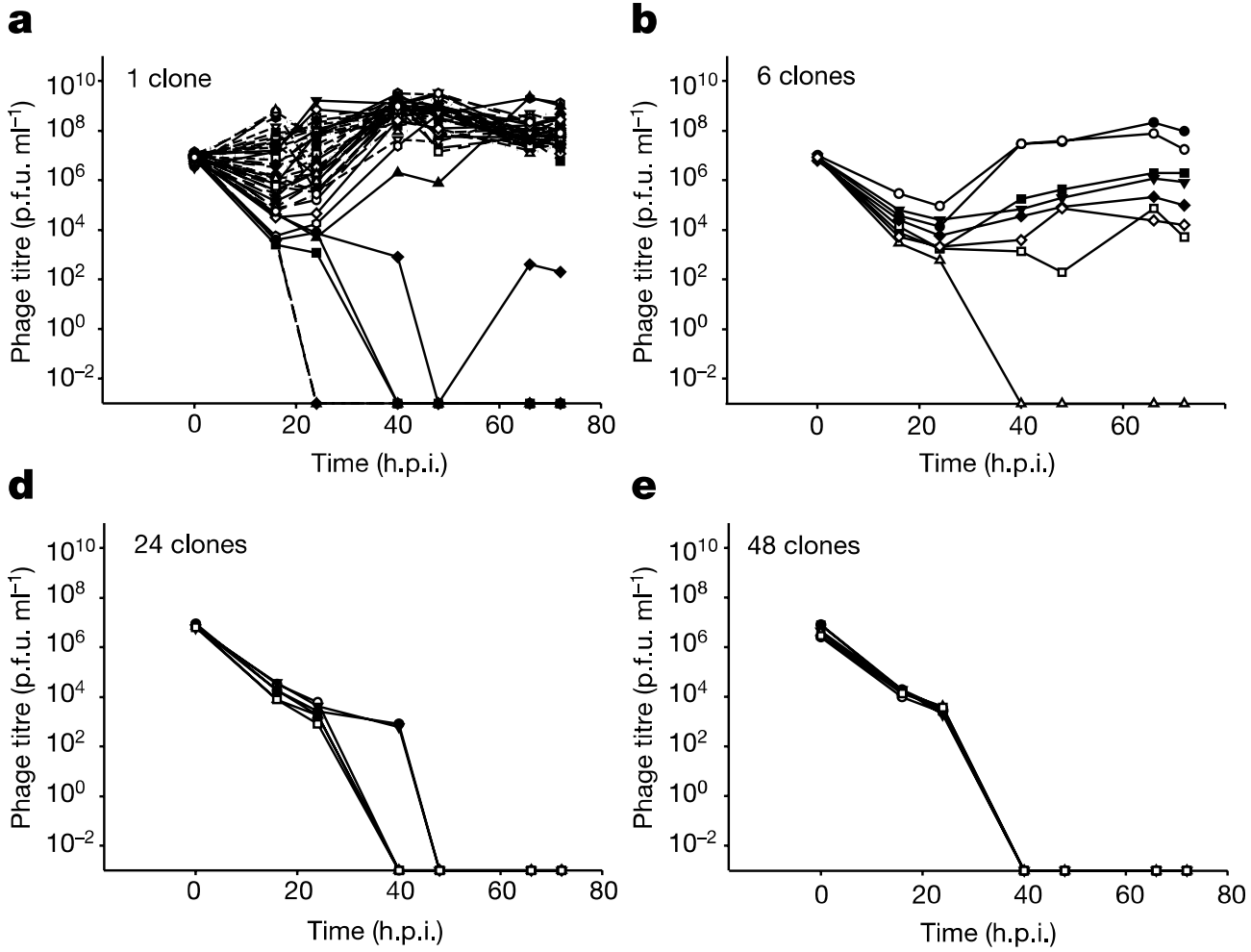
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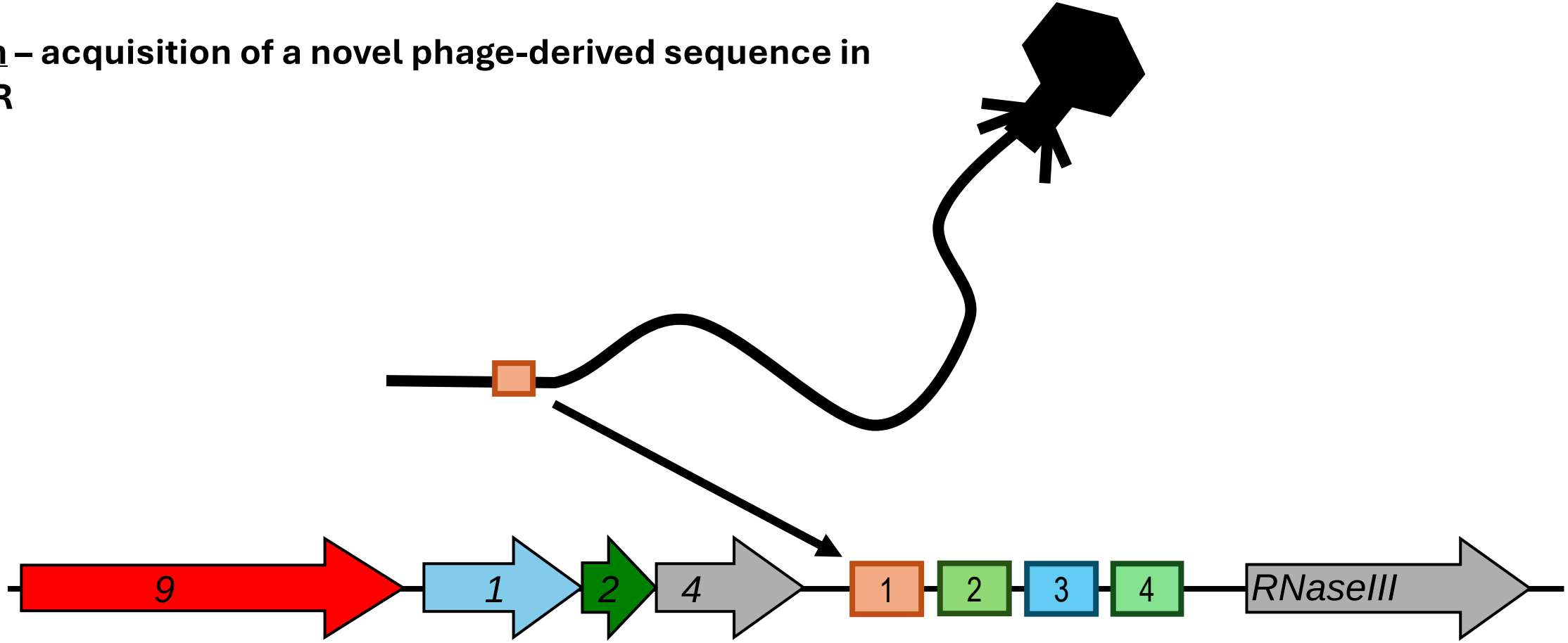
-- with CRISPR the phage go extinct

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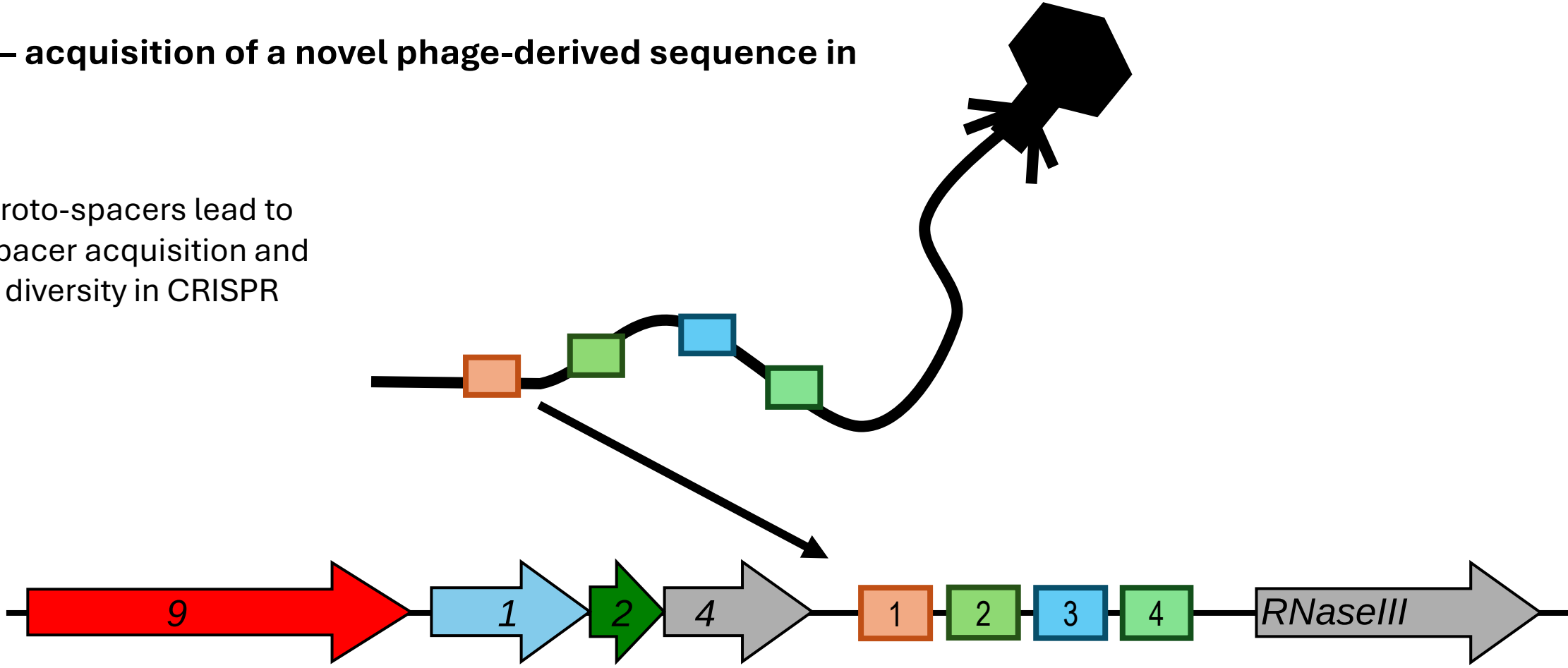
As the number of strains increases (thereby increasing spacer diversity), phage die out because they can't evolve resistance to all spacers

<u>Adaptation</u> – acquisition of a novel phage-derived sequence in the CRISPR



<u>Adaptation</u> – acquisition of a novel phage-derived sequence in the CRISPR

Diverse proto-spacers lead to diverse spacer acquisition and therefore diversity in CRISPR spacers



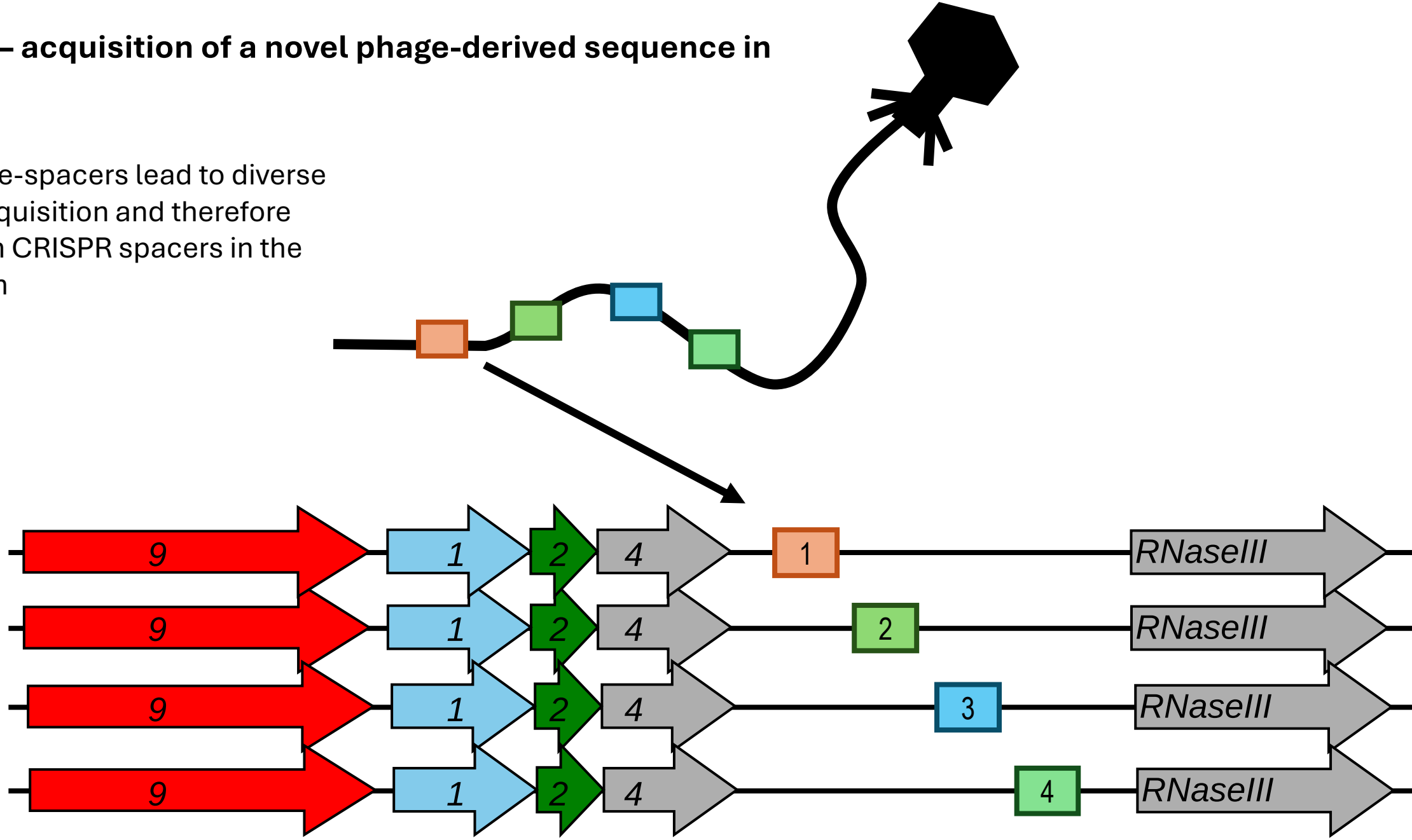
Diversity can arise within a single cell if it has repeated exposure to the same phage

OR

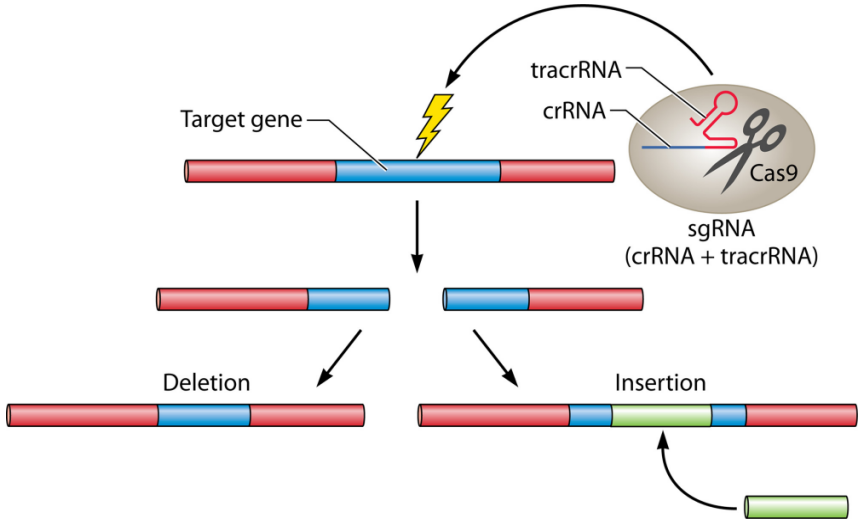
Diversity can arise in different cells, so that each cell carries different spacers

<u>Adaptation</u> – acquisition of a novel phage-derived sequence in the CRISPR

Diverse pre-spacers lead to diverse spacer acquisition and therefore diversity in CRISPR spacers in the population

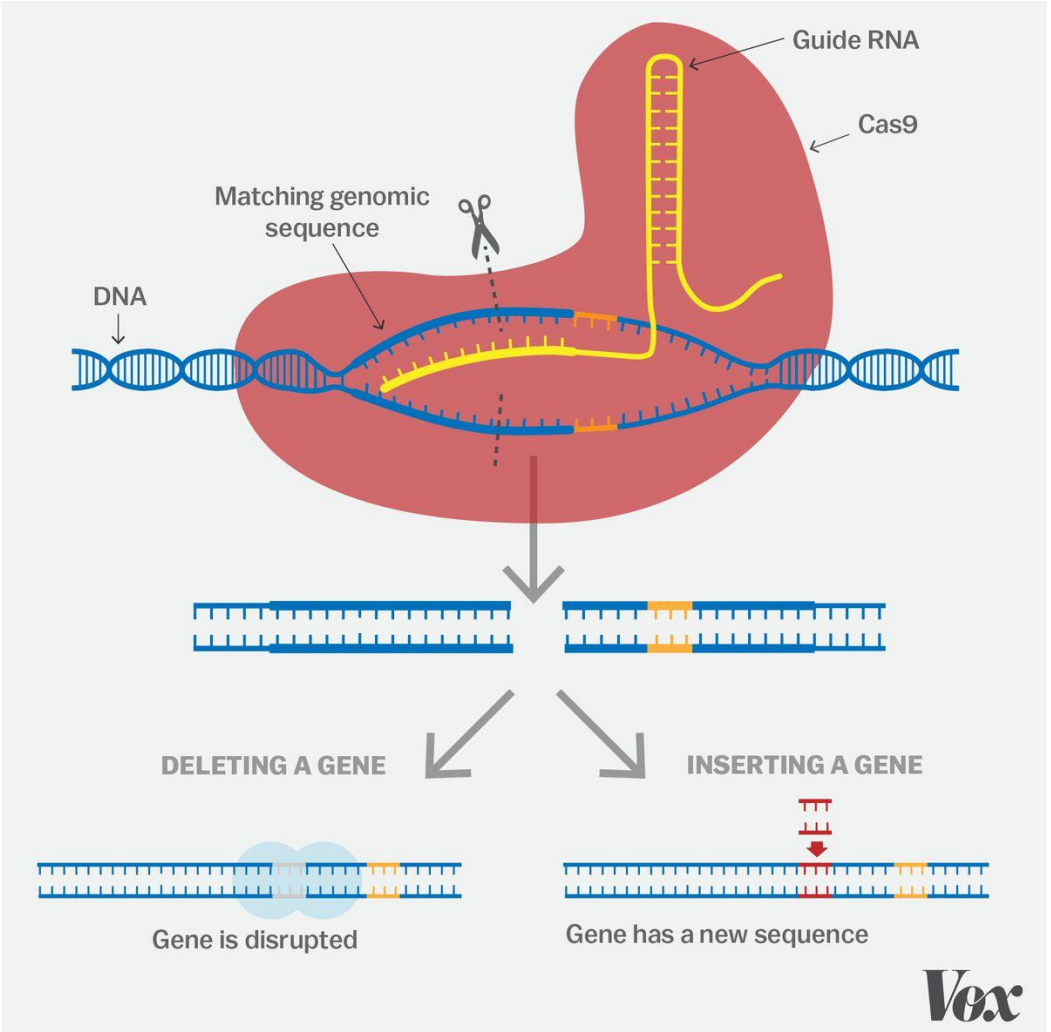


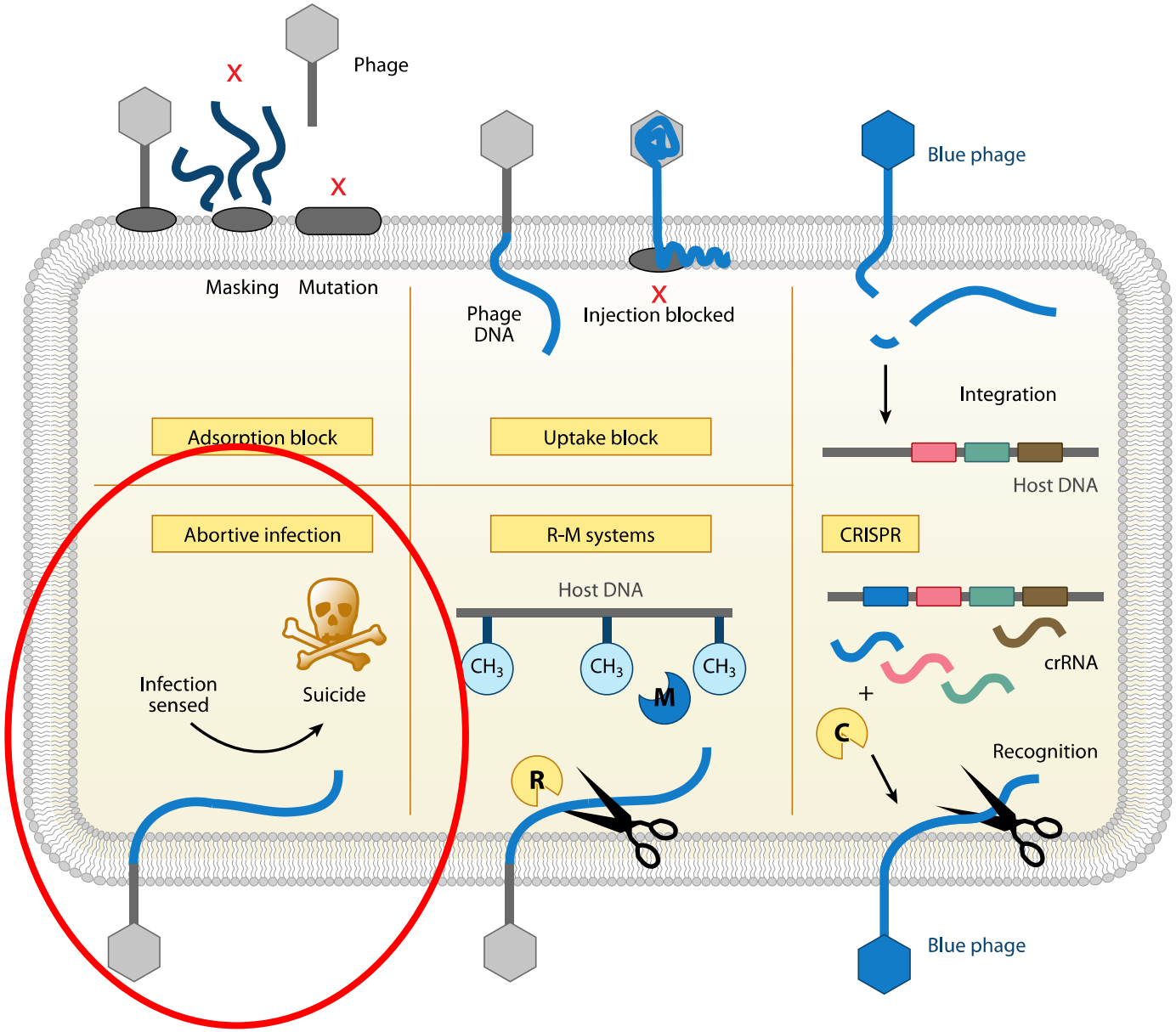
CRISPR-Cas9 for genome editing





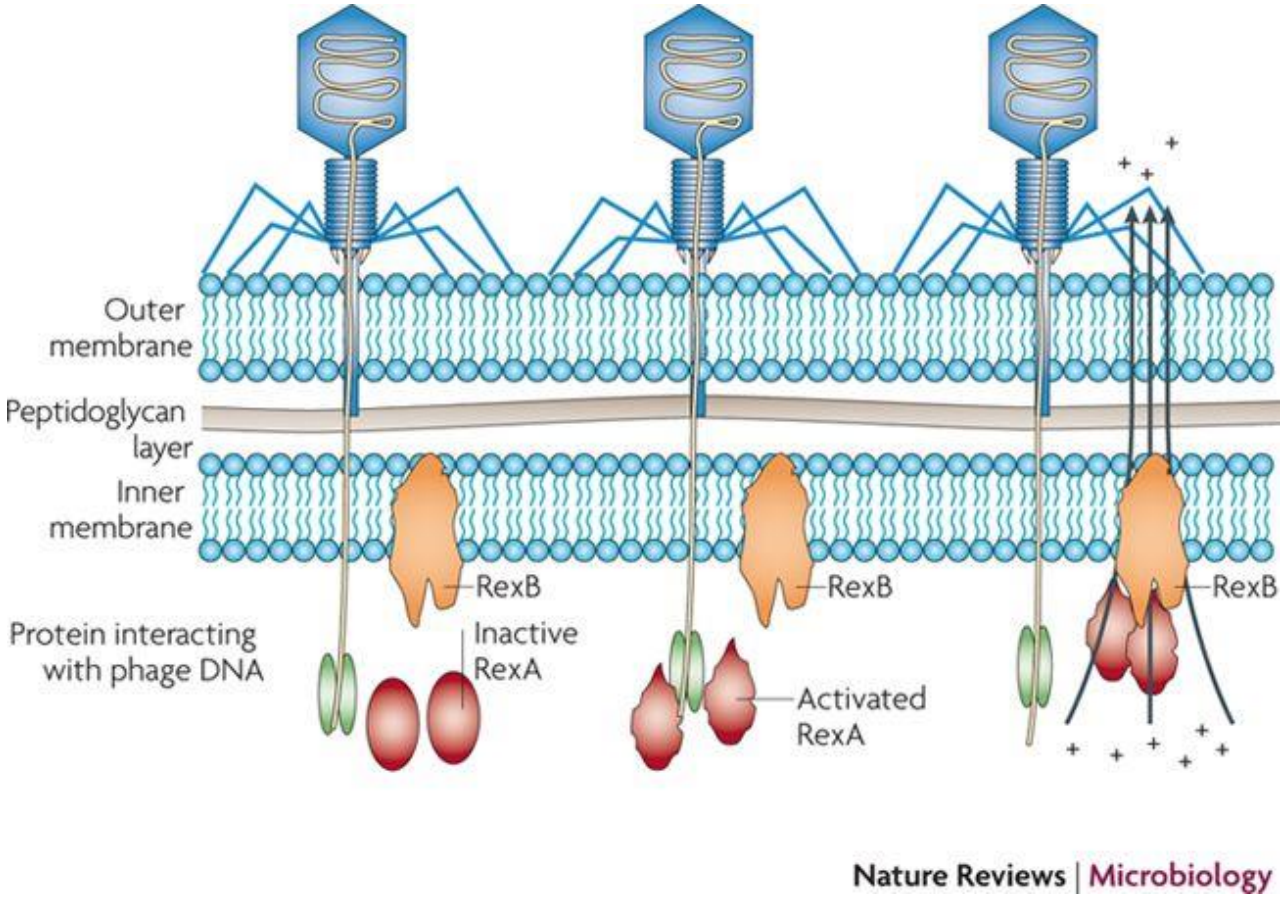
Jennifer Doudna and Emmanuelle Charpentier




Phage
X
X
Masking
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Phage DNA
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Injection blocked
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Adsorption block
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Abortive infection
R-M systems
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Infection sensed
Suicide
crRNA
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Recognition
Blue phage

And if all else fails.....

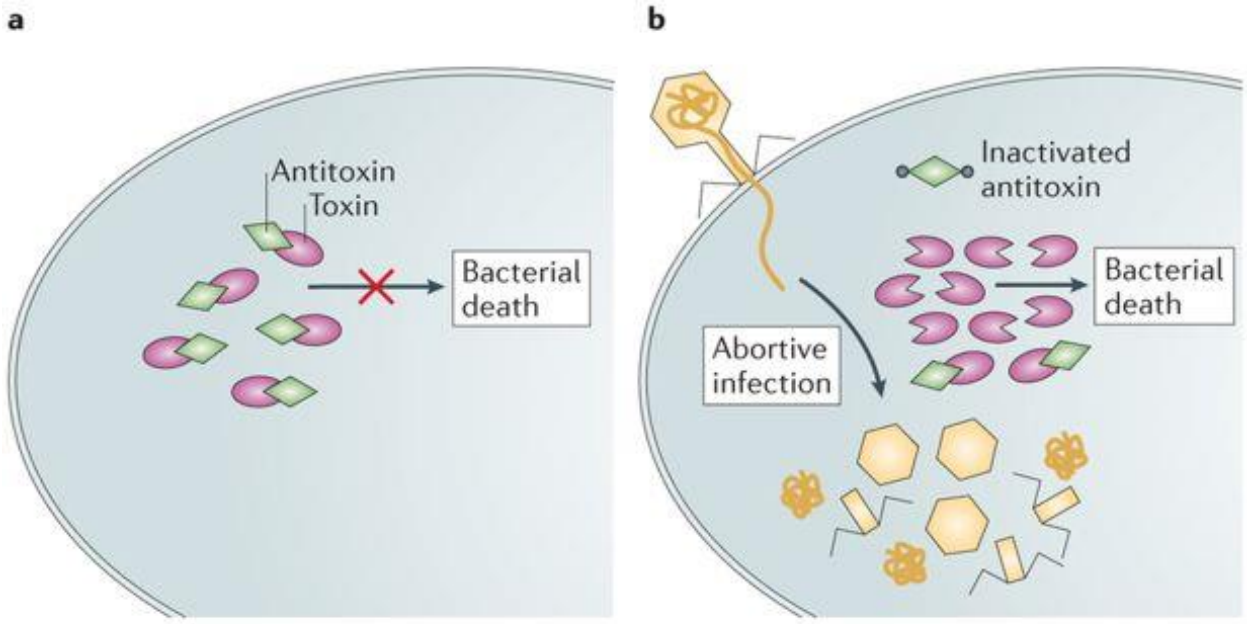
-- abortive infection and cellular suicide



- Activity depends on two proteins, RexA and RexB
- Phage DNA binds to RexA and thereby activates it
- Active RexA turns on RexB
- This creates pores in the cell, thereby leading to death

And if all else fails.....

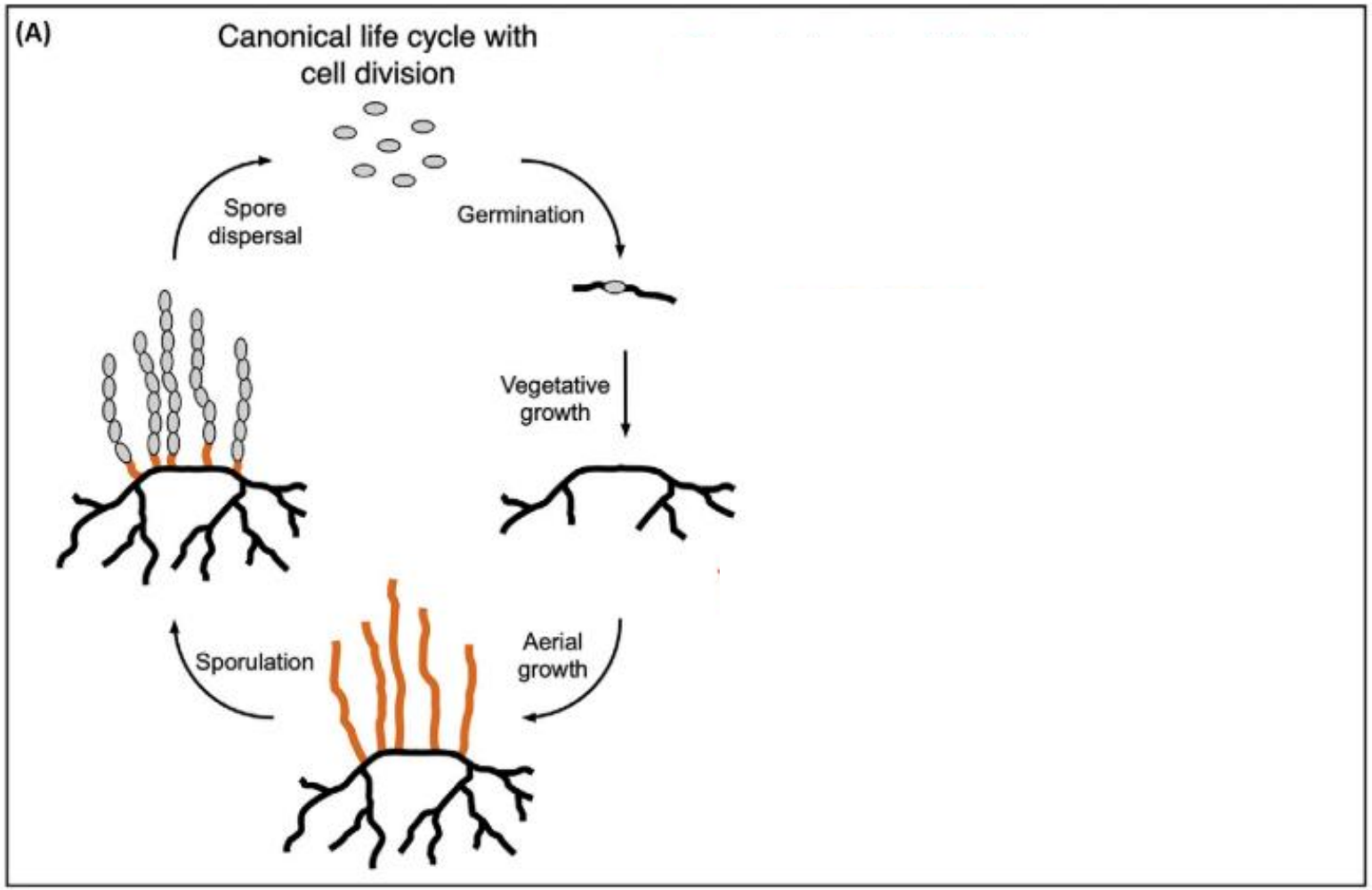
-- abortive infection and cellular suicide



- TA systems (toxin/anti-toxin)
- Cells produce both in equal amounts
- Phage inhibits the anti-toxin
- This leads to an increase in the amount of free toxin
- This leads to cell death

Recently discovered physical escape mechanism from phages

Lifecycle of *Streptomyces*



Ramijan, K. et al., 2018

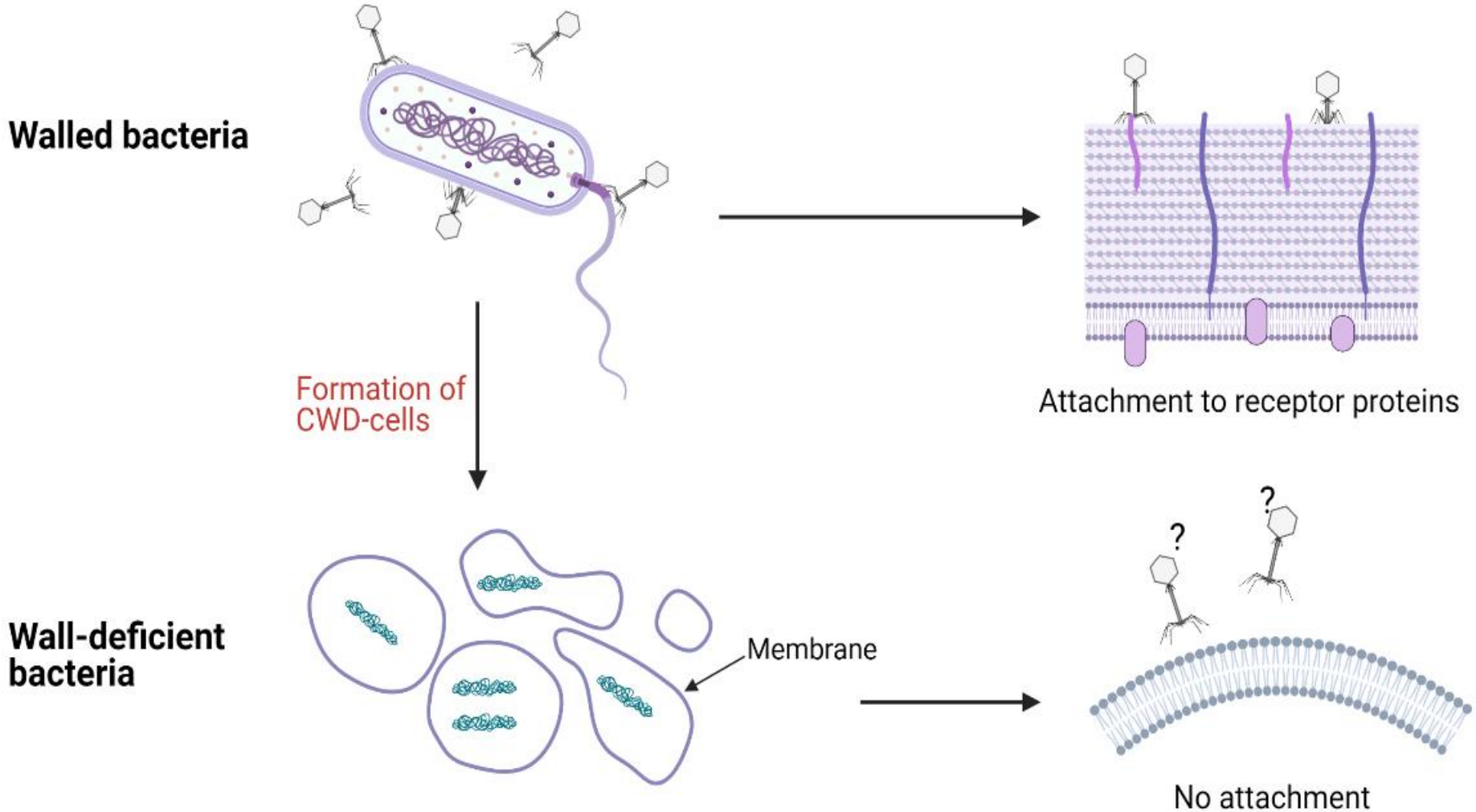
Disadvantage of wall-deficiency:

- Very fragile
- Need osmoprotective environment

Wall-deficient bacteria are resistant:

- Wall-targeting antibiotics
- Phages?

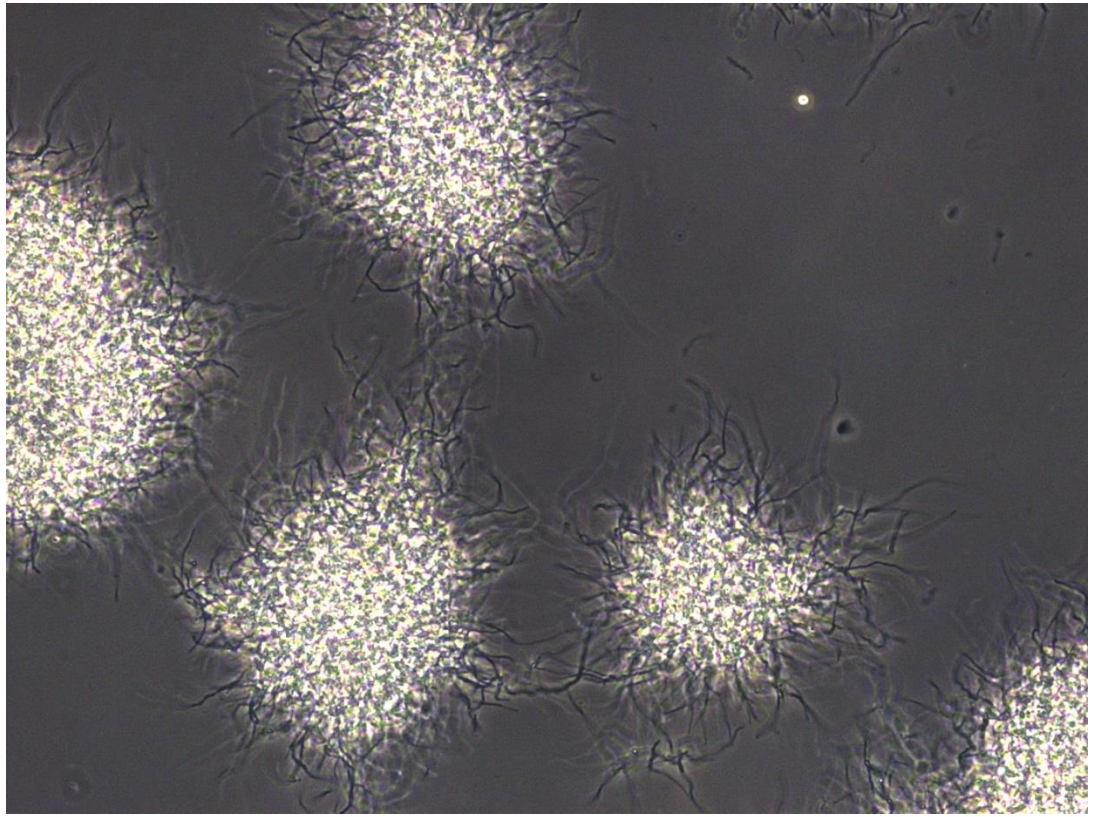
CWD as an escape mechanism against phages?



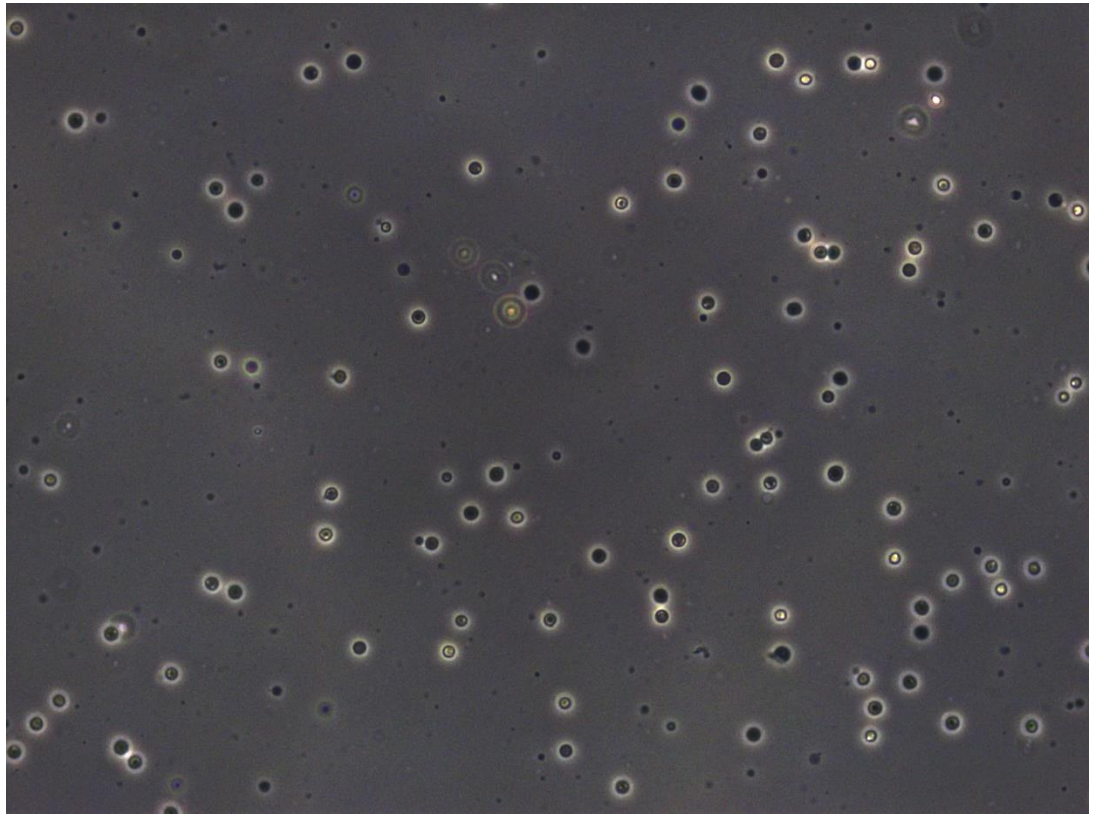
Ongenae, V et al. (2021).

Bacteria can shed their cell wall!

No phage



With phage

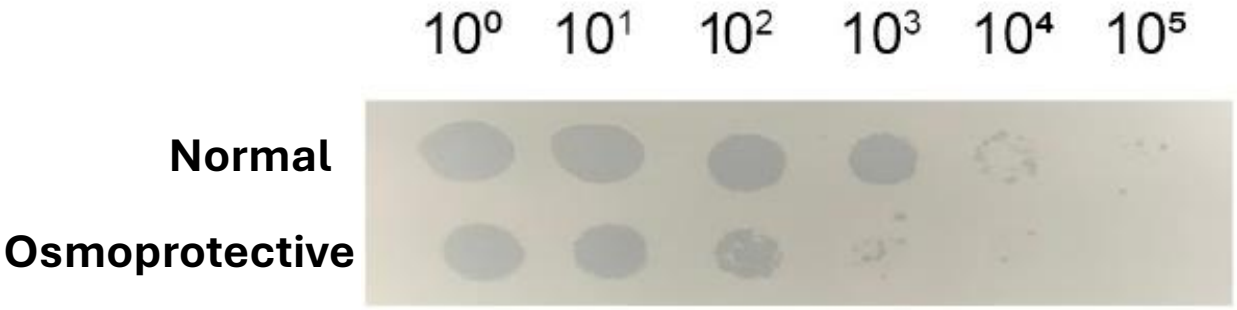


Medium has a high amount of sugar

Wall-deficient cells survive phage infection

- More colonies survive phage infection in osmoprotective medium

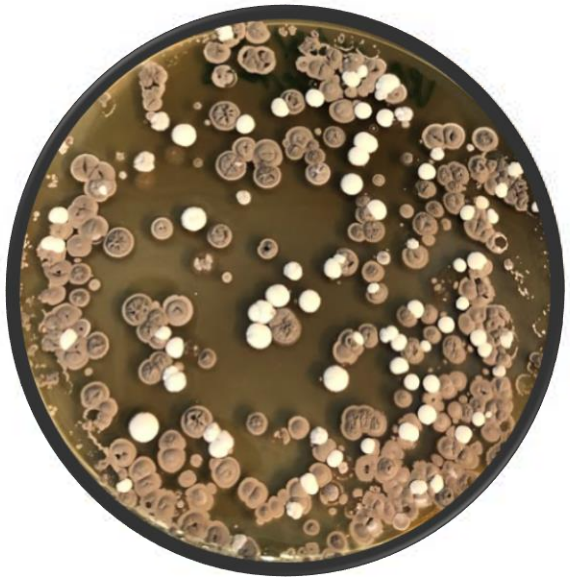
- Less phage production in osmoprotective medium



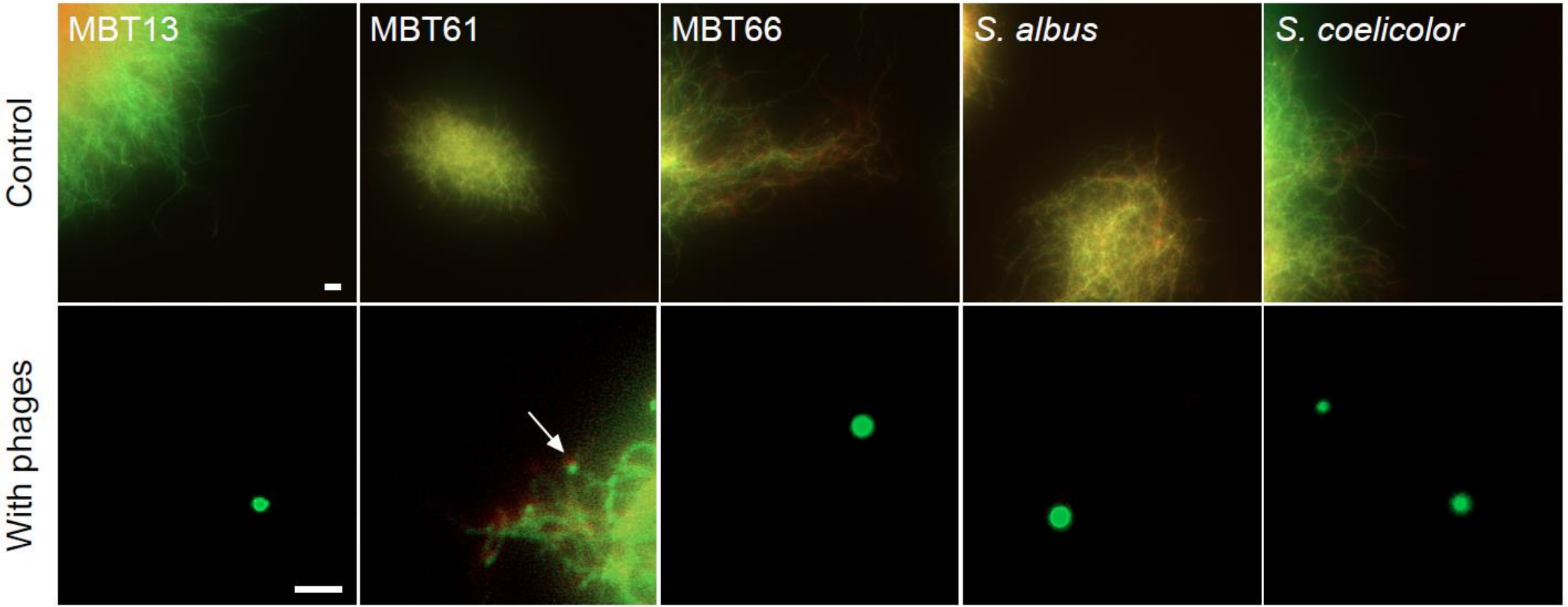
Normal medium



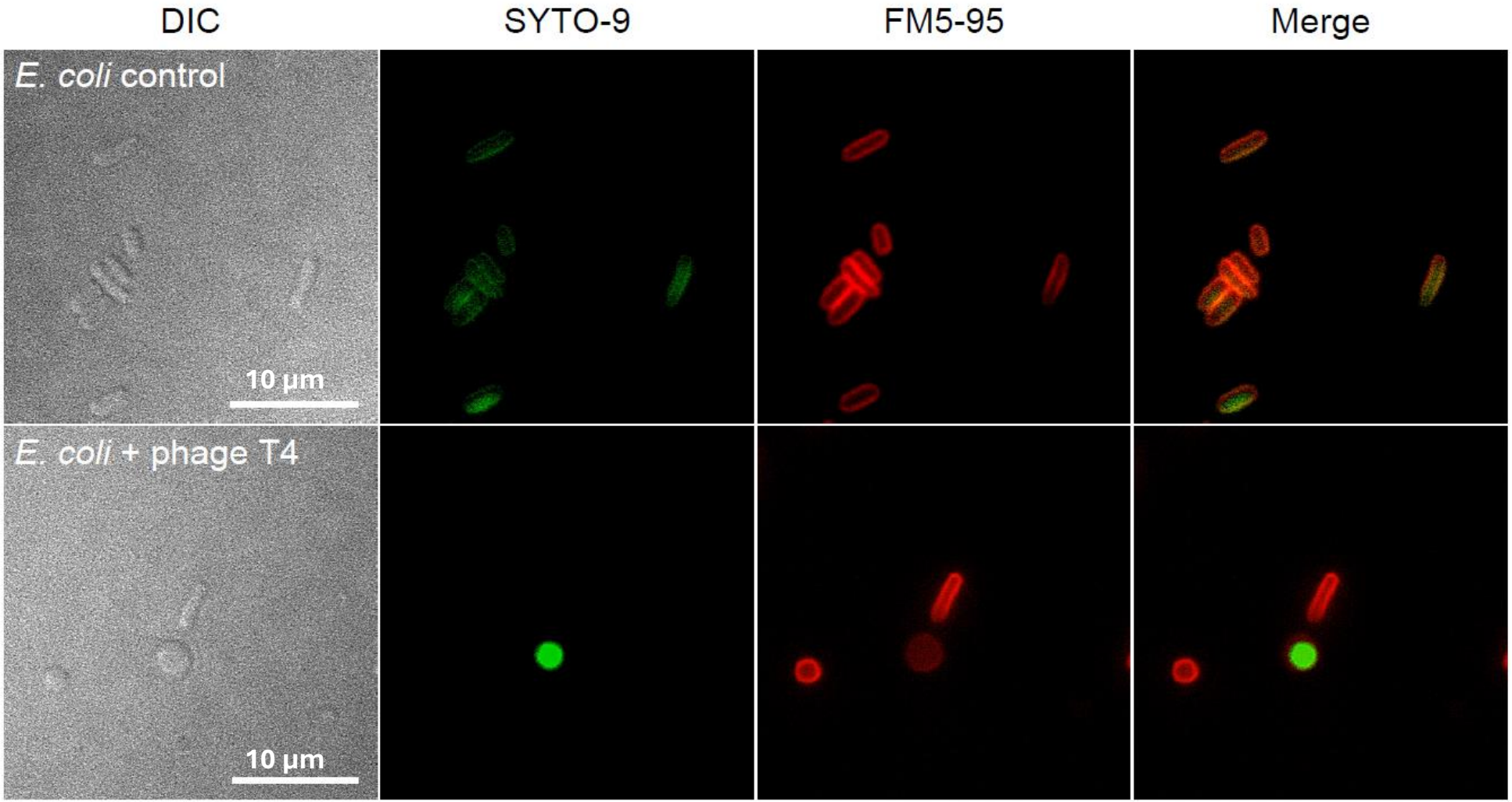
Osmoprotective medium



All tested *Streptomyces* strains formed wall-deficient cells after phage infection



E. coli can make CWD cells after phage infection

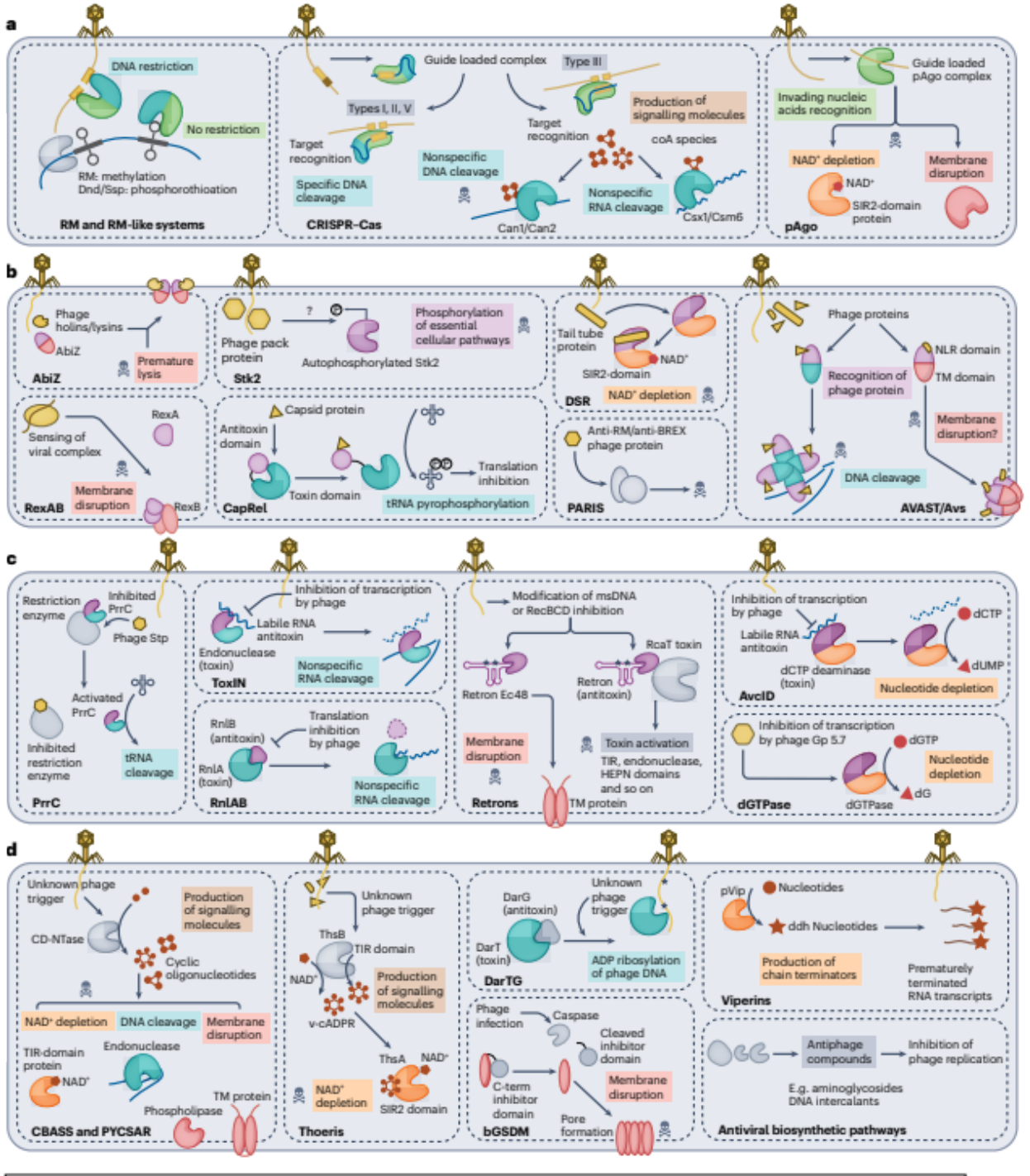


Green = DNA

Red = membrane



DefenseFinder



BREX
Zorya
Thoeris
Druantia
Hachiman
Shedu
Gabija
Septu
Lamassu
Kiwa
Wadjet
Cell wall-deficiency
Etc.

Take-home messages

- Bacteria are constantly under attack by bacteriophages
- Bacteria can evolve defense strategies
  - Hide/mutate receptors
  - Restriction modification
  - CRISPR
  - Abortive infection
  - And many more……
- There is a trade-off for being resistant to phage predation
- But phages can adapt as well! (Coevolution)

